

Integration over a Vanishing Cycle

P. Deligne

Introduction

Let k be a commutative field and let f be a formal power series in two indeterminates x and y over k :

$$f = \sum a_{n,m} x^n y^m.$$

Let $I(f)$ be the formal power series in one indeterminate t

$$I(f) = \sum a_{n,n} t^n.$$

The aim is to prove the following result.

Theorem 1. *If f represents an algebraic function of x and y , and if k has characteristic $p \neq 0$, then $I(f)$ represents an algebraic function of t .*

For a formal series in N indeterminates

$$g = \sum a_n x^n,$$

one defines similarly

$$I_N(g) = \sum a_{n,\dots,n} t^n,$$

and one has the following consequence.

Corollary 1. *If g is algebraic over the field $k(x_1, \dots, x_N)$, and if k has characteristic $p > 0$, then $I_N(g)$ is algebraic over $k(t)$.*

Under the stronger hypothesis $g \in k(x_1, \dots, x_N)$ this result is due to H. Furstenberg. The first purpose here is to give a geometric explanation of Furstenberg's result.

Suppose first that $k = \mathbb{C}$ and that the formal series f is convergent. Then $I(f)$ is convergent and has the integral representation

$$I(f)(t) = \frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt}. \quad (1)$$

Here $\pi : \mathbb{C}^2 \rightarrow \mathbb{C}$ is $(x, y) \mapsto xy$. The expression $(dx \wedge dy)/dt$ is a relative holomorphic differential on $\mathbb{C}^2 - \{(0, 0)\}$, relative to π ; it is also a section of the relative dualizing sheaf at $(0, 0)$, in the sense of Grothendieck duality. In the coordinate y on the fibre $xy = t$ one has

$$\frac{dx \wedge dy}{dt} = \frac{d(t/y) \wedge dy}{dt} = \frac{dy}{y},$$

up to the sign fixed by the chosen order of the variables.

If f converges in a polydisc of radius R , and if $|t| < R^2$, the cycle of integration is $|y| = r$, $|x| = |t|/r$, $xy = t$, with $|t|/R < r < R$. The integral is independent of r . Thus

$$\frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt} = \frac{1}{2\pi i} \int f(t/y, y) \frac{dy}{y},$$

and expansion into the series followed by the residue formula leaves only the terms with equal exponents in x and y .

The integration cycle is the Lefschetz vanishing cycle for π near $(0, 0)$: if $r = |t|^{1/2}$ the cycle visibly collapses as $|t| \rightarrow 0$. This explains the title.

Formula (1) suggests that the operation I has an intrinsic meaning for a morphism of formal schemes $\pi : X \rightarrow S$ over k with $X \simeq \mathrm{Spf} k[[x, y]]$, $S \simeq \mathrm{Spf} k[[t]]$, and π having a non-degenerate quadratic singularity. The two branches of $\pi^{-1}(0)$ are assumed to be defined over k , and an order between them is chosen. Thus we choose coordinates with $\pi(x, y) = xy$, with the first branch of $xy = 0$ being $x = 0$.

Let $\Omega_{X/k}^1$ be the Kähler differentials defined with respect to the adic topology. In coordinates $\Omega_{X/k}^1$ is freely generated by dx, dy , and $\Omega_{S/k}^1$ by dt . Let

$$\omega_{X/k} = \det \Omega_{X/k}^1, \quad \omega_{S/k} = \Omega_{S/k}^1, \quad \omega_{X/S} = \omega_{X/k} \otimes \pi^* \omega_{S/k}^{-1}.$$

In coordinates $\omega_{X/S}$ is generated by $(dx \wedge dy)/dt$. The operation I will be interpreted as a morphism

$$\pi_* \omega_{X/S} \longrightarrow \mathcal{O}_S,$$

or, for $S = \mathrm{Spf}(V)$, as a V -linear map

$$H^0(X, \omega_{X/S}) \longrightarrow V.$$

Section 1 provides this interpretation algebraically. The integration sign in (1) is replaced by Grothendieck's local trace. For a uniformizer t of S , the reduction modulo t^n is denoted with the index n . Let A_n be the complete local ring whose formal spectrum is the reduction X_n ; in coordinates $A_n = k[[x, y]]/(xy)^n$. It is convenient to replace X_n by the ordinary scheme $X'_n = \mathrm{Spec} A_n$. Let O be its closed point. Grothendieck's local trace is a morphism

$$\mathrm{Tr} : H_{\{O\}}^1(X'_n, \omega) \longrightarrow H^0(S_n, \mathcal{O}).$$

The role of the vanishing cycle is played by an element

$$v_n \in H_{\{O\}}^1(X'_n, \mathcal{O}).$$

In coordinates v_n comes from $\mathrm{Ext}_{A_n}^1(A_n/(x+y)^{2n-1}, A_n)$. In Section 2 these classes are related to an element of the Lie algebra of a Picard scheme. Section 3 globalizes the construction and proves the theorem and the corollary.

1 Algebraic analogue of the integral formula

Keep the notation above. The inclusion $X'_1 \subset X'_n$ is a homeomorphism, since X'_1 is defined by a nilpotent ideal in X'_n . The complement of the origin in X'_n has two connected components corresponding to the branches $x = 0$ and $y = 0$ of X'_1 . Let u_n be the section of the structure sheaf of $X'_n - \{O\}$ which is 1 on the first branch and 0 on the second.

Algebraically, the homomorphism

$$A_n = k[[x, y]]/((xy)^n) \rightarrow k[[x, y]]/(x^n) \times k[[x, y]]/(y^n)$$

induces an isomorphism after inverting $x + y$, and u_n is the element whose image is $(1, 0)$. Let v_n be the image of u_n under the boundary map

$$H^0(X'_n - \{O\}, \mathcal{O}) \longrightarrow H_{\{O\}}^1(X'_n, \mathcal{O}).$$

Applying $\mathrm{Ext}^\bullet(-, A_n)$ to

$$0 \rightarrow ((x + y)^N) \rightarrow A_n \rightarrow A_n/((x + y)^N) \rightarrow 0$$

gives a diagram showing that, for $N = 2n - 1$, v_n lies in the image of

$$\mathrm{Ext}_{A_n}^1(A_n/((x + y)^{2n-1}), A_n).$$

1.2. Proposition. For $f \in k[[x, y]]$ and for every n ,

$$\mathrm{Tr} \left(f \frac{dx \wedge dy}{dt}, v_n \right) = -I(f) \pmod{t^n}. \quad (1.2.1)$$

The trace is Grothendieck's local trace. We prove the formula when k is infinite; the general case follows by extension of scalars. Let $I'(f)$ denote the left-hand side of (1.2.1). It is a continuous $k[[t]]/(t^n)$ -linear map $A_n \rightarrow k[[t]]/(t^n)$. For $\lambda, \mu \in k^*$ let $\sigma(\lambda, \mu)$ be the automorphism

$$(x, y, t) \mapsto (\lambda x, \mu y, \lambda \mu t).$$

It preserves $(dx \wedge dy)/dt$ and v_n , hence

$$I'(\sigma(\lambda, \mu) f) = \sigma(\lambda, \mu)^* I'(f). \quad (1.2.2)$$

Applying this to monomials gives $I'(x^a y^b) = 0$ unless $a = b$. By continuity,

$$I'(f) = I(f)I'(1).$$

The scalar $I'(1)$ is independent of n , so it remains to compute it for $n = 1$.

For $n = 1$, $X'_1 = \mathrm{Spec} k[[x, y]]/(xy)$ is the union of two smooth formal arcs meeting transversely. Sections of ω are pairs of meromorphic differentials (α, β) on the two branches with at most simple poles at O and residues summing to zero. On the punctured curve arbitrary pole orders are allowed. The composite

$$H^0(X'_1 - \{O\}, \omega) \rightarrow H^1_{\{O\}}(X'_1, \omega) \xrightarrow{\mathrm{Tr}} k$$

is the negative of $(\alpha, \beta) \mapsto \mathrm{Res} \alpha + \mathrm{Res} \beta$. In these terms $(dx \wedge dy)/dt$ and u_1 give residue sum 1, proving (1.2.1).

2 Relation with the Picard functor

Let $S = \mathrm{Spec} k[[t]]$, let s be its closed point and η its generic point. Let $\pi : Y \rightarrow S$ be proper and flat, purely of relative dimension one, and let $O \in Y_s(k)$. Assume that the completion of Y at O is the formal singularity considered above. Thus Y is regular at O , O is an isolated non-smooth point of π , the singularity is non-degenerate quadratic, and the two branches are defined over k and ordered.

We may further suppose Y regular and connected. Let Y_η be the generic fibre over $k((t))$.

2.2. Lemma. The curve Y_η is absolutely irreducible and generically smooth over $k((t))$.

Indeed, the special fibre is connected and has a rational point, hence is absolutely connected. It is smooth near O away from O , so after a separable extension it has smooth rational points. The corresponding base change has a section; regularity then implies irreducibility, and absolute connectedness gives absolute irreducibility. The smooth locus of the generic fibre is non-empty and therefore dense.

Let C be a complete reduced generically smooth curve over a field K , and D a Cartier divisor on C . The generalized Jacobian $\mathrm{Pic}(C; D)$ represents invertible sheaves on C_T trivialized on D_T . It is smooth, and

$$\mathrm{Lie} \mathrm{Pic}(C; D) = H^1(C, \mathcal{O}(-D)). \quad (2.3.1)$$

If C is smooth, the map

$$j : C - D \rightarrow \text{Pic}(C; D), \quad u \mapsto [\mathcal{O}(u(T))] \quad (2.3.2)$$

relates Serre duality with the pullback of invariant differentials on the Picard scheme.

Returning to Y/S , let Z be a relative Cartier divisor disjoint from O , and let $\text{Pic}(Y; Z)^0$ be the relative generalized Picard scheme parametrizing line bundles on Y trivialized along Z and of degree zero on each geometrically irreducible component of each fibre. For each n one constructs morphisms

$$H^0(\mathcal{O}_T, \mathcal{O}_T) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(-Z_{n,T})), \quad (2.4.6)$$

$$H^0(\mathcal{O}_T, \mathcal{O}_T^*) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(1 \text{ on } Z_{n,T})^*). \quad (2.4.7)$$

The multiplicative version gives a morphism of $k[[t]]/(t^n)$ -group schemes

$$w_n : \mathbf{G}_m \longrightarrow \text{Pic}(Y_n, Z)^0. \quad (2.4.8)$$

For $m < n$, the reduction of w_n modulo t^m is w_m ; for $n = 1$ this is the morphism associated with the two branches.

Although one cannot take a direct limit to get a morphism $\mathbf{G}_m \rightarrow \text{Pic}(Y; Z)^0$, one can restrict to finite subgroup schemes. For every integer m one obtains

$$w : \mu_m \longrightarrow \text{Pic}(Y, Z)^0. \quad (2.4.9)$$

Passing to Lie algebras gives

$$dw : \text{Lie } \mathbf{G}_m \longrightarrow \text{Lie } \text{Pic}(Y; Z)^0, \quad (2.4.10)$$

and hence an element

$$dw(1) \in H^1(Y, \mathcal{O}(-Z)). \quad (2.4.11)$$

Let $\omega_{Y/S}$ be the relative dualizing sheaf and let φ be a section of $\omega_{Y/S}(Z)$. In formal coordinates near O ,

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

The restriction of φ to the generic fibre is paired with $H^1(Y_\eta, \mathcal{O}(-Z_\eta))$ by Serre duality.

2.5. Proposition. With the notation above,

$$I(f) = -(dw(1), \varphi). \quad (2.5.1)$$

The right-hand side lies in $k[[t]]$, not merely in $k((t))$. Reducing modulo t^n , the local trace and the global Serre trace are compatible; the proposition is reduced to the following lemma.

2.6. Lemma. The morphism from local cohomology at O to $H^1(Y_n, \mathcal{O}(-Z_n))$ sends v_n to $dw_n(1)$.

The image of v_n is also the image of 1 under (2.4.6). The assertion follows from the description of dw_n via dual numbers over $k[[t]]/(t^n)$ and the natural exact sequence comparing the additive and multiplicative constructions.

3 Proof of the theorem

Let S_0 be a smooth curve over k , let $s \in S_0(k)$, and let S be the completion at s . Let $\pi : Y_0 \rightarrow S_0$ be proper and flat, purely of relative dimension one, with Y its base change to S and with a point $O \in Y_s$ such that $(Y/S, O)$ is as above. We suppose, for simplicity, that Y_0 is regular and that π has geometrically connected fibres. Let ω_{Y_0/S_0} be the relative dualizing sheaf.

Let φ be a rational section of ω_{Y_0/S_0} , regular at O . In formal coordinates it can be written

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

3.2. Proposition. If k has characteristic $p > 0$, then $I(f)$ is algebraic over the field $k(S_0)$ of rational functions on S_0 .

After multiplying φ by a rational function on S_0 vanishing sufficiently at s , and after shrinking S_0 , we may assume that for a suitable relative Cartier divisor Z_0 disjoint from O , the form φ is a section of ω on $Y_0 - Z_0$. Let $\text{Pic}(Y_0; Z_0)^0$ be the relative generalized Picard scheme. Via Serre duality, φ corresponds to a section of the dual of the vector bundle

$$\text{Lie Pic}(Y_0; Z_0)^0 = R^1 \pi_* \mathcal{O}(-Z_0).$$

After base change to S we have $dw : \mathcal{O} \rightarrow \text{Lie Pic}(Y; Z)^0$, and Proposition 2.5 reduces Proposition 3.2 to the following key lemma.

3.3. Key Lemma. If $p > 0$, then dw is algebraic over S_0 .

In characteristic p , the inclusion $\mu_p \subset \mathbf{G}_m$ induces an isomorphism on Lie algebras:

$$\text{Lie}(\mu_p) \xrightarrow{\sim} \text{Lie}(\mathbf{G}_m).$$

Modulo t^n , one checks that dw is the differential of the morphism

$$w : \mu_p \rightarrow \text{Pic}(Y; Z)^0.$$

It remains to use the rigidity of μ_p : for every flat commutative group scheme P over S_0 , the functor $\text{Hom}(\mu_p, P)$ is represented by an etale scheme over S_0 . Applying this to $P = \text{Pic}(Y_0; Z_0)^0$, the morphism w is pulled back from a universal morphism over an etale cover of S_0 . Hence its differential, and therefore dw , is algebraic.

Let $k\{x_1, \dots, x_N\}$ be the henselization of the local ring of affine N -space at the origin. Its completion is $k[[x_1, \dots, x_N]]$, and an element of the completion is algebraic over $k(x_1, \dots, x_N)$ if and only if it lies in the henselization.

Proof of the theorem

If f is algebraic, there exists a surface Y_0 with a rational point O , an etale morphism $h : Y_0 \rightarrow \mathbf{A}^2$ sending O to the origin, and a section whose formal expansion at O is f . Let $S_0 = \mathbf{A}^1$ with coordinate t , and let π' be the composite

$$Y_0 \xrightarrow{h} \mathbf{A}^2 \xrightarrow{xy} \mathbf{A}^1 = S_0.$$

The differential form $f h^*((dx \wedge dy)/dt)$ is a rational section of the relative dualizing sheaf. After compactifying and resolving singularities, one obtains a proper flat regular model to which Proposition 3.2 applies. Therefore $I(f)$ is algebraic.

Proof of the corollary

Proceed by induction on N . The cases $N = 0$ and $N = 1$ are immediate, and $N = 2$ is the theorem. For $N > 2$, write

$$g = \sum g_{n,m} x_{N-1}^n x_N^m, \quad g_{n,m} \in k[[x_1, \dots, x_{N-2}]],$$

and put

$$I(g) = \sum g_{n,n} x_{N-1}^n.$$

Then

$$I_N(g) = I_{N-1}(I(g)). \quad (3.6.1)$$

If g is algebraic, the coefficients $g_{n,m}$ are algebraic over $k(x_1, \dots, x_{N-2})$. Applying the theorem over the fraction field of the henselization in the first $N - 2$ variables shows that $I(g)$ is algebraic, and induction proves the claim.

Degree estimates and remarks

The proof gives an estimate. With the notation of the geometric construction, let α be the degree of the curve S_0 over the t -line, let g be the genus of the normalization of the geometric generic fibre, and let $z = \max(0, z' - 1)$ where z' is the number of geometric points of the divisor Z . Then the degree d of the equation satisfied by $I(f)$ is bounded by

$$d < \alpha p^{g+z}. \quad (3.7.1)$$

For a formal algebraic series with integer coefficients, this implies that for almost all primes p , the degree d_p of the reduction of $I(f)$ modulo p satisfies

$$d_p < \alpha p^{g+z}. \quad (3.7.2)$$

Using μ_{p^m} when $p^m = 0$ on the base gives analogous information modulo powers of p , for almost all primes.

A direct proof for N variables would be preferable. Analytically, for a convergent series g one expects the formula

$$I(g)(t) = \int_{Z(t)} g \frac{dz_1 \cdots dz_N}{dt}, \quad (1')$$

where

$$Z(t) = \{(x_1, \dots, x_N) : x_1 \cdots x_N = t, |x_i| = r_i, \prod r_i = |t|\}.$$

A direct proof of this kind would be a test case for a p -adic theory of vanishing cycles.

Furstenberg's proof in the rational case shows that if g is one component of a column vector h with entries in $k[[x_1, \dots, x_N]]$ satisfying

$$h = Ah^p$$

for a square matrix A with polynomial entries, then $I(g)$ has the same property and is algebraic. This type of equation recalls F -crystals in crystalline cohomology. Since $T_p \text{ Pic}$ is an avatar of p -adic H^1 , the proof here and Furstenberg's proof are perhaps less distant from each other than they first appear.

A Signs

The treatment of signs in the duality formalism for coherent algebraic sheaves is delicate. We record the conventions used here. For a flat relative complete intersection morphism $f : X \rightarrow S$, purely of relative dimension n , the relative dualizing sheaf $\omega_{X/S}$ is related to the relative dualizing complex by

$$K_{X/S} = \omega_{X/S}[n]. \quad (\text{S.1})$$

When X is smooth over S , $\omega_{X/S} = \Omega_{X/S}^n$. If X is the fibre product over S of morphisms $f_i : X_i \rightarrow S$ of relative dimensions n_i , then

$$K_{X/S} = \boxtimes_i K_{X_i/S}. \quad (\text{S.2})$$

Equivalently, after choosing a total order on the index set,

$$\omega_{X/S} = \bigotimes_i \omega_{X_i/S}, \quad (\text{S.3})$$

with the sign change $(-1)^N$ when the order is changed, where N is the sum of $n_i n_j$ over inverted pairs.

For $f : X \rightarrow S$ proper as above, the trace is

$$\text{Tr} : Rf_* K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.4})$$

It induces

$$\text{Tr} : R^n f_* \omega_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.5})$$

For a section s of f there is a local trace

$$\text{Tr} : Rs^! K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.6})$$

The choices are fixed by requiring compatibility between global and local traces and by normalizing, for a smooth curve X over an algebraically closed field and a point $x \in X(k)$, the local trace

$$H_{\{x\}}^1(X, \Omega^1[1]) \rightarrow k \quad (\text{S.7})$$

to be the residue map Res_x .

With these conventions, for a smooth complete curve X and $x \in X(k)$, the boundary map

$$H^0(X - \{x\}, \Omega^1) \rightarrow H_{\{x\}}^1(X, \Omega^1)$$

composed with the trace is $-\text{Res}_x$. For products, if $X = X_1 \times X_2$ and $u_i \in H^{n_i}(X_i, \omega_{X_i})$, then

$$\text{Tr}(\text{pr}_1^* u_1 \smile \text{pr}_2^* u_2) = (-1)^{n_1 n_2} \text{Tr}(u_1) \text{Tr}(u_2).$$

Over $k = \mathbf{C}$, comparison with the transcendental trace identifies the algebraic trace with integration over $X(\mathbf{C})$, with the standard normalization by $(2\pi i)^{-n}$. For projective space $\mathbf{P}^n$ with homogeneous coordinates $T_0, \dots, T_n$ and $t_i = T_i/T_0$, the Čech class of

$$\frac{dt_1 \wedge \dots \wedge dt_n}{t_1 \dots t_n}$$

has trace $(-1)^{n(n+1)/2}$ with these conventions.

References

- [1] H. Furstenberg, Algebraic functions over finite fields, *J. Algebra* **7** (1967), 271–277.
- [2] M. Raynaud, Specialisation du foncteur de Picard, *Publ. Math. IHES* **38** (1970), 27–76.
- [3] M. Raynaud, *Anneaux locaux henseliens*, Lecture Notes in Mathematics 169, Springer, 1970.

The Lemma of Gabber

Pierre Deligne

The principal result of this exposé, Theorem 1.4, is due to O. Gabber.

1 Statements

1.1 In this exposé we systematically work with algebraic spaces, rather than only with schemes. This raises a problem of references. We shall allow ourselves to refer to an article or theorem proved for schemes and apply it to algebraic spaces whenever the same proof also proves the assertion for algebraic spaces. A proof becomes problematic when it uses the fact that every point has an affine neighbourhood and when “neighbourhood” cannot be replaced by “étale neighbourhood”.

This leads us to speak of an *abelian algebraic space* A over S , where one would ordinarily speak of an abelian scheme over S : this means a commutative S -algebraic space in groups

$$f : A \longrightarrow S$$

with f proper, flat, and of finite presentation, and with geometric fibres abelian varieties. We shall call A a *semi-abelian algebraic space* over S if it is a commutative S -algebraic space in groups $f : A \rightarrow S$, with f separated, flat, of finite presentation, and with geometric fibres extensions of abelian varieties by tori.

Note, however, the following points.

- (a) If S is the spectrum of a field, every algebraic space in groups over S is a scheme, and indeed is quasi-projective.
- (b) If S is a trait, every algebraic space in groups G over S , locally of finite type and separated over S , is a scheme. This is proved for the smooth case in Raynaud’s work on Néron models.
- (c) If S is normal integral, with generic point η , and if A is an abelian algebraic space over S , Grothendieck’s arguments show that every polarization of A_η extends uniquely to a polarization of A over S . It follows that A is projective over S , hence in particular that A is a scheme if S is a scheme.

An algebraic space $f : X \rightarrow S$, flat of finite presentation, separated over S , and with proper and connected geometric fibres, is proper over S . By localization on S and passage to limits, one reduces to the case where S is the spectrum of a noetherian local ring; by faithfully flat descent one may suppose the ring complete, and then one applies EGA III, first part, 5.5. In particular, a semi-abelian algebraic space over S all of whose fibres are abelian varieties is abelian over S .

Unless explicitly stated otherwise, all schemes and algebraic spaces considered below are quasi-compact and quasi-separated.

1.2 Let U be a dense open subspace of a normal algebraic space S , and let A be an abelian algebraic space over U . A *semi-abelian prolongation* of A over S is a semi-abelian algebraic space P over S together with an isomorphism

$$A \xrightarrow{\sim} P|_U.$$

Theorem 1.1 (Raynaud). *Under the preceding hypotheses, if A admits a semi-abelian prolongation, then that prolongation is unique up to unique isomorphism.*

Localization and passage to limits reduce the proof to the case where S is a noetherian scheme. If S is a trait and U is its generic point, every semi-abelian prolongation P of A coincides with the identity component N^0 of the Néron model N of A : since P is smooth with connected fibres, there is a morphism $P \rightarrow N^0$ inducing the identity on the generic fibre, and SGA 7 IX, 3.1 shows that it is an isomorphism. For normal noetherian S , two prolongations coincide on an open $V \supset U$ whose complement has codimension at least two. Raynaud's extension theorem then gives the result. A correction to this argument is supplied in the note at the end of the exposé.

1.3

Remark 1.2. *If P and Q are two semi-abelian algebraic spaces over S , with S normal and integral and with generic fibres P_η and Q_η abelian varieties over $k(\eta)$, the same argument gives*

$$\mathrm{Hom}_S(P, Q) \hookrightarrow \mathrm{Hom}_\eta(P_\eta, Q_\eta).$$

Remark 1.3. *The normality hypothesis on S is indispensable. Let S be a nodal cubic over an algebraically closed field, obtained from $\mathbf{P}^1$ by identifying 0 and ∞ in one point p ; let $U = S - \{p\}$; and let A be the constant elliptic curve E over U . To give a prolongation of A is to give a prolongation over $\mathbf{P}^1$ and a gluing isomorphism between the fibres at 0 and at ∞ . The prolongation over $\mathbf{P}^1$ is unique, namely $E \times \mathbf{P}^1$, but the gluing isomorphism can be chosen in at least two ways because $\mathrm{Aut}(E)$ contains $\{\pm 1\}$.*

The theorem should nevertheless remain true when U is schematically dense in S and S is only assumed geometrically unibranch. For general S , two prolongations inducing the same prolongations after the relevant normalizations should be isomorphic.

1.4 To avoid references that are not available, and because this case suffices for the intended application, we state Gabber's lemma under hypotheses that are probably unnecessarily restrictive.

Theorem 1.4 (Gabber's lemma). *Let U be a dense open subscheme of an excellent normal integral scheme S , and let A be an abelian algebraic space over U . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S' .

1.5 If there existed a moduli space for semi-abelian algebraic spaces, the semi-stable reduction theorem together with the valuative criterion of properness would show that this moduli object is proper over $\mathrm{Spec}(\mathbf{Z})$, and Gabber's lemma would follow immediately. But we are far from such a situation: the objects considered have automorphisms, sometimes infinitely many, and worse, a semi-abelian algebraic space over a complete discrete valuation ring is not determined by the associated formal algebraic space. If the special fibre is isomorphic to $\mathbf{G}_m$, the reduction modulo $\mathfrak{m}^k$ is isomorphic to that of $\mathbf{G}_m$ for every k by rigidity of tori, without A necessarily being a torus.

The bad behaviour of semi-abelian spaces with respect to formal completion comes from the fact that they are not necessarily proper. For curves the situation is better because one has the notion of a stable curve of genus g . Such curves are proper and, for $g \geq 2$, have only finitely many automorphisms. Because of finite automorphism groups, stable curves of genus $g \geq 2$ do not give a fine moduli space, but they do give a proper algebraic stack over $\mathrm{Spec}(\mathbf{Z})$ by the valuative criterion of properness and the stable reduction theorem for curves.

1.6

Lemma 1.5. *Let U be a quasi-compact dense open subspace of an algebraic space S , and let $X \rightarrow S$ be a family of stable curves of genus $g \geq 2$ parametrized by U . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of X over $f^{-1}(U)$ extends to a family of stable curves over S' .

Proof. Fix $g \geq 2$, and for brevity say simply “stable curve” for stable curve of genus g . Let $\mathcal{M}$ be the algebraic stack classifying stable curves. It is proper over $\mathrm{Spec}(\mathbf{Z})$. There is therefore a proper scheme S_0 over $\mathrm{Spec}(\mathbf{Z})$ and a surjective morphism $S_0 \rightarrow \mathcal{M}$; equivalently, S_0 is endowed with a family of stable curves $u_0 : X_0 \rightarrow S_0$ such that every stable curve over an algebraically closed field occurs as a fibre of X_0/S_0 .

Over $U \times S_0$ we have the two stable curves $\mathrm{pr}_1^* X$ and $\mathrm{pr}_2^* X_0$. Let

$$I = \mathrm{Isom}(\mathrm{pr}_1^* X, \mathrm{pr}_2^* X_0).$$

This algebraic space is finite over $U \times S_0$ and maps onto U . Let Γ be its schematic image in $U \times S_0$, and let $\bar{I}$ and $\bar{\Gamma}$ be the schematic closures inside $S \times S_0$:

$$\begin{array}{ccccc} I & \longrightarrow & \Gamma & \hookrightarrow & U \times S_0 \\ \downarrow & & \downarrow & & \downarrow \\ \bar{I} & \longrightarrow & \bar{\Gamma} & \hookrightarrow & S \times S_0. \end{array}$$

The algebraic space $\bar{I}$, endowed with the proper morphism

$$\bar{I} \longrightarrow \bar{\Gamma} \hookrightarrow S \times S_0 \longrightarrow S,$$

is the desired S' . The inverse image over $\bar{I}$ of the universal family X_0/S_0 extends the inverse image over I of X/U . $\square$

In Section 3 we shall demystify the use of algebraic stacks in this proof.

1.7 The principle of the proof of Gabber’s lemma is to reduce to the case of curves by using Jacobians. We record some preliminary remarks.

- (a) As explained in 1.1(a), A/U is automatically an abelian scheme.
- (b) Given $U \hookrightarrow S$, A/U , and $f : S' \rightarrow S$ as in Theorem 1.4, every proper surjective $f_1 : S'' \rightarrow S$ lying over S' again satisfies the conclusion. Applying Chow’s lemma and normalizing, if the theorem holds at all then it holds with $f : S' \rightarrow S$ projective and surjective, S' integral and normal.
- (c) It suffices to verify the theorem locally on S . If S is covered by finitely many open subspaces S_i , and over each S_i one has a solution $S'_i \rightarrow S_i$, one can find a projective surjective $S' \rightarrow S$, with $S'|_{S_i}$ lying over S'_i , with $f^{-1}(U)$ dense in S' , and with S' normal. The local prolongations glue by Theorem 1.1.
- (d) It suffices to verify the theorem after replacing U by a non-empty open subspace $V \subset U$: over integral normal S' , a prolongation of $A|_V$ is automatically a prolongation of A .

2 Proof of Gabber's lemma

Let $U \subset S$ and A/U be as in Theorem 1.4, and let d be the relative dimension of A over U . We may and shall assume $d \geq 1$. We seek an integral normal S' and a projective surjective morphism $f : S' \rightarrow S$ such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S' . In doing so we may shrink U and replace S by an integral normal scheme S_1 mapping projectively and surjectively to S , replace U by its inverse image, and replace A by its inverse image.

Let $\eta = \text{Spec } K$ be the generic point of U , and let A_η be the fibre of A/U at η . We may assume K infinite, otherwise $S = U = \text{Spec } K$. Choose a projective embedding

$$A_\eta \hookrightarrow \mathbf{P}^N.$$

In the Grassmannian of linear subspaces $H \subset \mathbf{P}^N$ of codimension $d - 1$, those for which $H \cap A_\eta$ is a smooth curve form the points of a dense open subset, hence have K -points. This density follows by iteration from the fact that a general hyperplane section is transverse. Choose such an H and put

$$T_\eta = A_\eta \cap H.$$

Over an algebraic closure $\overline{K}$ of K the weak Lefschetz theorem gives

$$H^0(A_{\overline{K}}, \mathbf{Z}_\ell) \longrightarrow H^0(T_{\overline{K}}, \mathbf{Z}_\ell), \quad H^1(A_{\overline{K}}, \mathbf{Z}_\ell) \hookrightarrow H^1(T_{\overline{K}}, \mathbf{Z}_\ell).$$

Thus $T_{\overline{K}}$ is connected and the morphism

$$\text{Alb}(T_\eta) = \text{Pic}^0(T_\eta) \longrightarrow A_\eta$$

from the Jacobian of T_η to A_η is an epimorphism. In particular T_η has genus $g \geq d$. Temporarily exclude the case $d = 1$; then $g \geq 2$.

After shrinking U there exists a smooth curve T_U of genus $g \geq 2$, embedded in A_U , such that

$$\text{Pic}^0(T_U) \twoheadrightarrow A_U.$$

Applying Lemma 1.5 to the curve T_U over $U \subset S$, and replacing S by the resulting modification (or by the normalization of one irreducible component), we may assume that T_U extends to a family of stable curves T over S .

By Artin's representability theorem, $\text{Pic}^0(T)$ exists as an algebraic space over S . Let

$$P := \text{Pic}^0(T)$$

be the open algebraic subspace in groups parametrizing invertible sheaves on T whose restriction to every irreducible component of every geometric fibre has degree zero. This algebraic space is separated and smooth of finite type over S . Its geometric fibres are extensions of abelian varieties by tori; hence P is a semi-abelian algebraic space over S .

We have therefore obtained, after the allowed modification of S and shrinking of U , a semi-abelian algebraic space P over S such that P_U is abelian and carries an epimorphism

$$u : P_U \rightarrow A.$$

Let $K_U = \text{Ker}(u)$. It is a subgroup algebraic subspace of P_U . After shrinking U if necessary, we may suppose it flat over U . By flattening by modification, a further modification of S permits us to suppose that the schematic closure K of K_U in P is flat over S ; then K is automatically a subgroup algebraic space of P .

For algebraic spaces in groups over a base S , the quotient by a flat closed subgroup algebraic space exists, is compatible with base change, is separated over S , and is flat if the original group space is flat. Let

$$N = P/K.$$

The geometric fibres of N are quotients of extensions of abelian varieties by tori, hence again extensions of abelian varieties by tori. Thus N is semi-abelian, and since $A = P_U/K_U$, it extends A and proves the theorem.

It remains only to handle the case $d = 1$. The same proof applies if one defines a stable curve of genus one as a pointed curve $X \rightarrow S$ with section in the smooth locus and with geometric fibres either elliptic curves or nodal cubics. Alternatively, one may define T_η as a hyperplane section of $A_\eta \times A_\eta$.

3 Exorcising stacks

3.1 Recall the definition of a groupoid in schemes. It is a diagram

$$S_2 \rightrightarrows S_1 \rightrightarrows S_0 \quad (3.1.1)$$

with face maps d_i , $i = 0, 1$ and $i = 0, 1, 2$ respectively, such that the square

$$\begin{array}{ccc} S_2 & \longrightarrow & S_1 \\ \downarrow & & \downarrow d_1 \\ S_1 & \xrightarrow{d_0} & S_0 \end{array}$$

is cartesian and such that, for every scheme T , the sets $\text{Hom}(T, S_i)$ define a groupoid. The objects are the elements of $\text{Hom}(T, S_0)$, the arrows are those of $\text{Hom}(T, S_1)$, the source and target of an arrow f are $d_1(f)$ and $d_0(f)$, and if the target of f is the source of g , the composition is $g \circ f = d_1\langle g, f \rangle$, where $\langle g, f \rangle \in \text{Hom}(T, S_2)$ is determined by $d_2\langle g, f \rangle = f$ and $d_0\langle g, f \rangle = g$. We denote this groupoid by $S_*(T)$. As T varies, the $S_*(T)$ form a prestack in groupoids.

The quasi-compact separated algebraic stacks are the stacks in groupoids obtained from such a prestack, for the étale topology, when S_0 is quasi-compact, $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$ is finite, and the $d_i : S_1 \rightarrow S_0$ are étale.

The quasi-compact separated algebraic spaces correspond to the particular case where $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$ is a closed immersion. Then $S_1 \subset S_0 \times S_0$ is an étale equivalence relation, and the algebraic space is the quotient S_0/S_1 .

3.2 Let $g \geq 2$, and say simply stable curve for stable curve of genus g . We explain how to construct a diagram (3.1.1) for the stack $\mathcal{M}$ of stable curves. If T_i is endowed with a stable curve $u_i : X_i \rightarrow T_i$ for $i = 1, 2$, write

$$T_1 \times_{\mathcal{M}} T_2 = \text{Isom}(X_1, X_2).$$

This represents the functor sending T to triples (a_1, a_2, φ) , where $a_i : T \rightarrow T_i$ and $\varphi : a_1^*X_1 \xrightarrow{\sim} a_2^*X_2$. This fibre product is finite over $T_1 \times T_2$.

The Hilbert scheme H of stable curves X endowed with a tricanonical embedding

$$X \hookrightarrow \mathbf{P}^{5g-6}$$

is smooth over $\mathcal{M}$ and maps onto $\mathcal{M}$. The group $\text{PGL}(5g-5)$ acts on H with finite reduced stabilizers. If S is a disjoint sum of smooth subschemes of H , transverse to the PGL -orbits, of complementary dimension and meeting all of them, then S is étale over $\mathcal{M}$ and maps onto it, hence supplies a diagram (3.1.1). The stack $\mathcal{M}$ is separated of finite type over $\text{Spec}(\mathbf{Z})$. Since deformations of stable curves are unobstructed, it is smooth; this also follows from the smoothness of H over $\text{Spec}(\mathbf{Z})$.

3.3 Let $n \geq 1$. Let $\mathcal{M}^0[n]$ be the algebraic stack classifying proper smooth curves X of genus g over bases S where n is invertible, together with a level- n structure

$$\alpha : \text{Pic}^0(X)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

If $\mathcal{M}^0$ is the open part of $\mathcal{M}$ corresponding to smooth curves, the forgetful morphism

$$\mathcal{M}^0[n] \longrightarrow \mathcal{M}^0[1/n]$$

makes $\mathcal{M}^0[n]$ a finite étale cover of $\mathcal{M}^0[1/n]$.

3.4

Definition 3.1. Let $\mathcal{M}_n$ be the normalization of $\mathcal{M}$ in $\mathcal{M}^0[n]$.

In the language of 3.1 this means the following. Choose S_0 étale over $\mathcal{M}$ and mapping onto it, whence a diagram S_* . Over each S_i there is a stable curve X_i , and these form a cartesian system

$$\begin{array}{ccccc} X_2 & \rightrightarrows & X_1 & \rightrightarrows & X_0 \\ \uparrow & & \uparrow & & \uparrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0. \end{array}$$

Let S_i^0 be the open subset over which X_i is smooth, and let $S_{i,n}[1/n]$ be the finite étale cover of $S_i^0[1/n]$ given by

$$\text{Isom}(\text{Pic}^0(X_i)_n, (\mathbf{Z}/n\mathbf{Z})^{2g}).$$

Let $S_{i,n}$ be the normalization of S_i in $S_{i,n}[1/n]$. Since normalization commutes with étale base change in this situation, the $S_{i,n}$ again form a diagram (3.1.1), defining $\mathcal{M}_n$. The diagram

$$\begin{array}{ccccc} S_{2,n} & \rightrightarrows & S_{1,n} & \rightrightarrows & S_{0,n} \\ \downarrow & & \downarrow & & \downarrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0 \end{array}$$

is cartesian. In particular, for $t \in S_{0,n}(T)$ with image $s \in S_0(T)$ one has an injection of automorphism groups

$$\text{Aut}(t) \hookrightarrow \text{Aut}(s). \tag{3.4.1}$$

3.5

Proposition 3.2. If $n \geq 3$, then $\mathcal{M}_n[1/n]$ is an algebraic space.

The essential point is the following lemma.

Lemma 3.3. Let X be a stable curve over an algebraically closed field k , and let $n \geq 3$ be invertible in k . Every automorphism of X acting trivially on $\text{Pic}^0(X)_n$ is the identity.

Proof. Write $\text{Pic}^0(X)$ as an extension of an abelian variety by a torus,

$$0 \longrightarrow T \longrightarrow \text{Pic}^0(X) \longrightarrow A \longrightarrow 0,$$

and let L be the character group of T . Every automorphism u of X has finite order. If it acts trivially on $\text{Pic}^0(X)_n$, it acts trivially on A_n and on T_n , hence on L/nL . By Serre's rigidity for A , and by the analogous direct argument for L , it acts trivially on A and on L , hence on T . Thus $u - 1$ sends $\text{Pic}^0(X)$ to T and is therefore zero; u acts trivially on $\text{Pic}^0(X)$, and the conclusion follows from the standard faithfulness result for stable curves. $\square$

Lemma 3.4. *Let S be normal, let $j : U \hookrightarrow S$ be a dense open subset, let A be an abelian algebraic space over U , let N be a semi-abelian prolongation of A over S , and let n be invertible on S .*

- (i) *The étale sheaf N_n of points killed by multiplication by n injects into j_*A_n .*
- (ii) *Every isomorphism over U from A_n to the constant sheaf $(\mathbf{Z}/n\mathbf{Z})^{2g}$ extends to a monomorphism*

$$N_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

Proof. The first assertion follows from the separatedness of N_n over S . Since S is normal, the constant sheaf satisfies

$$(\mathbf{Z}/n\mathbf{Z})^{2g} \simeq j_*(\mathbf{Z}/n\mathbf{Z})^{2g},$$

and the composite

$$N_n \longrightarrow j_*j^*N_n = j_*A_n \longrightarrow j_*(\mathbf{Z}/n\mathbf{Z})^{2g} = (\mathbf{Z}/n\mathbf{Z})^{2g}$$

is the required prolongation. □

Proof of the proposition. Let $T_1 = S_{1,n}[1/n]$. Since T_1 is finite and unramified over $T_0 \times T_0$, it suffices to verify that for every algebraically closed field k , the map

$$T_1(k) \longrightarrow T_0(k) \times T_0(k)$$

is injective. We may suppose n invertible in k , otherwise $T_1(k) = \emptyset$. Because $T_*(k)$ is a groupoid, it suffices to verify that for $t \in T_0(k)$ there is exactly one object of $T_1(k)$ over (t, t) , i.e. that $\text{Aut}(t)$ is reduced to the identity.

Let X denote the stable curve obtained by pullback of the universal curve. On $T_i^0 = S_{i,n}[1/n]$ there is by definition a cartesian system of isomorphisms

$$\text{Pic}^0(X_i)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

By Lemma 3.4 it extends over T_i to a monomorphism of étale sheaves. It follows that the morphism from the stack defined by T_* to that defined by S_* factors through the stack of stable curves X endowed with a monomorphism $\text{Pic}^0(X)_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}$. For $t \in T_0(k)$, the injection (3.4.1) has image consisting of automorphisms of the corresponding stable curve acting trivially on $\text{Pic}^0(X)_n$. By Lemma 3.3, such an automorphism is trivial. □

3.6

Corollary 3.5. *If n is divisible by two integers n_i , $i = 1, 2$, prime to one another and each at least 3, then $\mathcal{M}_n$ is an algebraic space.*

It suffices to verify that each $\mathcal{M}_n[1/n_i]$ is an algebraic space, i.e. that the corresponding objects have no non-trivial automorphisms. This follows from (3.4.1) and its factorization

$$\mathcal{M}_n[1/n_i] \longrightarrow \mathcal{M}_{n_i}[1/n_i] \longrightarrow \mathcal{M},$$

together with Proposition 3.5.

3.7 In 3.6, stack language gave an étale atlas defining an algebraic space $\mathcal{M}_n$. The valuative criteria of separatedness and properness, together with the semi-stable reduction theorem, show that this algebraic space is proper over $\text{Spec}(\mathbf{Z})$. One may use this $\mathcal{M}_n$ as the S_0 in the proof of Lemma 1.5. If one wants S_0 to be projective over $\text{Spec}(\mathbf{Z})$, one applies Chow's lemma to $\mathcal{M}_n$. In fact $\mathcal{M}_n$ is projective, by Mumford.

4 Obtaining schemes

In this section we show how to modify and complete the proof of Gabber's lemma so as to obtain schemes rather than algebraic spaces. Another approach was indicated to me by Michel Raynaud: he says he can prove that if X is an S -algebraic space in groups, smooth and separated over S , with connected fibres, then X becomes representable after base change to a suitable blow-up S'/S .

Let $g \geq 2$. For brevity again say stable curve for stable curve of genus g .

4.1 For a smooth curve X/S of genus g , autoduality of the Jacobian

$$\varepsilon : \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(X)^*$$

gives a canonical relatively ample invertible sheaf on $\mathrm{Pic}^0(X)$, namely

$$(1, \varepsilon)^*(\text{Poincaré sheaf}).$$

It is symmetric and is trivialized along the zero section. Its formation is compatible with arbitrary base change. Denote it by $\mathcal{L}(X/S)$.

4.2

Lemma 4.1. *It is possible, uniquely up to unique isomorphism, to attach to each stable curve X/S an invertible sheaf $\mathcal{L}(X/S)$ on $\mathrm{Pic}^0(X)$, trivialized along the zero section, compatible with all base change, and equal to the sheaf of 4.1 when X is smooth.*

Proof. It suffices to treat the universal case. With the notation of 3.4, choose X_*/S_* . We seek invertible sheaves $\mathcal{L}(X_i/S_i)$ on the $\mathrm{Pic}^0(X_i)$, trivialized along the zero section, extending the sheaves of 4.1 over the open subset where X_i is smooth and forming a cartesian system under the maps $S_i \rightarrow S_j$. Each S_i is smooth over $\mathrm{Spec}(\mathbf{Z})$, hence regular, and so is $\mathrm{Pic}^0(X_i)$, which is smooth over S_i . An extension exists by taking the determinant of any coherent extension. Correcting by the inverse image of its restriction to the zero section gives a trivialized extension. Since $\mathrm{Pic}^0(X_i)$ has geometrically irreducible fibres, such an extension is unique up to unique isomorphism, and these unique isomorphisms give the required compatibility. $\square$

4.3

Proposition 4.2. *For a stable curve X/S , the invertible sheaf $\mathcal{L}(X/S)$ of 4.2 is relatively ample. In particular, $\mathrm{Pic}^0(X)$ is quasi-projective over S and is a scheme if S is a scheme.*

As before it suffices to verify this in a universal case; one may suppose S regular and the open subset $S^0 \subset S$ over which X is smooth dense. If one already knew that $\mathrm{Pic}^0(X)$ were a scheme, the assertion would follow from Raynaud's theorem. Unfortunately that proof uses the scheme hypothesis essentially in order to construct, locally on S , relatively ample divisors as complements of affine open subsets.

4.4

Lemma 4.3. *Let X be a geometrically reduced connected curve over a separably closed field k . There exists a divisor D , contained in the smooth locus of X , of degree*

$$g = \dim H^1(X, \mathcal{O}_X),$$

such that

$$H^0(X, \mathcal{O}_X(D)) = k, \quad H^1(X, \mathcal{O}_X(D)) = 0.$$

Proof. The assertion on H^0 follows from the other assertion by Riemann-Roch. If ω is the dualizing sheaf, then $\dim H^0(X, \omega) = g$, and by duality $H^1(X, \mathcal{O}_X(D)) = 0$ is equivalent to $H^0(X, \omega(-D)) = 0$. The sheaf ω has no embedded component. We may therefore construct by induction divisors D_i of degree i , contained in the smooth locus, with

$$\dim H^0(X, \omega(-D_i)) = g - i,$$

by taking $D_0 = 0$ and $D_{i+1} = D_i + P_i$, where P_i is a smooth k -point at which not all sections of $H^0(X, \omega(-D_i))$ vanish. Take $D = D_g$. $\square$

4.5

Lemma 4.4. *Let $f : X \rightarrow S$ be projective, with geometrically reduced connected fibres. Locally for the étale topology on S , there is an open subset*

$$V \subset \mathrm{Pic}^0(X)$$

meeting every fibre and which is a scheme.

Proof. One reduces to the case where S is strictly local with closed point s , and it suffices to construct an open $V \subset \mathrm{Pic}^0(X)$ which is a scheme and meets the fibre over s . By Lemma 4.3, choose a divisor D_s of degree $g = \dim H^1(X_s, \mathcal{O}_{X_s})$ on X_s , contained in the smooth locus, and satisfying $H^1(X_s, \mathcal{O}(D_s)) = 0$. Extend it to a relative Cartier divisor D on X , possible because S is local henselian. Then

$$f_*\mathcal{O}(D) = \mathcal{O}_S, \quad R^1f_*\mathcal{O}(D) = 0.$$

Let V' be the open subset of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves $\mathcal{F}$ such that $R^1f_*\mathcal{F}(D) = 0$. For such $\mathcal{F}$, $f_*\mathcal{F}(D)$ is free of rank one and commutes with all base change. Let $V \subset V'$ be the open corresponding to those $\mathcal{F}$ for which the zero locus of a basis section of $f_*\mathcal{F}(D)$ is a relative Cartier divisor $D(\mathcal{F})$ contained in the smooth locus X^0 . By construction V contains the zero section. The map

$$\mathcal{F} \longmapsto D(\mathcal{F})$$

identifies V with an open subset of $\mathrm{Sym}^g(X^0)$, and hence V is a scheme. $\square$

4.6

Lemma 4.5. *Under the hypotheses of Lemma 4.4, $\mathrm{Pic}^0(X)$ is locally a scheme for the étale topology on S .*

Proof. Again suppose S strictly local. Put $P = \mathrm{Pic}^0(X)$. It is a smooth algebraic space over S , pure of relative dimension g . Let V be as in Lemma 4.4. It suffices to show that P is the union of the translates gV for $g \in P(S)$. Reducing to the case where S is essentially of finite type over $\mathbf{Z}$, one may count dimensions. Let P^n be the n -fold fibre product of P over S . For a geometric point t of S and a geometric point $\gamma = (\gamma_1, \dots, \gamma_n)$ of P^n over t , say that γ is allowed if the translates $\gamma_i V$ cover P_t . The allowed points form a constructible subset Y of P^n . If Z is its complement, then fibre by fibre Z has codimension at least $n - g$. For $n - g > \dim S$, the closure of Z cannot contain the whole special fibre, so one finds a section γ of $P^n - \overline{Z}$, and then P is the union of the $\gamma_i V$. $\square$

4.7 In Lemmas 4.4 and 4.5, the projectivity hypothesis on f may be replaced by properness: locally on S for the étale topology, f is automatically projective.

4.8

Proof of Proposition 4.2. The problem is local on S for the étale topology. By Lemma 4.5, one can apply Raynaud's theorem giving relative ampleness in the scheme case. $\square$

4.9 The proof of Raynaud's theorem seems to use the scheme hypothesis only through the conclusion of Lemma 4.4. This would allow one to avoid Lemma 4.5.

4.10

Proposition 4.6. *Gabber's lemma remains true with “semi-abelian algebraic space” in the conclusion replaced by “semi-abelian scheme.”*

Proof. Proceeding as in Section 2, one reduces to the case where there is a stable curve X over S , smooth over U , and an epimorphism over U

$$u : \mathrm{Pic}^0(X) \longrightarrow A$$

whose kernel K_U extends to a flat subgroup algebraic space K of $\mathrm{Pic}^0(X)$. It remains to show that

$$P := \mathrm{Pic}^0(X)/K$$

is a scheme.

Over U , Poincaré complete reducibility gives a morphism

$$v : A \longrightarrow \mathrm{Pic}^0(X)$$

such that uv is multiplication by an integer $n \geq 1$ on A . By the extension result of 1.3, v extends to a morphism, again denoted v , from P to $\mathrm{Pic}^0(X)$. By density of U in S , one still has $uv = n$. Since multiplication by n on P is quasi-finite, v is quasi-finite, hence quasi-affine. Since $\mathrm{Pic}^0(X)$ is a scheme, P is a scheme. $\square$

I thank M. Raynaud and O. Gabber for a critical reading of a first version of this text.

Note

(1).1 Let $S = \mathrm{Spec} A$ be affine, and let $U \subset S$ be an open subscheme which is a finite union of principal opens $U_i = \mathrm{Spec} A[f_i^{-1}]$. If F is a closed subscheme of U , defined in U_i by an ideal $I_i \subset A[f_i^{-1}]$, then the schematic closure $\overline{F}$ of F in S is the closed subscheme defined by

$$I = \bigcap_i \rho_i^{-1}(I_i),$$

where $\rho_i : A \rightarrow A[f_i^{-1}]$ is the canonical map. If $S' = \mathrm{Spec} A'$ is flat over S , then the inverse image of $\overline{F}$ in S' is again the schematic closure of the inverse image of F . Hence schematic closure commutes with étale localization and so makes sense for algebraic spaces.

More generally, for $f : X \rightarrow S$ quasi-compact, the closed schematic image $f(X)^-$ is the closed subscheme defined by the ideal of elements $a \in A$ whose inverse image on X is zero. For f the inclusion of F as above, this recovers schematic closure. The same base-change compatibility holds for schematic images.

(1).2 Let S be a noetherian algebraic space, and let X and Y be two algebraic spaces over S , each endowed with a section e . Let $V \subset X$ be open, put $U = e^{-1}(V) \subset S$, and let

$$f_V : V \longrightarrow Y$$

be an S -morphism sending e to e . For each n , f_V induces a morphism from the n -th infinitesimal neighbourhood of e in V to the corresponding neighbourhood in Y_U . Passing to the inverse limit gives a morphism

$$f_V^\wedge : X_U^\wedge \longrightarrow Y_U^\wedge$$

between formal completions along the sections.

Proposition 4.7. *Assume X smooth over S and Y separated of finite type.*

- (i) *If U contains all points of S of depth at most 1, then $f_V^\wedge$ extends uniquely to a morphism $f^\wedge : X^\wedge \rightarrow Y^\wedge$ between the formal completions of X and Y along the sections.*
- (ii) *Let Γ be the schematic closure in $X \times_S Y$ of the graph $\Gamma_V \subset V \times_S Y$ of f_V . If $f_V^\wedge$ extends to $f^\wedge : X^\wedge \rightarrow Y^\wedge$ and U contains all points of S of depth 0, then Γ is étale over X along the section e , and its formal completion there is the graph of $f^\wedge$.*

The proof is local on S for the étale topology; thus one may suppose S affine. Then the infinitesimal neighbourhoods of the section in X and Y are affine schemes, and (i) is proved as in Raynaud. Smoothness of X/S ensures that the infinitesimal neighbourhoods of e in X are finite and flat over S .

We use the following lemma for (ii).

Lemma 4.8. *Let $S = \operatorname{Spec} A$ with A a complete noetherian local ring, let*

$$X = \operatorname{Spec} R[[T_1, \dots, T_n]],$$

let e be the zero section, let V be an open subset of X such that $U = e^{-1}(V)$ contains all associated points of S , and let $F \subset V$ be a closed subscheme containing all infinitesimal neighbourhoods of $e(U)$ in V . Then $F = V$.

Proof. It suffices to show that X is the schematic closure of F in X . If

$$f = \sum a_I T^I$$

vanishes on F , then each coefficient a_I vanishes on $U \subset S$, hence is zero, and $f = 0$. $\square$

Proof of Proposition 4.7(ii). Let s be a point of S . Write $X_s^\wedge$ for the completion of X at $e(s)$, and let $V_s^\wedge$ and U_s be the inverse images of V and U . The morphism $f^\wedge$ induces $X_s^\wedge \rightarrow Y$, and f_V induces $V_s^\wedge \rightarrow Y$. These two morphisms coincide on a closed subscheme containing all infinitesimal neighbourhoods of $e(U_s)$, hence coincide by Lemma 4.8. The inverse image of Γ_V in $X_s^\wedge \times_S Y$ is therefore the graph of $f^\wedge|_{V_s^\wedge}$, whose schematic closure is the graph of $f^\wedge : X_s^\wedge \rightarrow Y$. Since schematic closure commutes with the flat base change $X_s^\wedge \times_S Y \rightarrow X \times_S Y$, it follows that after the base change $X_s^\wedge \rightarrow X$, the space Γ is étale over X . Hence Γ is étale over X at $e(s)$. $\square$

(1).3

Corollary 4.9. *Let G and H be algebraic spaces in groups over S , let U be an open subspace of S , and let*

$$f_U : G_U \longrightarrow H_U$$

be a morphism. Assume that G is smooth over S with connected fibres, that H is of finite type and separated over S , and that U contains all points of S of depth 0. If the formal completion of f_U along the zero sections extends to a morphism of formal completions

$$G^\wedge \longrightarrow H^\wedge,$$

then f_U extends uniquely to a morphism $G \rightarrow H$.

If U contains all points of depth at most 1, the existence of the formal extension is guaranteed by Proposition 4.7(i).

Proof. Uniqueness follows from the fact that U is schematically dense in S , hence G_U is schematically dense in G . For existence, let Γ_U be the graph of f_U , let $\Gamma \subset G \times_S H$ be its schematic closure, and let Γ^0 be the open subset of Γ where Γ is étale over G . By Proposition 4.7(ii), Γ^0 contains the zero section of $G \times_S H$. For every section γ of Γ , left translation by the element of $G_U \times_S H_U$ determined by $\gamma|_U$ sends Γ_U to itself. By transport of structure, left translation by γ sends Γ^0 to itself. After the flat base change Γ^0/S , this shows that Γ^0 is a subgroup algebraic space of $G \times_S H$.

The image of Γ^0 in G is an open subgroup algebraic space, hence is all of G because G/S has connected fibres. Since Γ^0/G is étale and separated, the function assigning to x the number of geometric points in the fibre over x is lower semicontinuous. It is at least one everywhere and equals one over G_U , hence is constantly one. Therefore $\Gamma^0 \rightarrow G$ is étale, surjective, and radicial, hence an isomorphism. The required extension is the composite

$$G \xleftarrow{\sim} \Gamma^0 \longrightarrow H.$$

□

(1).4 The preceding proposition should remain true if the hypothesis that G/S is smooth is weakened to flatness of finite type, and if the hypothesis that H is separated over S is omitted.

Bibliography

- [1] M. Artin, *The implicit function theorem in algebraic geometry*, in Proc. Colloquium on Algebraic Geometry, Tata Institute, 1969.
- [2] M. Artin, *Algebraization of formal moduli I*, in *Global Analysis, Papers in Honor of K. Kodaira*, Princeton University Press, 1969, pp. 21–71.
- [3] M. Artin and G. Winters, *Degenerate fibres and stable reduction of curves*, Topology 10 (1971), 373–383.
- [4] W. L. Chow, *On the projective embedding of homogeneous varieties*, in a symposium in honor of S. Lefschetz, Princeton University Press, pp. 122–128.
- [5] P. Deligne and D. Mumford, *The irreducibility of the space of curves of a given genus*, Publ. Math. IHES 36 (1969), 75–110.
- [6] D. Knutson, *Algebraic Spaces*, Lecture Notes in Math. 203, Springer, 1971.
- [7] J.-P. Murre, *Representation of unramified functors. Applications*, Séminaire Bourbaki 294, 1964–1965.
- [8] M. Raynaud, *Modèles de Néron*, C. R. Acad. Sci. 262 (1966), 413–416.
- [9] M. Raynaud, *Spécialisation du foncteur de Picard*, Publ. Math. IHES 38 (1970), 27–76.
- [10] M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes in Math. 119, Springer, 1970.
- [11] M. Raynaud and L. Gruson, *Critères de platitude et de projectivité*, Invent. Math. 13 (1971), 1–39.
- [12] J.-P. Serre, *Rigidité du foncteur de Jacobi d'échelon $n \geq 3$* , appendix to Exposé 17 of Séminaire Cartan 1960/61.

Sigla

EGA denotes *Éléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné. SGA denotes *Séminaire de géométrie algébrique du Bois-Marie*.

Heisenberg Group and Reality

P. Deligne

1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ) , the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$\tau : (x, y, \lambda) \mapsto (-x, -y, \lambda) \quad (1.2)$$

is an automorphism of H .

The coordinate system $x, y, \lambda : H \rightarrow \mathbb{C}^2 \times \mathbb{C}^*$ exhibits the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section, $\tau(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, by means of this section, a lift to a one-parameter subgroup of H , again denoted by D . The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$, with

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H , and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which as a set is identified with $\mathbb{R}^2 \times U^1$.

It is known, by the Stone-von Neumann theorem [2], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V for which $\lambda \in U^1$ acts by multiplication by λ . In the Schrödinger model of V , one has $V = L^2(\mathbb{R})$, and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (x, 0, 1) : f(z) &\mapsto f(z + x), \\ (0, y, 1) : f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of V one deduces a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V , compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [3], D. Shale [1]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathcal{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V , are the L^2 functions on $\mathbb{R}$ that extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\mathrm{Im} z| < A_1$, one has, for all $A_2 \in \mathbb{R}$,

$$|f(z)| \leq O(\exp(-A_2|z|)).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol} , one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V . If one did, the projective action of $\mathrm{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\mathrm{SL}(2, \mathbb{C})$ on V^{hol} ; since $\mathrm{SL}(2, \mathbb{C})$ is simply connected, this action would lift to an honest action of $\mathrm{SL}(2, \mathbb{C})$, and the projective action of $\mathrm{SL}(2, \mathbb{R})$ would lift to an honest action. This is not the case: only the double cover of $\mathrm{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

2

Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines of $\mathbb{C}^2$. For $D \in S$, lifted in H as in no. 1, let W_D be the space of holomorphic functions φ on H satisfying

$$\varphi(\lambda dh) = \lambda \varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by $g : \varphi(h) \mapsto \varphi(hg)$. For $E \neq D$, restriction to E is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \quad (2.1)$$

where $\mathcal{O}(E)$ denotes the space of holomorphic functions on E .

If R is a holomorphic representation of H , on which each $\lambda \in \mathbb{C}^*$ acts by λ , and if ω is a D -invariant linear form on R , then

$$w \mapsto \omega(hw) \quad (2.2)$$

is a morphism from R to W_D .

3. Example

Let V , as in nos. 1 and 2, be the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathcal{S} \subset V$ be the space of its C^∞ vectors, and let $\mathcal{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\text{Im } t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t : f \mapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t) dx$$

is defined on V , and even on $\mathcal{S}'$. The form ω_t is a holomorphic vector of the dual of V , fixed by D_t . In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t , the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2), ω_t defines

$$\Omega_t : V^{\text{hol}} \longrightarrow W_{D_t},$$

which extends to V and even to $\mathcal{S}'$. Let E be the x -axis. The map $r_E \Omega_t : V \rightarrow \mathcal{O}(E)$ sends $f \in V$ to

$$f(x, t) = r_E \Omega_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\left(\partial_t - \frac{1}{4\pi i} \partial_x^2 \right) f(x, t) = 0.$$

For real t , the form ω_t remains defined on $\mathcal{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $f \mapsto \int f(x) dx$. The homomorphism Ω_t remains defined on V^{hol} . For $t = 0$, $r_E \Omega_0$ is, in the Schrödinger model, the identity.

Assume $\text{Im } t > 0$. Let $\overline{D}_t$ be the complex conjugate of D_t . In the affine line $S - \{\overline{D}_t\}$, the “half-plane” $\{D_u \mid \text{Im } u > 0\}$ is a disk with center D_t . The map $r_{\overline{D}_t} \circ \Omega_t : V \rightarrow W_{D_t} \rightarrow \mathcal{O}(\overline{D}_t)$ gives the holomorphic model of V : the action of H is given by

$$\begin{aligned} d \in D_t : \quad & f(x) \mapsto f(x + d), \\ e \in \overline{D}_t : \quad & f(x) \mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned}$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d)\overline{f(\bar{d})}\exp(2\pi i\langle d, \bar{d} \rangle)$$

is invariant under $H_{\mathbb{R}}$; moreover $r_{\bar{D}_t} \circ \Omega_t$ identifies V with the space of f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i\langle d, \bar{d} \rangle)$, the commutator of d and $\bar{d}$, is

$$\exp(2\pi i\langle d, \bar{d} \rangle) = \exp(2\pi i(t - \bar{t})|z|^2).$$

4

Fix E . The isomorphisms r_E of (2.1) make it possible to regard the W_D for $D \neq E$ as a family of representations of H operating in the fixed space $\mathcal{O}(E)$, with the action of H depending holomorphically on D . Let us make explicit the dependence on E of r_E . Choose coordinates as in no. 1, with E the x -axis. Let E' be generated by $(1, u)$ and D by $(-t, 1)$. We use x as coordinate on both E and E' . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi i u x x')$. Hence $r_{E'} r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$, with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'}\left(\frac{x}{1+tu}\right) = \exp\left(\frac{\pi i u}{1+tu}x^2\right) f_E(x). \quad (4.1)$$

This formula is holomorphic in t : the W_D form a holomorphic vector bundle $\mathcal{W}$ of infinite dimension over S , on which H acts.

To understand (4.1), one may observe that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \neq D$ and of the coordinate z on E :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O(\exp(A|z|^2)). \quad (4.2)$$

Let $W_D^0 \subset W_D$ be defined by this condition. The W_D^0 form a holomorphic sub-bundle $\mathcal{W}^0$ of the bundle $\mathcal{W}$ in holomorphic representations of H .

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f :

- (i)_T There exists a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$, with initial condition $g(z, 0) = f(z)$.
- (ii)_T $|f(z)| \leq O(\exp(|z|^2/4T))$.
- (iii)_T $\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|dz \wedge d\bar{z}| < \infty$.

If $T < T_1$, each of (i)_T, (ii)_T, (iii)_T implies each of (i)_T, (ii)_T, (iii)_T.

Proof. (i)_T $\Rightarrow$ (ii)_T. The heat equation remains satisfied by the derivatives of g , and therefore

$$\partial_z^{2n} g = \partial_t^n g, \quad \partial_z^{2n+1} g = \partial_t^n (\partial_z g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \quad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for $\partial_z^{2n+1}g$. The Taylor series of f then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!} \frac{1}{T_2^n} |z|^{2n}\right).$$

One has

$$\frac{1}{2n+1} 2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n} \frac{1}{T^n} \frac{|z|^{2n}}{n!}\right),$$

and this sum is $\exp(|z|^2/4T)$.

$(ii)_{T_1} \Rightarrow (i)_T$. It suffices to take g given by the kernel of the heat equation:

$$g(z, t) = \int_{\sqrt{t}\mathbb{R}} f(z+x) \frac{1}{2\sqrt{\pi t}} \exp(-x^2/4t) dx.$$

Changing $\sqrt{t}$ to $-\sqrt{t}$ changes the orientation of the cycle $\sqrt{t}\mathbb{R}$ and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: $(ii)_{T_1} \Rightarrow (iii)_T$, and $(iii)_{T_1} \Rightarrow (ii)_T$ is verified by writing $f(z)$ as the mean of its values on a disk with center z and radius $1/|z|$, and estimating that mean by Cauchy–Schwarz.

6

In homage to the Stone-von Neumann theorem, the bundle $\mathcal{W}$ over S is, projectively, equipped with a holomorphic connection for which the action of H is horizontal.

Let us specify “projectively.” On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L endowed with an isomorphism $L^{\otimes 2} \simeq \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity ± 1 . This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathcal{W}$, but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \quad (6.1)$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and if ∇ is a connection on $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$, then its transport to $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$ of square roots of $\mathcal{O}(-1)$ does not depend on the choice of a , which can be changed locally only by a constant ± 1 . Thus the notion of a “connection on $\mathcal{W}(-\frac{1}{2})$ ” is well defined independently of the global existence, or the choice, of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D . For a local section s of $\mathcal{W}(-\frac{1}{2})$, the problem is to define $\nabla_t s$ in the fiber of $\mathcal{W}(-\frac{1}{2})$ at D . We proceed as follows: after making an auxiliary choice u , one defines $\nabla_t^u s$ for s a local section of $\mathcal{W}$ or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u , with dependence in u of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where a takes opposite values for $\mathcal{W}$ and for $\mathcal{O}(-\frac{1}{2})$. Hence, for $s = s_1 s_2$, with s_1 a local section of $\mathcal{W}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u .

Let $D \in S$ be a line of $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t : D \rightarrow \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D)^{\otimes 2}$: to $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$ one associates $t(d) = \langle u, d \rangle v'$, where u is any representative of u' . The auxiliary datum will be a lift $\tilde{t} : D \rightarrow \mathbb{C}^2$ of t , or equivalently a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it $\tilde{t} = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let δ_z be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of H . For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y \rangle \delta_{(0,0,1)}. \quad (6.2)$$

Let $\mathcal{U}_1$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta_{(0,0,1)} = 1$. In $\mathcal{U}_1$, (6.2) implies that, for $d \in D$,

$$\left[-\frac{\delta_u^2}{4\pi i}, \delta_d \right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d \rangle \delta_u = \langle u, d \rangle \delta_u = \delta_{t(d)}. \quad (6.3)$$

Let us compute to first order around D . For $\varepsilon^2 = 0$ and $\varphi(g)$ in W_D , so $\delta_d \varphi = 0$, one has

$$\delta_{d+\varepsilon t(d)} \left[\varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = \varepsilon \left[\delta_{t(d)} \varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = 0,$$

because $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$, by (6.3). To first order around D , the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle $\mathcal{W}$. If $\varphi(g, D')$ is a local section of $\mathcal{W}$, this permits one to define the derivative $\nabla_t^u \varphi$ in $\mathcal{W}_D$ by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on u . If u is replaced by $u + x$ with $x \in D$, then in $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u \delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u \delta_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For $\varphi \in W_D$, $\delta_x \varphi = 0$, and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

whence

$$\frac{1}{4\pi i} \delta_{u+x}^2 \varphi = \frac{1}{4\pi i} \delta_u^2 \varphi + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of u also makes it possible to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d \rangle u$ to obtain a homogeneous function of degree 1 on D . When u is changed into $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2} \nabla_t^u(s) s^{-3/2},$$

and, for the dependence on u , s now being a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x \rangle s. \quad (6.9)$$

The dependences on u in (6.8) and (6.9) cancel to provide a derivation ∇_t for $\mathcal{W}(-\frac{1}{2})$. Since the connection ∇ is defined in terms of right-invariant vector fields on H , it commutes with the action of H .

7

Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let E be the x -axis. On S , $\mathcal{O}(1)$ has a section s with a simple zero at E . It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1} . By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{W}$. We calculate it.

Let D_t be the line generated by $(-t, 1)$. By r_E of (2.1), a local section of $\mathcal{W}$ is identified with a function $f(z, t)$ holomorphic in z and t . We compute its derivative with respect to t . In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1, 0), d \rangle (1, 0).$$

The corresponding derivation of $\mathcal{O}(1)$ annihilates s , so that simply

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through $s \in W_D^0(-\frac{1}{2})$ there passes a horizontal local section of $\mathcal{W}(-\frac{1}{2})$. It is automatically a section of $\mathcal{W}^0(-\frac{1}{2})$. It exists in a neighborhood of D that is larger the slower the growth of s . Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection ∇ on $\mathcal{W}^0(-\frac{1}{2})$ defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\frac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of $\mathcal{W}^0(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{W}^0)$ on S has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{W}^0)$.

First proof. Divide S into two hemispheres S_1 and S_2 , and choose $\mathcal{O}(-\frac{1}{2})_i$ on S_i . On $S_1 \cap S_2$ one has locally an isomorphism $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$, defined up to sign, but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$: the monodromy is -1 . If there existed global horizontal projective sections of $\mathbb{P}(\mathcal{W}^0)$, there would exist horizontal sections f_1, f_2 of $\mathcal{W}^0(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would coincide up to sign. On $S_1 \cap S_2$

this would make it possible to normalize $\alpha : \mathcal{O}(-\frac{1}{2})_1 \rightarrow \mathcal{O}(-\frac{1}{2})_2$ by $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x -axis and D the y -axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D . By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{W}^0)$ passing through f , it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon|z|^2)). \quad (8.1)$$

If E' is another line, then by (4.1) the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon|z|^2)) \quad (8.2)$$

with suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$; hence in all directions except those for which az^2 is purely imaginary, g decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf, g decreases exponentially quadratically in all directions, hence is zero, and therefore $f = 0$.

9

In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. The commutator of d and $\bar{d}$ is

$$\langle d, \bar{d} \rangle = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \bar{d} \rangle > 1 \iff \operatorname{Im}(t) > 0.$$

The lines D such that, for d generating D , one has $\langle d, \bar{d} \rangle > 1$, therefore form a disk in S , called the disk attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ for which $\langle \cdot, \cdot \rangle$ is real on L form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, with $\operatorname{SL}(2, \mathbb{R})$ as the stabilizer of $\mathbb{R}^2$. Each defines a real structure $H_L = L \cdot U^1$ on H , hence a complex conjugation σ_L with fixed points LU^1 .

The disks in S likewise form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, the stabilizer of the one associated to $H_{\mathbb{R}}$ being $\operatorname{SL}(2, \mathbb{R})$. The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S .

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

10

The calculations of no. 3 admit the following interpretation. Let X be an open disk of S , corresponding to a real structure L , whence real forms H_L of H and $\operatorname{SL}(L)$ of $\operatorname{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ has an H_L -invariant pre-Hilbert structure; its completion $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$ is the irreducible unitary representation of H_L . One has

$$\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\frac{1}{2})^{\nabla=0}).$$

The double cover $\mathrm{SL}(L)^\sim$ of $\mathrm{SL}(L)$ acts on $\mathcal{O}(-\frac{1}{2})|_X$ compatibly with its action on $\mathcal{O}(-1)$; hence $\mathrm{SL}(L)^\sim$ acts on $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$. This is the action that normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)$ on H_L .

The subrepresentation $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ may be interpreted as the space of real analytic vectors for the action of $\mathrm{SL}(L)^\sim$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

11

Everything preceding generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle \cdot, \cdot \rangle$ real on $\mathbb{R}^{2n}$, replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle)),$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \bar{d} \rangle| > 1$ for $d \neq 0$ in D . The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ must be replaced by $\det(D)^{1/2}$.

12. Remark

Beside representations of H in which $\mathbb{C}^*$ acts by multiplication by λ , consider likewise representations in which the action is by λ^{-1} . One defines, as in no. 2, a holomorphic bundle $\mathcal{W}_-^0$ of such representations over S , and on $\mathcal{W}_-^0(-\frac{1}{2})$ one again has a connection, given by a heat equation.

On a connected simply connected open subset U of S , consider $W = \mathcal{W}^0(-\frac{1}{2})$, its connection, and the action of H . In $\Gamma(U, W)^{\nabla=0}$, for each $x \in U$ there is a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(e)$ for $f \in \mathcal{W}_x^0$. For each $x \notin U$, there exists a vector $v(x)$ fixed by D_x ; in the Schrödinger model, and for x corresponding to $E =$ the x -axis, it is the constant 1. It is unique up to a factor.

Fix the real structure $H_{\mathbb{R}}$, whence a real structure $S_{\mathbb{R}} \subset S$ (a circle), and a decomposition of $S - S_{\mathbb{R}}$ into two disks U_1, U_2 . Then:

- (a) The complex conjugate of $\Gamma(U_1, W)^{\nabla=0}$ is $\Gamma(U_2, W_-)^{\nabla=0}$.
- (b) The Hermitian form on $\Gamma(U_1, W)^{\nabla=0}$, invariant under $H_{\mathbb{R}}$, is reinterpreted as a bilinear pairing between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, invariant under H .

Question. Let C be a Jordan curve in S , separating $S - C$ into two open sets U_1 and U_2 . Does there exist a unique separating H -invariant bilinear form pairing $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, induced by a perfect duality between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$?

References

- [1] D. Shale, *Linear symmetries of free Boson fields*, Trans. Amer. Math. Soc. 103 (1962), 149–167.
- [2] J. von Neumann, *Die Eindeutigkeit der Schrödingerschen operatoren*, Ann. Mat. Pura Appl. (4) 104 (1931), 570–578.
- [3] A. Weil, *Sur certains groupes d'opérateurs unitaires*, Acta Math. 111 (1964), 143–211.

The Tame Symbol

Pierre Deligne

Working English translation. This is a faithful mathematical working translation preserving the article's numbered structure, formulas, and constructions. The accompanying French TeX layer is a cleaned working alignment layer.

1. Introduction

Let f and g be two meromorphic functions on a Riemann surface S . The *tame symbol* of f and g at $x \in S$ is

$$(f, g)_x = (-1)^{v_x(f)v_x(g)} \left(\frac{f^{v_x(g)}}{g^{v_x(f)}} \right) (x), \quad (1.1)$$

where $v_x(f)$ denotes the valuation of f at x , namely the order of the zero, or the negative of the order of the pole. The quotient appearing in (1.1) has valuation zero and hence can be evaluated at x . The definition is due to J. Tate; in the language of local symbols for the multiplicative group it gives, for compact S , the product formula

$$\prod_{x \in S} (f, g)_x = 1, \quad (1.2)$$

where almost all factors are equal to 1. This generalizes Weil reciprocity

$$f(\operatorname{div} g) = g(\operatorname{div} f)$$

for functions with disjoint divisors. If K is the fraction field of a discrete valuation ring with residue field k , formula (1.1) still defines a pairing

$$K^* \otimes K^* \longrightarrow k^*,$$

and, up to sign, this map can be interpreted in K -theory as the composite of the product $K_1(K) \otimes K_1(K) \rightarrow K_2(K)$ with the boundary $K_2(K) \rightarrow K_1(k)$.

Let D be the region in $\mathbf{C}$ bounded by a C^∞ curve Γ , and let f, g be meromorphic functions on a neighbourhood of D , without zeros or poles on Γ . The starting point of this article was a formula expressing

$$\prod_{x \in D} (f, g)_x$$

as the monodromy along Γ of a flat line bundle on Γ defined by the restrictions of f and g to Γ . Concretely, this is an integral formula for that product.

Following a suggestion of O. Gabber, the essential point is the construction, in the derived category of sheaves of abelian groups on a Riemann surface S , of a morphism of degree one

$$\mathcal{O}^* \otimes^L \mathcal{O}^* \longrightarrow \mathbf{C}^*[1]. \quad (1.3)$$

This morphism is given by an explicit formula at the level of complexes resolving $\mathcal{O}^*$ and $\mathbf{C}^*$. It therefore has a local interpretation: to two invertible holomorphic functions f and g it attaches a $\mathbf{C}^*$ -torsor, equivalently a flat line bundle, denoted (f, g) .

In arbitrary dimension there is an analogous morphism, with values in the complex

$$\mathcal{O}^* \xrightarrow{d\log} \Omega^1,$$

which is quasi-isomorphic to $\mathbf{C}^*$ in dimension one but not in higher dimension. It attaches to two invertible holomorphic functions f, g a line bundle (f, g) with connection, not necessarily integrable. In fact this morphism is the one defining the product in Beilinson-Deligne cohomology, $\mathbf{Z}(1) \otimes \mathbf{Z}(1) \rightarrow \mathbf{Z}(2)$; the corresponding interpretation in terms of Hodge theory is described in Section 3.

The same constructions exist on C^∞ manifolds, where $\mathcal{O}^*$ is replaced by the sheaf of invertible C^∞ functions and $\mathcal{O}^* \rightarrow \Omega^1$ by its differentiable analogue. Less regularity is also possible.

For a commutative complex analytic group G , (1.3) admits the generalization

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1], \quad (1.4)$$

which attaches a flat G -torsor $(f, g]$ to an invertible holomorphic function f and a holomorphic map g with values in G on a Riemann surface. Its monodromy is analogous to the local symbols of class field theory. In particular, if $\mathcal{L}$ is an invertible sheaf trivialized at infinity and g is a holomorphic map to G , one obtains a symbol $(\mathcal{L}, g] \in G$ with $(\mathcal{O}(x), g] = g(x)$.

For the circle S^1 , the map

$$(f, g) \longmapsto \text{monodromy}(f, g)$$

identifies, in a sense made precise below, the multiplicative group of invertible functions on S^1 with its own Cartier dual. This autoduality is skew-symmetric. It is an analytic analogue of the Contou-Carrere result that the functor sending a commutative ring A to $(A[[T]][T^{-1}])^*$ is its own Cartier dual. A twisted version with coefficients in a local system of free finite-rank $\mathbf{Z}$ -modules on S^1 is also proved.

If S is a Riemann surface with boundary a circle, the boundary values of invertible holomorphic functions form a maximal isotropic subspace of the group of invertible smooth functions on the boundary. Variants are proved for several boundary circles and for local systems. These results are parallel to the autoduality of the Jacobian, and of Prym varieties.

Throughout, Tate twists such as $\mathbf{Z}(1) = 2\pi i \mathbf{Z}$ are needed to make the statements natural and independent of the choice of a square root i of -1 in $\mathbf{C}$.

The last section applies these tools to a preliminary construction needed for holomorphic conformal field theories attached to certain orbifolds of the form “torus modulo a finite group”. Let B be an even positive definite bilinear form on $\mathbf{Z}^n$, and let S be an oriented circle. From B and an auxiliary datum one obtains a central extension of the group of smooth maps $S \rightarrow \mathbf{R}^n/\mathbf{Z}^n$ by $\mathbf{C}^*$, and a compatibility with gluing along surfaces with boundary.

1 The main construction

Let X be a complex analytic variety. We write $\mathcal{O}$ for the sheaf of holomorphic functions, $\mathcal{O}^*$ for the invertible functions, and $G(\mathcal{O})$ for the sheaf of holomorphic maps to a commutative complex Lie group G . The exponential sequence for G is written in the form

$$0 \longrightarrow G(\mathbf{C}) \longrightarrow G(\mathcal{O}) \xrightarrow{d\log} \Omega^1 \otimes \text{Lie}(G) \longrightarrow 0 \quad (2.1.1)$$

when X is one-dimensional and G is locally represented in logarithmic coordinates.

For a complex $[A \rightarrow B]$ placed in degrees $n, n+1$ we write $[A \rightarrow B]_n$. The fundamental morphism is obtained from the elementary formula

$$[\mathcal{O}^* \rightarrow \Omega^1]_0 \otimes^L G(\mathcal{O}) \longrightarrow G[1]. \quad (2.2.1)$$

At the level of local logarithms, it is given by the expression

$$(f, g) \longmapsto \exp \left(\frac{1}{2\pi i} \int \log(f) d \log(g) \right), \quad (2.2.2)$$

with the usual correction terms on overlaps. Passing to cohomology, (2.2.1) gives a product

$$H^i(X, \mathcal{O}^*) \times H^j(X, G(\mathcal{O})) \longrightarrow H^{i+j+1}(X, G). \quad (2.3.1)$$

For $i = j = 0$ this attaches to local functions f, g the torsor $(f, g]$.

The construction may be written using Čech cocycles. Let $\{U_i\}$ be an open cover, choose branches $\log f_i$ of f , and represent g locally. The change of branch $\log f_i - \log f_j \in 2\pi i \mathbf{Z}$ supplies the transition functions of a G -torsor by applying these integral changes to the periods of g . This Čech description is often more explicit than the derived-category one, but it is the same construction.

On a C^∞ manifold the construction is identical after replacing holomorphic functions by smooth invertible functions and holomorphic differentials by smooth one-forms. For a Riemann surface S , since $[\mathcal{O}^* \rightarrow \Omega^1]$ is quasi-isomorphic to $\mathbf{C}^*$, the construction becomes a pairing

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1] \quad (2.6.1)$$

which yields flat torsors.

Let $x \in S$, let f be invertible on $S - \{x\}$ with prescribed divisor at x , and let $g : S - \{x\} \rightarrow G$. The monodromy of $(f, g]$ around x is the local symbol. For $G = \mathbf{G}_m$ it is the tame symbol (1.1). For $G = \mathbf{G}_a$ one obtains the residue pairing

$$(f, g]_x = \text{res}_x(g d \log f). \quad (2.8.1)$$

For $G = \mathbf{G}_m$ we write (f, g) instead of $(f, g]$. The symbol is bimultiplicative and satisfies the product formula on compact Riemann surfaces.

Let $\mathcal{L}$ be an invertible sheaf on S trivialized at infinity. Applying the construction to a divisor representing $\mathcal{L}$ gives a global symbol

$$(\mathcal{L}, g] \in G, \quad (2.10.1)$$

characterized by $(\mathcal{O}(x), g] = g(x)$ and by multiplicativity in divisors. The case of cohomology with compact support gives a parallel construction when $i = j = 1$.

There is also a relative version. For a family of Riemann surfaces and holomorphic functions varying over a base, the torsors $(f, g]$ vary functorially and are compatible with base change. The local monodromy formula survives in families.

2 The case of the multiplicative group

Take $G = \mathbf{G}_m$. The product (2.2.1) becomes the product in Deligne-Beilinson cohomology

$$\mathbf{Z}(1)_{\mathcal{D}} \otimes^L \mathbf{Z}(1)_{\mathcal{D}} \longrightarrow \mathbf{Z}(2)_{\mathcal{D}}, \quad (3.1.1)$$

where $\mathbf{Z}(1) = 2\pi i\mathbf{Z}$ and $\mathbf{Z}(2) = (2\pi i)^2\mathbf{Z}$. Explicitly, the line bundle with connection (f, g) is represented locally by logarithms of f and g and by the connection one-form

$$\nabla = d + \frac{1}{2\pi i} \log(f) d \log(g). \quad (3.1.2)$$

When X is a Riemann surface the connection is flat and its monodromy is the tame symbol.

The Čech formula reads as follows. Choose logarithms $u_i = \log f$ and $v_i = \log g$ on U_i . On intersections the differences $u_i - u_j$ and $v_i - v_j$ lie in $2\pi i\mathbf{Z}$. The transition functions of (f, g) and the local connection forms are assembled from

$$\frac{1}{2\pi i} u_i dv_i, \quad \exp\left(\frac{1}{2\pi i} (u_i - u_j) v_j\right), \quad (3.2.1)$$

with the standard cocycle correction on triple intersections. The same formula with parameters gives the relative torsor.

The construction is bimultiplicative. There are canonical isomorphisms

$$(f_1 f_2, g) \simeq (f_1, g) \otimes (f_2, g), \quad (3.7.1)$$

$$(f, g_1 g_2) \simeq (f, g_1) \otimes (f, g_2). \quad (1)$$

It is skew-symmetric in the sense that there is a canonical isomorphism

$$(f, g) \otimes (g, f) \simeq \mathbf{1}. \quad (3.8.1)$$

For $f = -1$ the canonical trivialization in (3.8.1) must be treated with the sign convention encoded in the Tate twist; otherwise one misses the contribution of the chosen square root of -1 .

By bilinearity, one may extend (f, g) to sections of a local system. Let A be a local system of free finite-rank $\mathbf{Z}$ -modules, $B : A \otimes A \rightarrow \mathbf{Z}$ a bilinear form, and let f be a section of $\mathcal{O}^* \otimes A$. The expression

$$(f, f)_B \quad (3.10.1)$$

is defined by combining the pairings of the components with coefficients given by B . If B is symmetric and even, the construction is independent of ordering. If B is symmetric but not necessarily even, the parity term

$$B(x, x) \pmod{2} \quad (3.11.1)$$

controls the additional sign.

Let S be an oriented circle and A a local system of free finite-rank $\mathbf{Z}$ -modules on S . Choosing $i \in \mathbf{C}$ identifies $\mathbf{Z}$ with $\mathbf{Z}(1)$ by $1 \mapsto 2\pi i$. The exponential sequence gives exact sequences of sheaves on S :

$$0 \longrightarrow A \longrightarrow \mathcal{C}^\infty \otimes A \xrightarrow{\exp(2\pi i \cdot)} (\mathcal{C}^\infty)^* \otimes A \longrightarrow 0, \quad (3.13.1)$$

$$0 \longrightarrow A \longrightarrow \mathcal{C} \otimes A \longrightarrow \mathbf{C}^* \otimes A \longrightarrow 0. \quad (2)$$

These sequences will be used in the duality theorem of Section 4.

3 Duality

This section gives analytic analogues of the autoduality of the Jacobian and of the Contou-Carrere symbol. For an oriented C^∞ circle S the exponential sequence

$$0 \longrightarrow 2\pi i \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0 \quad (4.1.1)$$

and the complex

$$0 \longrightarrow \mathbf{C}^* \longrightarrow \mathcal{O}^* \xrightarrow{d\log} \Omega^1 \longrightarrow 0 \quad (4.1.2)$$

identify the group of smooth invertible functions, up to the correct completion, with its Cartier dual.

More generally, for a local system A of free $\mathbf{Z}$ -modules, the pairing of Section 3 gives a perfect pairing between the group of invertible smooth functions with values in A and the corresponding dual group with values in $A^\vee(1)$. The duality is skew-symmetric and functorial in A .

One treats the relevant cohomology groups as topological abelian groups. For a locally convex topological vector space the dual means the continuous character group. Exactness is understood in the topological sense. The pairings are computed by integrating logarithms and logarithmic differentials along the circle; in a local coordinate x one encounters expressions of the form

$$\int_S a(x) d\log f(x) \quad (4.6.1)$$

and, for the skew form, expressions of the form

$$\int_S \log f d\log g. \quad (4.6.2)$$

If A is a local system on S , the exact sequence

$$0 \rightarrow H^0(S, A) \rightarrow H^0(S, \mathcal{O} \otimes A) \rightarrow H^0(S, \Omega^1 \otimes A) \rightarrow H^1(S, A) \rightarrow 0 \quad (4.7.1)$$

serves to write the pairing explicitly and to show its non-degeneracy. In the basic example $S = U(1)$ and $A = \mathbf{Z}$, the group of invertible smooth functions decomposes into winding number and exponentials of smooth functions; the duality exchanges these two parts as expected.

Let S now be a compact Riemann surface with boundary. Boundary values of holomorphic invertible functions form an isotropic subgroup for the boundary pairing. The image is maximal isotropic. This is the analytic counterpart of the assertion that the Jacobian of a compact Riemann surface is principally polarized.

With coefficients in a local system A , and a positive definite bilinear form B on A , the boundary pairing is written

$$\langle f, g \rangle_B = (f, g)_B. \quad (4.16.1)$$

The proof shows that $\log(f, f)_B$ is the L^2 -norm of a harmonic representative. Hence for positive definite B the corresponding isotropic condition is also a positivity statement. For the annulus one obtains a Fourier-expansion model: if the two boundary components are related by the modulus of the annulus, the pairing is expressed as a convergent sum over Fourier modes.

4 Construction of central extensions

Let S be a compact Riemann surface with boundary and A a local system on S endowed with an even positive definite symmetric bilinear form B . The objective is to construct central extensions of groups of boundary values by $\mathbf{C}^*$, using the torsors and gerbes associated with the symbol.

We use the language of torsors and gerbes. A G -torsor is a sheaf locally isomorphic to G with its simply transitive G -action. A gerbe banded by G is a stack locally non-empty and locally connected, with automorphism group G ; it represents classes in $H^2(-, G)$. The torsor associated with (f, g) is thus the H^1 incarnation of the symbol, and the gerbe construction is its H^2 analogue.

A central extension of a sheaf of groups L by G is a G -torsor over L equipped with a multiplication lifting that of L , with G central. Equivalently, it is given locally by a two-cocycle, with the usual coboundary equivalence. In categorical language, the same data are encoded by a gerbe with a compatible group law.

Following an idea of Grothendieck, the auxiliary data needed to construct the extension are kept in a sheaf $\mathcal{H}^*$. A section consists of a lattice with bilinear form and of the choices required to make the signs and Tate twists coherent. If $k = (\mathbf{C}, s)$ is such an auxiliary datum, one defines an extension

$$1 \longrightarrow \mathbf{C}^* \longrightarrow \tilde{L}_k \longrightarrow L \longrightarrow 1. \quad (5.12.1)$$

The construction is local on the boundary and compatible with restriction to boundary components.

If $k' = (A', B')$ and $k'' = (A'', B'')$ are two data, the extension associated with $k' + k''$ is obtained from those associated with k' and k'' by the contracted product. The commutator in the extension is the symbol obtained from the boundary pairing:

$$[\tilde{f}, \tilde{g}] = (f, g)_B. \quad (5.16.1)$$

For a circle lying on the boundary of a surface, the data of Section 5.1 therefore provide a central extension of the boundary loop group. These extensions are the analytic input for the construction of holomorphic conformal field theories attached to torus orbifolds.

References

- [1] A. Beilinson, Higher regulators and values of L -functions.
- [2] J. Giraud, Cohomologie non abelienne.
- [3] J. Tate, Residues of differentials on curves.

Real Mixed Hodge Structures

Pierre Deligne

English working translation; formulas, filtrations, and numbering retained.

The real mixed Hodge structures of §2 form a rigid abelian tensor category, and the functor “underlying real vector space” is a fiber functor. The functor Gr^W is another one. We compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposite filtrations

1.1

This paragraph is a variation on §1.2 of P. Deligne, *Théorie de Hodge II*, Publ. Math. IHES 40, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \bar{F}$. Here “filtration” means finite filtration; W is increasing, and F and $\bar{F}$ are decreasing. The condition of opposition is

$$\mathrm{Gr}_n^W \mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q A = 0 \quad \text{for } n \neq p + q.$$

For a bigraded object $B = \bigoplus B^{p,q}$, we write $W, F_W, \bar{F}_W$ for the three opposite filtrations

$$W_n B = \bigoplus_{p+q \leq n} B^{p,q}, \quad F_W^i B = \bigoplus_{p \geq i} B^{p,q}, \quad \bar{F}_W^i B = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, saying that $W, F, \bar{F}$ are opposite on A is equivalent to saying that, for each n , $\mathrm{Gr}_n^W A$ has a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations induced by F and $\bar{F}$ on $\mathrm{Gr}_n^W A$ are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, p \geq i} A_W^{p,q}, \quad \bar{F}^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, q \geq i} A_W^{p,q}.$$

Put

$$A_F^{p,q} := W_{p+q} A \cap F^p A \cap \left(\bar{F}^q A + \sum_{i>0} W_{p+q-i} A \cap \bar{F}^{q-i+1} A \right),$$

and, interchanging the roles of $F, \bar{F}$ and of p, q ,

$$A_{\bar{F}}^{p,q} := W_{p+q} A \cap \bar{F}^q A \cap \left(F^p A + \sum_{i>0} W_{p+q-i} A \cap F^{p-i+1} A \right).$$

For $p + q = n$, the projection from $W_n A$ to $\mathrm{Gr}_n^W A$ induces isomorphisms

$$A_F^{p,q} \simeq A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \simeq A_W^{p,q}.$$

It follows that the $A_F^{p,q}$, respectively the $A_{\bar{F}}^{p,q}$, form a bigrading of A . The sums of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

are bifiltered isomorphisms

$$a_F : (A; W, F) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, F_W), \quad a_{\bar{F}} : (A; W, \bar{F}) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, \bar{F}_W).$$

Lemma 1.1. *The automorphism $d = a_F a_{\overline{F}}^{-1}$ of $\mathrm{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r < p \\ s < q}} A_W^{r,s}. \quad (1.1.1)$$

Proof. Passing to the associated graded, the isomorphisms a_F and $a_{\overline{F}}$ induce the identity on $\mathrm{Gr}^W A$. Thus $\mathrm{Gr}^W(d) = 1$. For a subset $I \subset \mathbb{Z}^2$ and a bigraded object B , write $B_I = \bigoplus_{(r,s) \in I} B^{r,s}$. Let

$$I(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } r < p)\},$$

$$J(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } s < q)\},$$

$$K(p,q) = \{(r,s) : r+s \leq p+q \text{ and } r < p \text{ and } s < q\}.$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects, the first subobject is $A_{FI(p,q)}$ and the second is $A_{\overline{F}J(p,q)}$. Hence $dA_W^{p,q}$ lies in the intersection corresponding to $I(p,q) \cap J(p,q) = K(p,q) \cup \{(p,q)\}$. Since $\mathrm{Gr}^W(d) = 1$, this gives exactly (1.1.1). $\square$

Proposition 1.1. *The functor $A \mapsto \mathrm{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \overline{F}$ to the category of objects of $\mathcal{A}$ endowed with a finite bigrading and with an automorphism d satisfying (1.1.1).*

The isomorphism $a_F : A \rightarrow \mathrm{Gr}^W A$ respects W , sends F to F_W , and sends $\overline{F}$ to $d^{-1}\overline{F}_W$. Thus the functor Gr^W has the inverse

$$s : (B = \bigoplus B^{p,q}, d) \mapsto (B, W, F_W, d^{-1}\overline{F}_W),$$

with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \mathrm{Gr}^W A$. Conversely, starting from a bigraded object B with an automorphism d , define $W, F, \overline{F}$ as above. If d satisfies (1.1.1), the evident map $B \rightarrow \mathrm{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \mathrm{Gr}^W \circ s$.

Remark 1.1. *If 2 is invertible in the ground field, then d has a unique square root $d^{1/2}$ still satisfying (1.1.1). Put*

$$a := d^{1/2} a_F = d^{-1/2} a_{\overline{F}}.$$

Then $a_{\overline{F}} = d^{1/2} a$ and $a_F = d^{-1/2} a$; thus a sends F , respectively $\overline{F}$, to

$$d^{1/2}(F_W), \quad d^{-1/2}(\overline{F}_W)$$

on $\mathrm{Gr}^W A$. The isomorphism a is the same for $(A; W, F, \overline{F})$ and for $(A; W, \overline{F}, F)$.

1.4

The constructions are compatible with tensor products. If

$$\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{A}$$

is a biadditive exact functor, it is extended to filtered objects by

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a A_1 \otimes F^b A_2 \subset A_1 \otimes A_2.$$

The associated-graded functor is compatible with $\otimes$, and, for bifiltered objects, the canonical isomorphism

$$\mathrm{Gr}^W \mathrm{Gr}^F(A_1 \otimes A_2) \simeq \mathrm{Gr}^W \mathrm{Gr}^F(A_1) \otimes \mathrm{Gr}^W \mathrm{Gr}^F(A_2)$$

sends the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2 . For vector spaces this follows from the fact that every bifiltration underlies a bigrading. Hence, if $(W_i, F_i, \overline{F}_i)$ are three opposite filtrations on A_i , then $(W_1 \otimes W_2, F_1 \otimes F_2, \overline{F}_1 \otimes \overline{F}_2)$ are again opposite on $A_1 \otimes A_2$. The bigradings, the isomorphisms $a_F, a_{\overline{F}}$, and the automorphism d are all compatible with $\otimes$.

1.5

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k , and let $\mathcal{C}$ be the category of vector spaces endowed with three opposite filtrations $W, F, \overline{F}$. This is a rigid abelian tensor category with $\text{End}(1) = k$. It has the fiber functor ω_0 , the underlying vector space, hence it is neutral Tannakian. It also has the fiber functor $\text{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\overline{F}}$ are isomorphisms of fiber functors $\omega_0 \rightarrow \omega_W$. Put

$$\mathcal{G} = \text{Aut}(\omega_W).$$

The functorial automorphism d defines a k -point of the pro-algebraic group $\mathcal{G}$. The functorial bigrading of $\text{Gr}^W(V)$ by the $V^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow \mathcal{G}.$$

It is a section of the projection

$$\text{pr} : \mathcal{G} \longrightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion of the category of bigraded vector spaces in $\mathcal{C}$. We normalize α by

$$\alpha(\lambda, \mu)v = \lambda^p \mu^q v \quad (v \in V^{p,q}).$$

By Proposition 1.1, $\mathcal{G}$ is also the algebraic group of automorphisms of the fiber functor on the category of bigraded vector spaces endowed with an automorphism d satisfying (1.1.1). We now compute it when k has characteristic zero.

Let $\mathfrak{l}$ be the Lie algebra over k freely generated by elements

$$D^{i,j} \quad (i, j < 0),$$

with the unique bigrading in which $D^{i,j}$ has bidegree (i, j) . The $W_n \mathfrak{l}$ are ideals, and the quotients $\mathfrak{l}/W_n \mathfrak{l}$ are nilpotent Lie algebras. Let $\mathcal{U}$ be the pro-algebraic group which is the projective limit of the unipotent groups

$$\exp(\mathfrak{l}/W_n \mathfrak{l}).$$

Let $\mathbb{G}_m \times \mathbb{G}_m$ act on $\mathfrak{l}$ so that (λ, μ) acts on bidegree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects W and therefore gives an action of $\mathbb{G}_m \times \mathbb{G}_m$ on $\mathcal{U}$.

Construction 1.1. *The group $\mathcal{G}$ is the semidirect product*

$$\mathcal{G} = (\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U}.$$

Indeed, giving an action of this semidirect product on a finite-dimensional vector space V is equivalent to giving an action of $\mathbb{G}_m \times \mathbb{G}_m$, hence a bigrading, together with a compatible nilpotent action of $\mathfrak{l}$. Equivalently, it is the datum of endomorphisms $D^{i,j}$ of bidegree (i, j) , or of their sum

$$D = \sum D^{i,j},$$

with

$$D(V^{p,q}) \subset \bigoplus_{r < p, s < q} V^{r,s}.$$

Writing $d = \exp(D)$, this is precisely the datum of a bigrading and of an automorphism satisfying (1.1.1). The construction is compatible with tensor products, giving the required isomorphism

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U} \xrightarrow{\sim} \mathcal{G} = \text{Aut}(\omega_W).$$

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space endowed with an increasing filtration W , whose complexification $V_{\mathbb{C}}$ is endowed with a decreasing filtration F , such that the filtration $W_{\mathbb{C}}$, the filtration F , and its complex conjugate $\overline{F}$ form a system of three opposite filtrations.

Via the dictionary between real vector spaces and complex vector spaces with a real structure, giving a real mixed Hodge structure is equivalently giving a complex vector space with three opposite filtrations $W, F, \overline{F}$ and with a complex conjugation respecting W and interchanging F and $\overline{F}$. By Proposition 1.1, the functor Gr^W identifies this category with the category of bigraded complex vector spaces

$$V = \bigoplus V^{p,q}$$

equipped with an automorphism d satisfying (1.1.1) and a complex conjugation σ interchanging $V^{p,q}$ and $V^{q,p}$, for which the complex conjugate of d is d^{-1} :

$$\sigma d \sigma^{-1} = d^{-1}.$$

Let $\mathcal{H}$ be the category of real mixed Hodge structures. It is a rigid abelian tensor category with $\mathrm{End}(1) = \mathbb{R}$. It has the fiber functor ω_0 , the underlying real vector space, hence is neutral Tannakian. It also has the fiber functor $\mathrm{Gr}^W =: \omega_W$. With the notation of Remark 1.1, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \mathrm{Gr}^W(V_{\mathbb{C}})$$

respects the real structures on both sides, since it depends symmetrically on F and $\overline{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, put $k = \mathbb{C}$, and let $\mathcal{G}_{\mathbb{R}}$ be the real form of $\mathcal{G}$ defined by the following complex conjugation:

- (a) on $\mathbb{G}_m \times \mathbb{G}_m$, it is $(\lambda, \mu) \mapsto (\overline{\mu}, \overline{\lambda})$;
- (b) on $\mathcal{U}$, it is the one compatible with the complex conjugation of $\mathfrak{l}$ given by

$$D^{i,j} \longmapsto -D^{j,i}.$$

As in §1.1, giving an action of $\mathcal{G}_{\mathbb{R}}$ on a real vector space V is equivalent to giving:

- (a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another;
- (b) an automorphism d of $V_{\mathbb{C}}$ such that $\overline{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{i < p, j < q} V^{i,j}.$$

Proposition 2.1. *The fiber functor ω_W induces an equivalence between $\mathcal{H}$ and the category of representations of $\mathcal{G}_{\mathbb{R}}$. Thus*

$$\mathcal{G}_{\mathbb{R}} = \mathrm{Aut}(\omega_W).$$

The inverse functor is

$$(V, \text{ action of } \mathcal{G}_{\mathbb{R}}) \longmapsto (V, \text{ bigrading of } V_{\mathbb{C}}, d) \longmapsto (V, W, F),$$

where W is the real filtration whose complexification is the weight filtration attached to the bigrading of $V_{\mathbb{C}}$, and

$$F = d^{-1/2}(F_W)$$

on the complexification. The isomorphism a gives the functorial identification

$$\phi \circ \mathrm{Gr}^W \xrightarrow{\sim} \mathrm{Id}.$$

Action of the Braid Group on a Category

Full working English translation

P. Deligne

Abstract

We describe how to define, by generators and relations, an action of the braid group on a category. More generally, we describe how to define a functor from a generalized positive braid monoid, corresponding to a finite Coxeter group, to a monoidal category. Applications are given to constructions of Bondal (exceptional systems) and of Broué–Michel (correspondences on flag manifolds).

Contents

0. Introduction	1
1 Actions	4
2 Contractibility	10

0. Introduction

An action of a monoid M (not necessarily having a unit) on a category C is the data of functors $T(f)$, for $f \in M$, and isomorphisms of functors

$$c_{f,g} : T(f) \circ T(g) \longrightarrow T(fg)$$

which make the diagrams

$$\begin{array}{ccc} T(f) \circ T(g) \circ T(h) & \longrightarrow & T(fg) \circ T(h) \\ \downarrow & & \downarrow \\ T(f) \circ T(gh) & \longrightarrow & T(fgh) \end{array} \quad (0.1)$$

commutative.

The actions of a monoid M on a category C form a category: a morphism $(T, c) \rightarrow (T', c')$ is the data of morphisms of functors $T(f) \rightarrow T'(f)$ which make the diagrams

$$\begin{array}{ccc} T(f)T(g) & \xrightarrow{c_{f,g}} & T(fg) \\ \downarrow & & \downarrow \\ T'(f)T'(g) & \xrightarrow{c'_{f,g}} & T'(fg) \end{array}$$

commutative.

Suppose that M is given by generators and relations, $M = \langle G \mid R \rangle$. If to each generator $g \in G$ one assigns a functor $T(g) : C \rightarrow C$, and if each relation $r_1 = r_2$ in R gives an equality of composite functors, then by composing the $T(g)$, $g \in G$, one obtains an action of M on C for which the $c_{f,g}$ are identity isomorphisms. This construction is of no interest. The useful

question is the following. Consider the category of systems of the following type: (a) functors $T(g) : C \rightarrow C$ for $g \in G$; (b) isomorphisms between composite functors for each relation $r_1 = r_2$ in R ; (c) compatibilities of the type (0.1), to be specified. How should (c) be specified so that the restriction functor from the category of actions of M on C to the category of systems (a),(b) satisfying (c) is an equivalence of categories? In other words, when is it “the same thing” to give an action of M on C as to give a system (a),(b) satisfying (c)?

In this article we treat the case in which M is the monoid B_n^+ of strictly positive braids (1.2). We use a system of generators indexed by the non-trivial elements of the symmetric group S_n . For $b \in B_n^+$, let $E(b)$ be the set of decompositions of b as a product of generators. This set has a natural order, and our main tool is the theorem that the geometric realization of the ordered set $E(b)$ is contractible. One may think of this theorem as a uniqueness result, up to homotopy, for the decomposition of b as a product of generators. In proving the contractibility of $E(b)$, our methods are those of Deligne (1972) and hence, indirectly, those of Garside (1969).

Our methods apply equally well to generalized strictly positive braid monoids (1.4), corresponding to finite Coxeter groups. Moreover, $f \mapsto T(f)$ may take values in an arbitrary monoidal category $\mathcal{M}$, rather than in $\text{Hom}(C, C)$.

At the end of this introduction we give the two applications which motivated us. The reader may prefer to read them after §2.

Application 1. Let T be a triangulated category and let

$$W_1 \subset \cdots \subset W_N = T$$

be a filtration of T by triangulated subcategories, with W_n thick (Verdier (1963), p. 13) in W_{n+1} for $1 \leq n < N$. Put $W_0 := 0$.

Following Bondal–Kapranov (1989), 4.1, say that the filtration W is right admissible if the following equivalent conditions are satisfied for $1 \leq n < N$: (i) the inclusion of W_n into W_{n+1} has a right adjoint; (ii) the quotient functor $W_{n+1} \rightarrow W_{n+1}/W_n$ has a right adjoint; (iii) every object Y of W_{n+1} occurs in a distinguished triangle (X, Y, Z) with X in W_n and Z in the right orthogonal of W_n in W_{n+1} . For the equivalence of these conditions, see Bondal–Kapranov (1989), §1, or Verdier (1963), Ch. 1, §2, no. 6. The filtration W is left admissible if it is right admissible in the opposite triangulated category: in (i),(ii), replace “right adjoint” by “left adjoint”. It is admissible if it is both left and right admissible.

Write

$$\text{Gr}_n^W(T) := W_n/W_{n-1} \quad (1 \leq n \leq N).$$

For W right admissible, write

$$\begin{aligned} i_{n*} : W_n &\rightarrow T && \text{for the inclusion functor,} \\ i_n^! : T &\rightarrow W_n && \text{for its right adjoint,} \\ j_{n*} : \text{Gr}_n^W(T) &\rightarrow T && \text{for the composite of } i_{n*} \text{ and the right adjoint} \\ &&& \text{of } W_n \rightarrow W_n/W_{n-1} = \text{Gr}_n^W(T), \\ j_n^! : T &\rightarrow \text{Gr}_n^W(T) && \text{for the composite of the quotient functor and } i_n^!. \end{aligned}$$

As explained in Bondal–Kapranov (1989), 4.3, giving a right admissible filtration W is equivalent to giving a sequence $(B_i)_{1 \leq i \leq N}$ of strictly full triangulated subcategories of T which generate T , with B_j right orthogonal to B_i for $i < j$, i.e. $\text{Hom}(X, Y) = 0$ for $X \in B_i$ and $Y \in B_j$. Passing from W to (B_i) : B_n is the essential image of $\text{Gr}_n^W(T)$ by the functor $j_{n!}$. Passing from (B_i) to W : W_n is the triangulated subcategory of T generated by the B_i with $i \leq n$, i.e. the smallest strictly full triangulated subcategory containing these B_i .

For a permutation σ of $\{1, \dots, N\}$, the right mutation $\sigma_d W$, or simply σW , of W is defined by

$$(\sigma W)_n := \text{the triangulated subcategory generated by the } B_i \text{ with } \sigma i \leq n.$$

Equivalently, it is the category of objects K of T for which $j_i^! K = 0$ whenever $\sigma i > n$. The equivalences

$$\mathrm{Gr}_n^W T \xleftarrow{\sim} B_n \xrightarrow{\sim} \mathrm{Gr}_{\sigma n}^{\sigma W} T \quad (0.2)$$

give an equivalence of $\mathrm{Gr}_n^W T$ with $\mathrm{Gr}_{\sigma n}^{\sigma W} T$.

For W left admissible, the left mutation $\sigma_s W$ is the right mutation in the opposite triangulated category. If W is admissible, the right mutation $\sigma_d W$ is still left admissible, and $(\sigma^{-1})_s \sigma_d W = W$.

A filtration W is ∞ -admissible if it is admissible, as are all its iterated left or right mutations. Bondal–Kapranov (1989), §§3,4 give convenient criteria for verifying ∞ -admissibility.

The following example explains the notation chosen above.

Example 1. Let X be a topological space and let

$$F_1 \subset \cdots \subset F_N = X$$

be a filtration of X by closed subspaces. Put $F_0 := \emptyset$ and $S_n := F_n - F_{n-1}$, $n \in I := \{1, \dots, N\}$. The filtration W of $T := D^+(X)$ by the thick subcategories $D^+(F_n)$ is admissible. The functors i_{n*} and $i_n^!$ (respectively j_{n*} and $j_n^!$) identify with the functors of the same name attached to the inclusion $i_n : F_n \hookrightarrow X$ (respectively $j_n : S_n \hookrightarrow X$).

Factor the inclusion j_n of S_n into X as the inclusion maps

$$g_{n,i} : F_n - F_i \longrightarrow F_n - F_{i-1} \quad (n \geq i > 0)$$

and i_n . Let σ be a permutation of $\{1, \dots, N\}$, and let σW be the corresponding right mutation of W . The subcategory $(\sigma W)_n$ of $D^+(X)$ is the category of objects K such that $j_i^! K = 0$ for $\sigma i > n$. The filtration σW is left admissible; the right orthogonal of $(\sigma W)_{\sigma n-1}$ in $(\sigma W)_{\sigma n}$ is the essential image of $D^+(S_n)$ by the composite functor of the $g_{n,i}!$ for $\sigma i < \sigma n$, or the $g_{n,i*}$ for $\sigma i > \sigma n$, followed by $i_{n*} = i_n!$. This composite functor induces the already considered equivalence from $D^+(S_n) = \mathrm{Gr}_n^W(T)$ to $\mathrm{Gr}_{\sigma n}^{\sigma W}(T)$.

Let T be a triangulated category and let $(T_i)_{i \in I}$ be a family with N elements of triangulated categories. Consider exceptional systems E of the following type (a),(b),(c):

- (a) a total order on I , identified with a bijection $\tau : I \rightarrow \{1, \dots, N\}$;
- (b) an ∞ -admissible filtration $W : W_1 \subset \cdots \subset W_N = T$ of T ;
- (c) equivalences of triangulated categories $\varphi_i : \mathrm{Gr}_{\tau(i)}^W(T) \rightarrow T_i$ for $i \in I$.

Exceptional systems form a category: there is a morphism from E to E' only if the corresponding orders coincide; in that case a morphism is a system of isomorphisms of functors $\varphi_i \rightarrow \varphi'_i$.

Let σ be a permutation of $\{1, \dots, N\}$. The right mutation σE of $E = (\tau, W, (\varphi_i))$ is the exceptional system defined by $\sigma \circ \tau$, by σW , and by the equivalences

$$\mathrm{Gr}_{\sigma\tau(i)}^{\sigma W}(T) \xleftarrow{(0.2)} \mathrm{Gr}_{\tau(i)}^W(T) \xrightarrow{\varphi_i} T_i.$$

The construction $E \mapsto \sigma E$ is an autoequivalence of the category of exceptional systems; its inverse is given by a left mutation.

Let $\ell(\sigma)$ be the number of pairs (i, j) such that $i < j$ and $\sigma(j) < \sigma(i)$. If $\ell(\sigma_1 \sigma_2) = \ell(\sigma_1) + \ell(\sigma_2)$, one checks that $\sigma_1 \sigma_2 E$ is canonically isomorphic to $\sigma_1(\sigma_2 E)$, with the compatibility for a triple product $\sigma_1 \sigma_2 \sigma_3$ if the lengths add. Applying 1.5 and 1.10, one obtains an action of the braid group on the category of exceptional systems.

This makes precise Bondal–Kapranov (1989), 4.10, which defines by mutations the action of the braid group on the set of ∞ -admissible filtrations and proves the equivalence of $\mathrm{Gr}_n^W(T)$

with $\mathrm{Gr}_{\bar{b}n}^{bW}(T)$, where $b \mapsto \bar{b}$ is the projection from the braid group to the symmetric group. We obtain an equivalence

$$[b] : \mathrm{Gr}_n^W(T) \longrightarrow \mathrm{Gr}_{\bar{b}n}^{bW}(T)$$

well-defined up to a unique isomorphism, an isomorphism $[b_1 b_2] \xrightarrow{\sim} [b_1] \circ [b_2]$, and a compatibility for a triple product.

Variant. Suppose the triangulated category T is k -linear, for k a commutative field, and that the k -vector spaces

$$\mathrm{Hom}^*(X, Y) := \bigoplus_n \mathrm{Hom}(X, Y[n])$$

are finite-dimensional. If each category T_i is the bounded derived category of finite-dimensional vector spaces over k , i.e. the category of finite-dimensional graded vector spaces, the notion of exceptional system reduces to that of an exceptional sequence: a sequence $X_1, \dots, X_N$ of objects of T such that $\mathrm{Hom}^*(X_i, X_j) = 0$ for $i < j$ and $\mathrm{Hom}^*(X_i, X_i) = k$, which generate T . More precisely, one should consider a finite family $(X_i)_{i \in I}$, a total order $\leq$ on I , and require that the X_i generate T , that $\mathrm{Hom}^*(X_i, X_j) = 0$ for $i < j$, and that $\mathrm{Hom}^*(X_i, X_i) = k$. The corresponding exceptional system is defined by the order $\leq$, identified with a bijection τ as in (a), and by the filtration W , where W_n is generated by the X_i for $\tau(i) \leq n$. The equivalence φ_i is characterized by sending X_i to k in degree 0. According to Bondal–Kapranov (1989), 3.2 and 4.11, the filtration W is automatically ∞ -admissible.

The action of the braid group on the category of exceptional systems therefore gives an action on the category of exceptional sequences.

Application 2. Let G be a split reductive group over a field k and let $\mathcal{B}$ be the variety of its Borel subgroups. Let $\mathcal{M}$ be the category of schemes over $\mathcal{B} \times \mathcal{B}$, viewed as correspondences from $\mathcal{B}$ to $\mathcal{B}$. Composition of correspondences makes $\mathcal{M}$ a monoidal category. Let W be the Weyl group, identified with the set of relative positions of Borel subgroups, i.e. with the orbits of $G(k)$ on $\mathcal{B}(k) \times \mathcal{B}(k)$. For $w \in W$, write $X(w)$ for the variety of pairs (B', B'') of Borel subgroups in relative position w . In this description of W , for $e \in W$ the identity element, $X(e)$ is the diagonal. The distinguished generators s_i correspond to those orbits X whose closure is X union the diagonal, and the group law is characterized by

$$X(w) = X(w') \circ X(w'') \quad \text{if } w = w'w'' \text{ and } \ell(w) = \ell(w') + \ell(w''). \quad (0.3)$$

The distinguished generators of W make it a finite Coxeter group. Let B^+ be the corresponding generalized positive braid monoid. Broué and Michel (1995) deduce from (0.3) that one may extend $w \mapsto X(w)$ to a construction which attaches to $b \in B^+$ an isomorphism class of schemes $X(b)$ over $\mathcal{B} \times \mathcal{B}$, with $X(b) \simeq X(b') \circ X(b'')$ for $b = b'b''$. Our results show that $X(b)$ is in fact defined up to a unique isomorphism.

1 Actions

1.1. As explained in the introduction, the actions of a monoid M on a category C form a category. To say that it is “the same thing” to give an action of M on C as to give something else means the construction of an equivalence of the category of actions with the category of that something else.

1.2. The monoid without unit B_n^+ of strictly positive braids on n strands, $n \geq 2$, is the monoid generated by generators s_i ($1 \leq i \leq n-1$) subject to the relations

$$s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \quad (1.2.1)$$

$$s_i s_j = s_j s_i \quad \text{for } j \geq i+2. \quad (1)$$

It is known that it injects into the braid group B_n , defined by the same generators and relations.

1.3. To define an action of B_n^+ on a set E , it suffices to give maps $T(s_i) : E \rightarrow E$ satisfying (1.2.1).

To define an action of B_n^+ on a category C , it does not suffice to give functors $T(s_i) : C \rightarrow C$ and isomorphisms of functors

$$T(s_i)T(s_{i+1})T(s_i) \xrightarrow{\sim} T(s_{i+1})T(s_i)T(s_{i+1}), \quad (1.3.1)$$

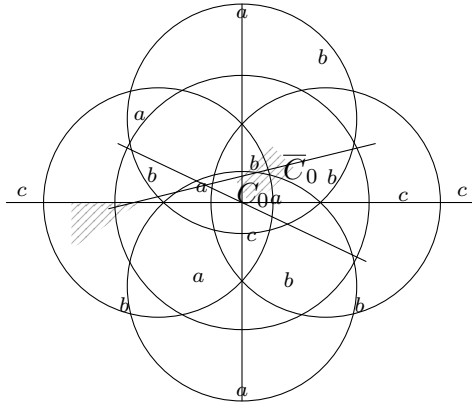
$$T(s_i)T(s_j) \xrightarrow{\sim} T(s_j)T(s_i) \quad \text{for } j \geq i + 2. \quad (2)$$

In the case of B_4^+ , one must moreover require the following condition. Write a, b, c for $T(s_1), T(s_2), T(s_3)$. In the following cycle of twelve functors, each is linked to the next by the isomorphisms (1.3.1) or their inverses, applied to the indicated subcomposites in the original notation:

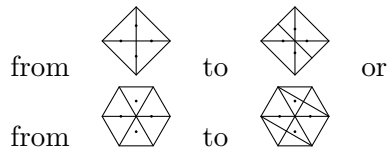
$$\begin{array}{cccc} acbcab & abcbab & abcaba & abacba \\ babcba & bacbca & bcabac & bcbabc \\ cbcabc & cbacbc & cbabcb & cabacb. \end{array} \quad (1.3.2)$$

The composite of these isomorphisms must be the identity.

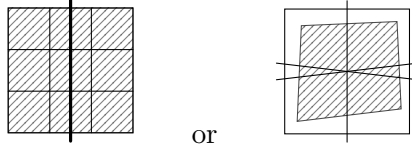
Here is the geometric meaning of (1.3.2). Consider the sphere S^2 , cut into spherical triangles (chambers) by the root hyperplanes of the root system A_3 . A regular tetrahedron inscribed in S^2 gives a decomposition of S^2 into four spherical triangles; the decomposition considered is its barycentric subdivision. Call one of the triangles the fundamental chamber, and attach to each edge a label a, b , or c , invariantly under the Weyl group, with the edges of a triangle having distinct labels and with b opposite the right angle. In stereographic projection, sending one vertex to infinity:



The shaded chamber is the fundamental chamber C_0 , and the other shaded chamber is its opposite $\overline{C}_0$. Each word in (1.3.2) corresponds to a sequence G of adjacent chambers, a gallery, going from the fundamental chamber C_0 to the opposite chamber $\overline{C}_0$ and crossing exactly once each of the root hyperplanes: in the figure, the two lines and the four circles. The word gives the sequence of labels of the edges crossed. The cycle of words (1.3.2) gives a cycle of galleries. One passes from a gallery G to the next by a modification of one of the following types, or by two disjoint modifications of the first type:



Attach to G the path from the center of C_0 to the center of $\overline{C}_0$ obtained as the sequence of geodesics going from the center of one triangle to the center of the next. When passing from one gallery to the next, one has a homotopy between the corresponding paths:



For the sequence of galleries (1.3.2), these successive homotopies sweep out S^2 , giving rise to the generator of $\pi_1(\Omega S^2) = \pi_2(S^2)$.

Theorem 1.5 below makes it possible to verify that the compatibility condition (1.3.2) is sufficient. More precisely, one has a restriction functor from the category of actions of B_4^+ on C to the category of triples of functors $T(s_1), T(s_2), T(s_3)$ endowed with isomorphisms (1.3.1) satisfying (1.3.2): to T one attaches the $T(s_i)$ and the isomorphisms

$$\begin{aligned} T(s_i)T(s_{i+1})T(s_i) &\xrightarrow{\sim} T(s_i s_{i+1} s_i) = T(s_{i+1} s_i s_{i+1}) \xleftarrow{\sim} T(s_{i+1})T(s_i)T(s_{i+1}), \\ T(s_i)T(s_j) &\xrightarrow{\sim} T(s_i s_j) = T(s_j s_i) \xleftarrow{\sim} T(s_j)T(s_i). \end{aligned}$$

This restriction functor is an equivalence of categories.

1.4. A more convenient description is obtained by replacing (1.2.1) by a redundant system of generators and relations. Consider the more general case of the generalized strictly positive braid monoid B^+ corresponding to a finite Coxeter group W with distinguished generating set R . The usual braids correspond to $W = S_n$, with R the set of transpositions $(i, i+1)$. Let ℓ be the length function $W \rightarrow \mathbb{N}$ defined by the generating system R .

Write $\text{prod}(m, a, b)$ for a product of m alternating factors $abab \dots$. The group W is generated by the $i \in R$, subject to relations

$$\text{prod}(m_{ij}, i, j) = \text{prod}(m_{ij}, j, i) \quad (i, j \in R, i \neq j), \quad (1.4.1)$$

$$i^2 = e \quad (i \in R). \quad (1.4.2)$$

The corresponding monoid B^+ is generated by generators $\tilde{i}$, $i \in R$, subject only to the relations (1.4.1). It injects into the generalized braid group B defined by the same generators and relations (1.4.1) (Deligne (1972), 4.14(ii)). The projection $\tilde{i} \mapsto i$ from B^+ to W admits, on the complement of the identity element e , a canonical set-theoretic section τ , characterized by

$$\tau(i) = \tilde{i} \quad (i \in R), \quad (1.4.3)$$

$$\tau(w'w'') = \tau(w')\tau(w'') \quad \text{if } \ell(w'w'') = \ell(w') + \ell(w''). \quad (1.4.4)$$

(Bourbaki, Lie IV, 1.5, Proposition 5). This section provides a new presentation of B^+ :

$$\text{generators: } \tau(w) \quad (w \in W, w \neq e), \quad (1.4.5)$$

with relations (1.4.4). Note that (1.4.5), i.e. (1.4.4), makes it possible to write each $\tau(w)$ as a product of $\tau(i)$, $i \in R$, and implies (1.4.1) for the $\tau(i)$: the two sides of (1.4.1) are minimal decompositions of the same element of W .

Our main result is the following.

Theorem 1.1 (1.5). *It is the same thing (1.1) to give an action of B^+ on C as to give:*

(1.5.1) *for $w \in W$, $w \neq e$, a functor $T(w) : C \rightarrow C$;*

(1.5.2) *for w', w'' with $\ell(w'w'') = \ell(w') + \ell(w'')$, an isomorphism of functors*

$$(1.5.2)_{w', w''} : T(w')T(w'') \longrightarrow T(w'w'');$$

(1.5.3) these isomorphisms satisfying, for $\ell(w'w''w''') = \ell(w') + \ell(w'') + \ell(w''')$, the commutativity of the diagram

$$\begin{array}{ccc} T(w')T(w'')T(w''') & \longrightarrow & T(w'w'')T(w''') \\ \downarrow & & \downarrow \\ T(w')T(w''w''') & \longrightarrow & T(w'w''w'''). \end{array} \quad (1.5.3)$$

In this statement, “it is the same thing” presupposes that the data (1.5.1),(1.5.2) satisfying (1.5.3) form a category. A morphism $T_1 \rightarrow T_2$ is the data of morphisms of functors $T_1(w) \rightarrow T_2(w)$, $w \in W - \{e\}$, which make the diagrams

$$\begin{array}{ccc} T_1(w')T_1(w'') & \longrightarrow & T_1(w'w'') \\ \downarrow & & \downarrow \\ T_2(w')T_2(w'') & \longrightarrow & T_2(w'w'') \end{array}$$

commute whenever w', w'' are as in (1.5.2). We shall only construct the quasi-inverse to the restriction functor, and leave to the reader the verification that it is indeed a quasi-inverse. The difficult part of the proof is the contractibility theorem 1.7, proved in §2.

Let data (1.5.1),(1.5.2) satisfying (1.5.3) be given, and construct an action of B^+ on C which extends (1.5.1),(1.5.2).

Consider the set $E(b)$ of ways of writing $b \in B^+$ as a product of $\tau(w)$. More precisely, let $L(W - \{e\})$ be the free monoid generated by the $w \in W - \{e\}$. It maps to B^+ by $w \mapsto \tau(w)$, and $E(b) \subset L(W - \{e\})$ is the inverse image of b . The datum (1.5.1) gives, for each $A \in L(W - \{e\})$, a composite functor $T(A)$, extending (1.5.1) and such that $T(AB) = T(A)T(B)$.

If A_1 and A_2 are of the form

$$B : w' : w'' : C \quad \text{and} \quad B : (w'w'') : C \quad (1.5.4)$$

with $\ell(w'w'') = \ell(w') + \ell(w'')$, the datum (1.5.2) gives an isomorphism

$$T(B) : (1.5.2)_{w', w''} : T(C) : T(A_1) \longrightarrow T(A_2). \quad (1.5.5)$$

The essential point is to verify the following.

Lemma 1.2 (1.6). *The isomorphisms (1.5.5) generate a transitive system of isomorphisms among the $T(A)$, $A \in E(b)$.*

Proof of 1.6 $\Rightarrow$ 1.5. Define $T(b)$ to be the common value of the $T(A)$, $A \in E(b)$. More precisely, it is the projective limit of the system of the $T(A)$: one has isomorphisms $T(b) \xrightarrow{\sim} T(A)$, $A \in E(b)$, which make the diagrams

$$\begin{array}{ccc} & & T(A_1) \\ & \nearrow & \downarrow (1.5.5) \\ T(b) & & \\ & \searrow & T(A_2) \end{array}$$

commute.

For $b, c \in B^+$, $B \in E(b)$ and $C \in E(c)$, the composite word BC belongs to $E(bc)$. Moreover, if $u : T(B_1) \rightarrow T(B_2)$ is an isomorphism (1.5.5), then $u T(C)$ is an isomorphism (1.5.5) for bc ;

likewise after interchanging the roles of B and C . This allows one to define the isomorphism $T(b)T(c) \rightarrow T(bc)$ as the one making the diagrams

$$\begin{array}{ccc} T(b)T(c) & \longrightarrow & T(bc) \\ \downarrow & & \downarrow \\ T(B)T(C) & \longrightarrow & T(BC) \end{array}$$

commutative. This construction, functorial in the data (1.5.1),(1.5.2) satisfying (1.5.3), is the desired quasi-inverse to the restriction functor.

Proof of 1.6. Let $\leq$ be the order relation on $E(b)$ generated by $A_1 \leq A_2$ for A_1, A_2 as in (1.5.4). In other words, $A \leq B$ if B is deduced from A by factoring A as $A_1 \cdots A_k$, each A_i lying in $L(W - \{e\})$ and being of the form $w_{i,1} : \cdots : w_{i,m}$ with $\ell(w_{i,1} \cdots w_{i,m}) = \sum_j \ell(w_{i,j})$, and by replacing A_i by the product $w_{i,1} \cdots w_{i,m}$ in W . The following theorem is proved in 2.3 and 2.4.

Theorem 1.3 (1.7). *The geometric realization $|E(b)|$ of the ordered set $E(b)$ is contractible.*

The geometric realization of $E(b)$ is that of the category attached in the usual way to $E(b)$: its set of objects is $E(b)$, and $\text{Hom}(x, y)$ is reduced to a single element if $x \leq y$ and is empty otherwise. See Quillen (1973), §1, p. 89.

Let us show that 1.6 follows from the connectedness and simple connectedness of $|E(b)|$. The condition (1.5.3) assures that, for $A \leq B$, the composites of isomorphisms (1.5.5) give a unique isomorphism

$$\varphi_{B,A} : T(A) \longrightarrow T(B)$$

with $\varphi_{C,B}\varphi_{B,A} = \varphi_{C,A}$ for $A \leq B \leq C$.

By Quillen (1973), §1, Proposition 1, p. 90, the category of coverings of $|E(b)|$ is equivalent to the category of functors from the category $E(b)$ to the category (Sets, is) of sets and bijections. If $|E(b)|$ is connected and simply connected, every functor from $E(b)$ to (Sets, is) is therefore isomorphic to a constant functor: given sets $T(x)$ for $x \in E(b)$ and a system of bijections $\varphi_{y,x} : T(x) \rightarrow T(y)$ for $x \leq y$ satisfying $\varphi_{z,y}\varphi_{y,x} = \varphi_{z,x}$, these bijections generate a transitive system of bijections. The same result follows for T with values in a category $\mathcal{X}$ and the $\varphi_{x,y}$ isomorphisms: apply the preceding result to $\text{Hom}(A, T(x))$ for A in $\mathcal{X}$.

1.8. Variant: units. A unital action of a monoid with unit (M, e) on a category C is an action T of M such that $T(e)$ is an autoequivalence. There is then a unique isomorphism

$$T(e) \xrightarrow{\sim} \text{Id} \tag{1.8.1}$$

such that $c_{e,f} : T(e)T(f) \rightarrow T(f)$ and $c_{f,e} : T(f)T(e) \rightarrow T(f)$ are induced by (1.8.1). If M^+ is obtained from a monoid M by adjoining a unit, it is the same thing to give an action of M or a unital action of M^+ .

An action of a group G is a unital action of G .

Proposition 1.4 (1.9). *Let M be a monoid with unit satisfying the left or right Ore condition, and let G be its group of fractions. It is the same thing (1.1) to give an action of G on a category C as to give a unital action of M on C by autoequivalences.*

Proof. For definiteness, assume the left Ore condition.

For $f \in G$, let L_f be the category whose objects are pairs $(x, y) \in M \times M$ such that $f = x^{-1}y$, and whose arrows $a : (x, y) \rightarrow (x', y')$ are elements a of M such that $x' = ax$ and $y' = ay$. The Ore condition assures that L_f is filtered. For $a : (x, y) \rightarrow (x', y')$, one has an isomorphism

$$\varphi_a : T(x)^{-1}T(y) = T(x)^{-1}T(a)^{-1}T(a)T(y) = T(x')^{-1}T(y'),$$

and $\varphi_{ba} = \varphi_b \varphi_a$. Since the φ_a are isomorphisms and L_f is filtered, the inductive limit $T(f)$ of the $T(x)^{-1}T(y)$ exists, and $T(x)^{-1}T(y) \xrightarrow{\sim} T(f)$.

For $f, g \in G$, let $L_{f,g}$ be the category of triples (x, y, z) such that $f = x^{-1}y$ and $g = y^{-1}z$; an arrow $a : (x, y, z) \rightarrow (x', y', z')$ is an element a of M such that $x' = ax$, $y' = ay$, and $z' = az$. The Ore condition again assures that this category is filtered and that the functors (x, y) , (y, z) , (x, z) from $L_{f,g}$ to L_f , L_g , and L_{fg} are cofinal. Define $c_{f,g} : T(f)T(g) \rightarrow T(fg)$ as the inductive limit of the isomorphisms

$$T(x)^{-1}T(y)T(y)^{-1}T(z) \longrightarrow T(x)^{-1}T(z).$$

To verify (0.1), write f, g, h in the forms $x^{-1}y$, $y^{-1}z$, and $z^{-1}t$. □

1.10. By Deligne (1972), the generalized positive braid monoid obtained from B^+ (1.4) by adjoining an identity satisfies the Ore condition both on the left and on the right. By 1.8, 1.9, and 1.5, it is the same thing to give an action on a category of the corresponding generalized braid group as to give data (1.5.1), (1.5.2), (1.5.3) for which the $T(w)$ are autoequivalences.

1.11. Variant. Let $\mathcal{M}$ be a monoidal category and consider the following categories (A) and (B).

(A) The category of systems consisting of:

(A.1) a function $B^+ \rightarrow \text{Ob } \mathcal{M}$, $b \mapsto T(b)$;

(A.2) isomorphisms $T(b')T(b'') \xrightarrow{\sim} T(b'b'')$ satisfying

(A.3) the commutativity of the square

$$\begin{array}{ccc} T(b')T(b'')T(b''') & \longrightarrow & T(b'b'')T(b''') \\ \downarrow & & \downarrow \\ T(b')T(b''b''') & \longrightarrow & T(b'b''b'''). \end{array}$$

(B) The category of systems consisting of:

(B.1) a function $W - \{e\} \rightarrow \text{Ob}(\mathcal{M})$, $w \mapsto T(w)$;

(B.2) isomorphisms $T(w')T(w'') \xrightarrow{\sim} T(w'w'')$ for $\ell(w'w'') = \ell(w') + \ell(w'')$, satisfying

(B.3) the commutativity of the square

$$\begin{array}{ccc} T(w')T(w'')T(w''') & \longrightarrow & T(w'w'')T(w''') \\ \downarrow & & \downarrow \\ T(w')T(w''w''') & \longrightarrow & T(w'w''w''') \end{array}$$

whenever $\ell(w'w''w''') = \ell(w') + \ell(w'') + \ell(w''')$.

The restriction functor from (A) to (B) is an equivalence. The proof is the same as that of 1.5.

The same conclusion holds if $\mathcal{M}$ is a Picard category, i.e. a monoidal category with unit in which every object is invertible, and if in (A) one replaces B^+ by the generalized braid group B . The proof is the same as in 1.9 and 1.10.

2 Contractibility

2.1. Let V be a finite-dimensional real vector space and let $\mathcal{M}$ be a finite set of homogeneous hyperplanes of V , called walls. Assume that the connected components of $V - \bigcup_{M \in \mathcal{M}} M$, called chambers, are open simplicial cones. We shall use the terminology of Bourbaki (Lie V, §1), namely walls, chambers, faces, facets, and the terminology of Deligne (1972) recalled below.

Two chambers A and B are called adjacent if $A \neq B$ and A and B have a face in common. A gallery of length n , $n \geq 0$, with source A and target B , is a sequence of chambers C_i , $0 \leq i \leq n$, with $A = C_0$, C_i adjacent to C_{i+1} for $0 \leq i < n$, and $C_n = B$. The composite GG' of two galleries $G = (C_0, \dots, C_n)$ and $G' = (C'_0, \dots, C'_m)$, defined if $C_n = C'_0$, is $(C_0, \dots, C_n = C'_0, \dots, C'_m)$. For two chambers A and B , the minimal galleries from A to B are those of minimal length, i.e. length the number $d(A, B)$ of walls separating A and B . They cross exactly once each wall separating A from B . Equivalence of galleries is the finest equivalence relation compatible with composition for which the minimal galleries from a chamber A to a chamber B are equivalent. Equivalent galleries have the same length. More precisely, they cross each wall the same number of times. The minimal galleries from A to B form an equivalence class, denoted $u(A, B)$, and if

$$d(A, B) + d(B, C) = d(A, C),$$

then $u(A, B)u(B, C) = u(A, C)$ for composition of equivalence classes of galleries.

Let A, B, C be three chambers. The following conditions are equivalent (Deligne (1972), 1.4): (a) $d(A, B) + d(B, C) = d(A, C)$; (b) there exists a minimal gallery from A to C passing through B ; (c) B lies in the intersection of the half-spaces bounded by a wall and containing A and C . Write $D(A, C)$ for the set of chambers B satisfying these conditions, the chambers between A and C . For fixed A , the relation “ B is between A and C ”, equivalently $D(A, B) \subset D(A, C)$, is an order relation among chambers.

By the central technical result 1.9 of Deligne (1972), for every equivalence class g of galleries with source A , if g begins with $u(A, C)$, i.e. if there exists g' such that $g = u(A, C)g'$, then the class g' is unique. Moreover, there is an optimal chamber C such that g begins with $u(A, C)$: for g to begin with $u(A, B)$ it is necessary and sufficient that B be between A and C .

2.2. Let g be an equivalence class of galleries from A to B , let $\ell(g)$ be their length, and let $E(g)$ be the set of ways of writing g as a composite of classes $u(C, D)$. More precisely, $E(g)$ is the set of sequences of chambers (C_i) , $0 \leq i \leq n$, such that $C_0 = A$, $C_n = B$, $C_i \neq C_{i+1}$ for all i , and

$$g = \text{composite of the } u(C_i, C_{i+1}). \quad (2.2.1)$$

The condition (2.2.1) implies

$$\sum d(C_i, C_{i+1}) = \ell(g). \quad (2.2.2)$$

Order $E(g)$ as follows:

$$(C_0, \dots, C_n) \leq (D_0, \dots, D_m)$$

if there is a strictly increasing map $\varphi : \{0, \dots, m\} \rightarrow \{0, \dots, n\}$ such that $\varphi(0) = 0$, $\varphi(m) = n$, $D_j = C_{\varphi(j)}$, and

$$d(C_{\varphi(j)}, C_{\varphi(j+1)}) = \sum_{\varphi(j) \leq i < \varphi(j+1)} d(C_i, C_{i+1}).$$

The map φ is then unique. If $(C_0, \dots, C_n) \leq (D_0, \dots, D_m)$, one passes from the decomposition $g = \prod u(C_i, C_{i+1})$ to the decomposition $g = \prod u(D_j, D_{j+1})$ by replacing the consecutive factors $u(C_i, C_{i+1})$, for $\varphi(j) \leq i < \varphi(j+1)$, by their composite $u(D_j, D_{j+1})$.

2.3. Example. Let $W \subset GL(V)$ be a finite group generated by reflections. Assume that the invariant subspace V^W is 0. Take as walls the mirrors, i.e. fixed hyperplanes, of the reflections in W . It is known that the chambers of V are simplicial cones, permuted simply transitively by W (Bourbaki, Lie V, 3.9, Prop. 7 and 3.2, Th. 1).

Choose a chamber C_0 and call it the fundamental chamber. For every chamber A , let $g(A)$ be the unique element of W which sends C_0 to A . For two chambers A and B , the relative position $w(A, B)$ of A and B is the element w of W such that $g(A)$ sends (C_0, wC_0) to (A, B) : one has $w(A, B) = g(A)^{-1}g(B)$ and $w(A, B)w(B, C) = w(A, C)$.

The group W , equipped with the set R of reflections with respect to the mirrors containing the faces of C_0 , is a Coxeter group (Bourbaki, Lie V, 3.2, Theorem 1).

Let $G = (A_0, \dots, A_n)$ be a gallery of length $n > 0$. For $0 \leq i < n$, let $r(i) \in R$ be the relative position of A_i and A_{i+1} . Attach to G the element

$$b(G) = \widetilde{r(0)} \cdots \widetilde{r(n-1)} \quad (2.3.1)$$

of B^+ (1.4). The image of $b(G)$ in W is the relative position of A_0 and A_n . For a composite gallery, one has

$$b(G'G'') = b(G')b(G''). \quad (2.3.2)$$

The definition (2.1) of equivalence of galleries mimics the presentation (1.4.5) of B^+ , and one verifies that (2.3.1) induces a bijection, still denoted b :

$$\left\{ \begin{array}{c} \text{equivalence classes of galleries of fixed source} \\ \text{and of positive length} \end{array} \right\} \xrightarrow{\sim} B^+. \quad (2.3.3)$$

Moreover, the map

$$E(g) \longrightarrow E(b(g)), \quad (C_0, \dots, C_n) \longmapsto w_0 : \dots : w_{n-1},$$

where w_i is the relative position of C_i and C_{i+1} , is an isomorphism of ordered sets. Theorem 1.7 is therefore a particular case of the following statement.

Theorem 2.1 (2.4). *With the notation of 2.2, the geometric realization $|E(g)|$ of the ordered set $E(g)$ is contractible.*

Proof. We prove 2.4 by induction on the length $\ell(g)$ of g . If $\ell(g) = 0$, then $A = B$ and the set of sequences $E(g)$ is reduced to the sequence (A) . Thus $|E(g)|$ is contractible. Let ℓ be an integer > 0 , suppose 2.4 known for $\ell(g) < \ell$, and prove 2.4 for $\ell(g) = \ell$.

Let g be an equivalence class of galleries of length ℓ from A to B ; let C be the optimal chamber (2.1) such that g begins with $u(A, C)$; and let S be the set of walls of A which separate A from C . Since $\ell > 0$, $A \neq C$ and S is non-empty.

For $s \in S$, let $E_s(g) \subset E(g)$ be the set of sequences (C_i) in $E(g)$ such that the wall s separates A from C_1 . The set $E(g)$ is the union of the $E_s(g)$. Each $E_s(g)$ is a final segment of $E(g)$. With the induced order, one therefore has, for geometric realizations,

$$|E(g)| = \bigcup |E_s(g)|. \quad (2.4.1)$$

For a non-empty subset T of S , let $E_T(g)$ be the intersection of the $E_t(g)$ for $t \in T$, equipped with the induced order. Then

$$|E_T(g)| = \bigcap |E_t(g)|. \quad (2.4.2)$$

Lemma 2.2 (2.5). *Each $|E_T(g)|$ is contractible.*

Proof. Let F_T be the facet of A open in the intersection P_T of the walls $t \in T$, and let A_T be the chamber opposite to A in the star of F_T . In the notation 1.6(ii) of Deligne (1972), $A_T := A \cdot \Delta(P_T)$. The chamber A_T is the least (2.1) of the chambers separated from A by the walls $t \in T$. In particular, A_T lies between A and C . Let h be the unique equivalence class of galleries from A_T to B such that $g = u(A, A_T)h$. One has $\ell(h) < \ell(g)$.

Let $f : E(h) \rightarrow E_T(g)$ be the increasing map

$$(C_0 = A_T, \dots, C_n) \mapsto (A, C_0, \dots, C_n). \quad (2.5.1)$$

If (C_i) lies in $E(g)$, then C_1 lies in $D(A, C)$, and for (C_i) to lie in $E_T(g)$ it is necessary and sufficient that C_1 lie in $D(A_T, C)$. The map f therefore induces an isomorphism of $E(h)$ with an initial segment of $E_T(g)$.

Let $f_* : E_T(g) \rightarrow E(h)$ be the increasing map

$$(C_0 = A, C_1, \dots, C_n) \mapsto \begin{cases} (C_1, \dots, C_n), & C_1 = A_T, \\ (A_T, C_1, \dots, C_n), & C_1 \neq A_T. \end{cases} \quad (2.5.2)$$

For $x \in E(h)$ and $y \in E_T(g)$, one has

$$f(x) \leq y \iff x \leq f_*(y). \quad (2.5.3)$$

Viewing $E(h)$ and $E_T(g)$ as categories, and f, f_* as functors (cf. 1.7), (2.5.3) says that (f, f_*) is an adjoint pair of functors. By Quillen (1973), §1, Corollary 1 to Proposition 2, p. 92, the map

$$|f| : |E(h)| \longrightarrow |E_T(g)|$$

is a homotopy equivalence; 2.5 follows from the induction hypothesis. $\square$

The non-empty intersections of the $|E_s(g)|$ being contractible ((2.4.2) and 2.5), $|E(g)|$ has the same homotopy type as the nerve of its finite closed covering by the $|E_s(g)|$. Since all intersections are non-empty, this nerve is a simplex and $|E(g)|$ is contractible. $\square$

Remark 2.6. Let $\mathcal{C}$ be a 2-category whose objects are indexed by the chambers. Just as 1.7 gave 1.5, Theorem 2.4 gives, by the same proof, that it is the same thing (1.1) to give (A) or (B) as follows.

(A) (i) For g an equivalence class of galleries of positive length from A to B , a 1-arrow $T(g) : X_A \rightarrow X_B$;

(ii) for $g = g'g''$, an isomorphism $T(g'')T(g') \rightarrow T(g)$, these isomorphisms making the diagrams

$$\begin{array}{ccc} T(g''')T(g'')T(g') & \longrightarrow & T(g''g''')T(g') \\ \downarrow & & \downarrow \\ T(g''')T(g'g'') & \longrightarrow & T(g'g''g''') \end{array}$$

commutative.

(B) (i) For two distinct chambers A and B , a 1-arrow $T_{B,A} : X_A \rightarrow X_B$;

(ii) for three distinct chambers A, B, C , with C between A and B , an isomorphism

$$T_{C,B}T_{B,A} \xrightarrow{\sim} T_{C,A},$$

these isomorphisms making the diagrams

$$\begin{array}{ccc} T_{D,C}T_{C,B}T_{B,A} & \longrightarrow & T_{D,B}T_{B,A} \\ \downarrow & & \downarrow \\ T_{D,C}T_{C,A} & \longrightarrow & T_{D,A} \end{array}$$

commutative when they are defined, i.e. when $d(A, D) = d(A, B) + d(B, C) + d(C, D)$.

Application 1 from the introduction is more conveniently expressed in this language: the set of total orders on I is identified with the set of chambers of a root system of type A_n , for $n = |I| - 1$. For each total order $\leq$, corresponding to a chamber C , let X_C be the category of corresponding exceptional systems, and let $\text{Hom}(X_C, X_D)$ be the category of functors from X_C to X_D . What Bondal constructs is a datum of type (B).

References

- [1] A. I. Bondal, M. M. Kapranov, *Representable functors, Serre functors and mutations*, Math. USSR Izvestiya 63, 1183–1205 (1989) (in Russian); AMS translation 35, 519–541 (1990).
- [2] M. Broué, J. Michel, *Sur certains éléments réguliers des groupes de Weyl et les variétés de Deligne–Lusztig associées*, in *Finite reductive groups, related structures and representations* (Proceedings of the October 1994 Luminy conference), Basel–Boston–Stuttgart: Birkhäuser, 1996.
- [3] P. Deligne, *Les immeubles des groupes de tresses généralisés*, Invent. Math. 17, 273–302 (1972).
- [4] F. A. Garside, *The braid group and other groups*, Quart. J. Math. Oxford, 2nd series 20, 235–254 (1969).
- [5] D. Quillen, *Higher algebraic K-theory I*, Battelle Institute Conference 1972, pp. 85–142, Lecture Notes in Math. 341, Springer, 1973.
- [6] J. L. Verdier, *Catégories dérivées (état 0) [texte de 1963]*, in *Cohomologie étale* (SGA 4 $\frac{1}{2}$), Lecture Notes in Math. 569, Springer, 1977, pp. 262–311.

Some Master Ideas in the Work of A. Grothendieck

Pierre Deligne

Abstract

This article attempts to explain four fundamental mathematical concepts created by Grothendieck: schemes, topoi, the six operations, and motives.

In *Récoltes et semailles* (third part), Grothendieck writes:

Take, for example, the task of proving a theorem which still remains hypothetical (to which, for some, mathematical work would seem to reduce). I see two extreme approaches to going about it. One is that of the hammer and chisel, when the problem posed is seen as a large nut, hard and smooth, whose inside one must reach, the nourishing flesh protected by the shell. The principle is simple: one places the edge of the chisel against the shell, and one strikes hard. If need be, one begins again at several different places, until the shell breaks — and one is satisfied. [...]

I could illustrate the second approach while keeping the image of the nut which has to be opened. The first parable which came to my mind just now is that one plunges the nut into an emollient liquid, simply water, why not; from time to time one rubs it so that the liquid penetrates better, and for the rest one lets time do its work. The shell softens over the weeks and months; when the time is ripe, a pressure of the hand suffices, the shell opens like that of an avocado ripened to the point! Or again, one lets the nut ripen under the sun and the rain and perhaps also under the winter frosts. When the time is ripe, it is a delicate shoot issuing from the substantial flesh which will have pierced the shell as if in play; or, to speak more exactly, the shell will have opened of itself to let it pass. [...]

The reader who is at all familiar with some of my work will have no difficulty recognizing which of these two modes of approach is “mine.”

A little farther on [ibid., p. 554, note (***)], Grothendieck puts forward four examples: Riemann–Roch, the structure of the prime-to-the-characteristic π_1 for curves, rationality of L -functions for schemes of finite type over a finite field, and the semi-stable reduction theorem for abelian varieties. I remember my amazement, in 1965–66, after Grothendieck’s lecture [SGA 5] proving the base-change theorem for $Rf_!$: dévissages, dévissages, nothing seems to happen and yet at the end of the lecture a clearly non-trivial theorem is there.

Many of Grothendieck’s ideas have become so familiar to us, and are so perfectly adapted to their object, that we forget that they were far from obvious at their birth; we even forget their author. My aim in this article is to describe four of these ideas: schemes, topoi, the six operations, and motives.

1 Schemes

The invention of schemes is the first of Grothendieck's ideas that comes to mind, perhaps because it was the most quickly accepted. Serre's address in Stockholm in 1962 begins: "I would like to describe here some of the recent developments in algebraic geometry. I should specify that I take this last term in the sense which it has acquired over the last few years: that of the theory of schemes." This acceptance was facilitated by the rapid publication, thanks to the collaboration of Dieudonné, of EGA.

The boldness of Grothendieck's definition is to accept that every commutative ring (with unit) A defines an affine scheme $\text{Spec}(A)$, that is, not to try to restrict oneself to a category of "good" rings (domains, reduced rings, noetherian rings, ...). This has a price. The points of $\text{Spec}(A)$ (prime ideals of A) do not have a manageable geometric meaning, and the structural sheaf $\mathcal{O}$ is not a sheaf of functions. When one has to construct a scheme, one does not in general begin by constructing its set of points.

Perhaps more important is the decision to build a relative theory, to which the omnipresent

$$X \longrightarrow S$$

in Grothendieck's lectures bears witness. The classical case of a variety defined over a field k becomes the special case $S = \text{Spec}(k)$. In a relative theory, with an S -scheme X , that is, with a morphism of schemes $f : X \rightarrow S$, one systematically considers the scheme X' deduced from X by a base change $u : S' \rightarrow S$, that is, the fiber product $X' := S' \times_S X$ and its projection f' onto S' :

$$\begin{array}{ccc} X' & \longrightarrow & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{u} & S. \end{array}$$

In the category of schemes, fiber products always exist: if allowing every commutative ring to define an affine scheme gives citizenship to strange schemes, allowing it supplies a category of schemes with good properties.

A property of X over S will be called geometric if it has good invariance properties under base change. Classical analogue: for a variety X defined over a field k , the set of points of X over an algebraically closed extension Ω of k (for example, Weil's universal domain) is regarded as "geometric", while the set of k -points is "arithmetic".

If X is a scheme over S , and $u : S' \rightarrow S$ is a morphism of schemes, an S' -point p of X is a morphism of S -schemes:

$$\begin{array}{ccc} & & X \\ & \nearrow p & \downarrow \\ S' & \xrightarrow{u} & S. \end{array}$$

One denotes by $X(S')$ the set of S' -points of X . It identifies with the set of sections of $X' \rightarrow S'$.

Example 1.1. Let X be the affine scheme over a field k defined by equations

$$P_\alpha(X_1, \dots, X_n) = 0 : \quad X = \text{Spec}(k[X_1, \dots, X_n]/(P_\alpha)).$$

Let k' be a k -algebra. The set $X(k') := X(\text{Spec}(k'))$ is the set of solutions in k'^n of the equations $P_\alpha = 0$. For k' , an algebraically closed extension Ω of k , this is the set $X(\Omega)$ that Weil regards as underlying X .

Example 1.2. The scheme GL_N over $\text{Spec}(\mathbb{Z})$ is such that for every commutative ring A (automatically a $\mathbb{Z}$ -algebra: $\text{Spec}(A)$ is a scheme over $\text{Spec}(\mathbb{Z})$), $GL_N(\text{Spec}(A))$ “is” $GL_N(A)$.

In these examples, the geometric intuition that one has of X , a scheme over S , is well reflected in $X(S')$. Better than in the underlying set of X . In the case of Example 1.2, this underlying set is not a group, whereas each $GL_N(S')$ is one. More precisely, it is useful to attach to the scheme X over S the contravariant functor from S -schemes to sets

$$S' \longmapsto \text{the set } X(S') \text{ of } S'\text{-points of } X.$$

In the category of S -schemes, this is simply the representable functor

$$h_X : S' \longmapsto \text{Hom}(S', X)$$

attached to X . By the Yoneda lemma, the functor $X \mapsto h_X$ is fully faithful. Rather than thinking of an S -scheme X as a ringed space, equipped with $X \rightarrow S$, with suitable properties, it is often convenient to think of it as a functor

$$S\text{-points} : (\text{Schemes}/S)^\circ \longrightarrow (\text{Sets}),$$

which has the virtue of being representable. When one wants to define a “fine moduli space”, the first step is to define the corresponding functor. Typically, defining this functor requires a relative theory.

Example 1.3. Let X be projective over a field k . Question: what does “moduli space of closed subschemes of X ” mean? For $u : S' \rightarrow \text{Spec}(k)$, let X' over S' be deduced from X by base change. Let $\mathcal{H}(S')$ be the set of closed subschemes Z' of X' which are flat over S' (flat of finite presentation, if one does not want to assume S' noetherian). Answer: it is a scheme $\text{Hilb}(X)$ representing the functor $S' \mapsto \mathcal{H}(S')$.

For this point of view to be viable, one must have methods for verifying when a functor is representable. Most of the lectures of FGA are devoted to this problem. A definitive solution, extending this work, was obtained by M. Artin [1969], at least if one agrees to replace the category of schemes by the more natural category of algebraic spaces. More natural: in the same sense in which the étale topology is more natural than the Zariski topology.

2 Topoi

Let $u : S' \rightarrow S$ be a morphism of schemes, the “base-change morphism”. Descent theory [Grothendieck FGA, Séminaire Bourbaki 190, 1959–60] considers problems of the following kinds. Descent of properties: let X be a scheme over S , and X' over S' deduced from X by base change. Suppose that X'/S' has a property P . Can one conclude that X/S satisfies P ? Descent of morphisms: let X, Y be over S , let X', Y' be deduced by base change, and let $g' : X' \rightarrow Y'$ be a morphism of S' -schemes. When does g' arise by base change from $g : X \rightarrow Y$? Descent of objects: let X' be over S' . What descent datum on S' is required in order to construct X over S from which X' is deduced by base change?

If S' is the disjoint union of the open subsets $(U_i)_{i \in I}$ of a covering of S , the base change to S' is essentially the restriction to each U_i , and the preceding problems are problems of localization on S and gluing. Gluing typically requires considering the pairwise intersections

$U_i \cap U_j$, and in terms of $u : S' \rightarrow S$, the disjoint union of the $U_i \cap U_j$ is simply the fiber product $S' \times_S S'$.

The theory of topoi makes it possible to transpose topological intuition into descent theory. For $u : S' \rightarrow S$, a base-change morphism of a type considered in descent theory, for example faithfully flat and quasi-compact (fpqc), base change from S to S' becomes a localization. A descent datum is the analogue of a gluing datum.

An antecedent of descent theory is Galois descent, corresponding to S , the spectrum of a field k , and S' , the spectrum of a Galois extension. Here, $S' \times_S S'$ is a sum of copies of S' indexed by $\text{Gal}(k'/k)$. The proofs, and in particular Galois theory, are however simpler in the more general framework of descent theory. In Cartier's phrase: Grothendieck proves Galois theory, and Galois descent, by Galois descent.

The tool that is topos theory enabled the construction of étale cohomology of schemes, and this is its best-known success. A sheaf on X for the Zariski topology is a contravariant functor from the category of Zariski open subsets of X to the category of sets, with a gluing condition for $(U_i)_{i \in I}$, a Zariski open covering of U . In the étale topology, the Zariski site considered above is replaced by the étale site: the category of Zariski open subsets is replaced by that of $f : U \rightarrow X$ étale over X , and coverings by the covering families $(U_i)_{i \in I}$, that is, a surjective morphism of schemes over X from the disjoint union of the U_i to U . An antecedent: Serre's introduction of the notion of an isotrivial principal homogeneous space (= locally trivial for a topology close to the étale topology). In his Kansas [1955] and Tôhoku [1957] papers, Grothendieck had shown that, once a category of sheaves is given, a notion of cohomology groups results. The étale topology thus supplies étale cohomology. That the H^1 groups obtained are reasonable is not surprising. The miracle is that the higher H^i are reasonable too.

For Grothendieck, the importance of topos theory goes far beyond the single case of the étale topology. The title given to SGA 4 bears witness to this: "Theory of topoi and étale cohomology of schemes."

Other applications:

- (A) *Crystalline cohomology*. Crystalline cohomology is the cohomology of the crystalline topos, and this definition makes its functoriality clear. The crystalline topos is nevertheless delicate to use, and it is often necessary to pass to an interpretation in terms of de Rham complexes.
- (B) *Rigid analytic spaces*. Tate's rigid analytic sheaves are coherent sheaves on a suitable ringed topos, and their cohomology is the corresponding cohomology.
- (C) *Foliations*. A variety X equipped with a foliation $\mathcal{F}$ defines a "quotient" $X/\mathcal{F}$, which is a topos locally isomorphic to that of sheaves on a variety. This point of view seems to have been eclipsed by that of Connes, who instead associates to $\mathcal{F}$ a non-commutative C^* -algebra.

I would also like to mention Grothendieck's use of "big" sites (topoi), notably for interpreting classifying spaces.

3 The six operations

The formalism of the six operations presupposes that of derived categories. For a history of the latter, I refer to Illusie's text [1990]. Basic idea: for all sorts of cohomology groups,

their definition supplies not only these groups, but also a complex K whose cohomology groups they are. Typically, this complex is not uniquely determined, but it is so up to quasi-isomorphism: for two variants K', K'' of K , one has K''' and morphisms $K' \leftarrow K''' \rightarrow K''$ inducing isomorphisms on cohomology. In the derived category, K thereby becomes unique up to unique isomorphism.

The six operations are functors between derived categories. They are the derived tensor product $\otimes^L$, the internal Hom $R\mathcal{H}om$ (giving rise to the local $\mathcal{E}xt^i$), and, for $f : X \rightarrow Y$ a morphism of schemes, two direct-image functors: Rf_* and $Rf_!$, and two inverse-image functors: Lf^* and $Rf^!$.

The formalism of these operations was first isolated in the cohomology of coherent sheaves [Grothendieck 1963]. Here, for direct images, one should consider only proper morphisms, for which $Rf_* = Rf_!$. The formalism provides a relative version of Serre duality, in the form of an adjunction between the functors Rf_* and $Rf^!$. It also provides a fixed-point formula more general than the “Woodshole fixed point formula”, which is contemporary with it (Summer Institute on Algebraic Geometry, Whitney estate, Woodshole, Massachusetts, 1964).

Miraculously, the same formalism applies in étale cohomology — with very different proofs. It again supplies a formalism of duality (Poincaré duality) and fixed-point formulae (Lefschetz formulae).

4 Motives

Let X be an algebraic variety over an algebraically closed field k . For each prime number ℓ prime to the characteristic, the étale topology supplies ℓ -adic cohomology groups $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$. If k is a subfield of $\mathbb{C}$, one has comparison isomorphisms

$$H^i(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} H_{\text{ét}}^i(X, \mathbb{Z}_\ell).$$

For k of characteristic > 0 , there is no functorial integral cohomology giving rise to such isomorphisms. Nevertheless, the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$ have, as ℓ varies, a “family resemblance”. For $i = 1$, and X projective and smooth, $\text{Pic}^0(X)$ plays the role of the nonexistent integral theory: it recovers the $H_{\text{ét}}^1(X, \mathbb{Z}_\ell)$ and is an object “over $\mathbb{Z}$ ”, in that groups of homomorphisms between abelian schemes are of finite type.

The theory of motives is first of all an attempt to find a substitute for the nonexistent integral cohomology, explaining the family resemblance among the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$, especially for X projective and smooth. One recognizes the Master’s hand in the idea that the problem is not to define what a motive is: the problem is to define the category of motives, and to bring out the structures it carries. These structures should make it possible to prove the Weil conjecture on the model of Serre [1960]. See Grothendieck [1969].

It was on this occasion that Grothendieck invented the notion of a Tannakian category, developed in Saavedra’s thesis [1972]. The aim was to formalize the notion of tensor product of motives, corresponding by the Künneth formula to the product of varieties.

Even if the theory of motives has not achieved its original aim, its influence has been great. I refer to the Seattle conference on motives [Jannsen et al., 1994] for a panorama of its applications.

Bibliography

A bibliography of Grothendieck is given at the beginning of his Festschrift [Cartier et al. 1990, p. xiii–xx].

Other sources: the introduction, and J. Dieudonné’s article “From functional analysis to the foundations of algebraic geometry”, in the same Festschrift (p. 1–14); the International Congress of Mathematicians addresses by J.-P. Serre (Stockholm 1962), “Algebraic geometry” (p. 190–496), J. Dieudonné (Moscow 1966), “The work of Alexander Grothendieck” (p. 21–24), and, for Riemann–Roch and Grothendieck groups, H. Cartan (Moscow 1966), “The work of Michael F. Atiyah” (p. 9–14).

ARTIN (M.)

[1969] Algebraization of formal moduli, I, in *Global Analysis* (Papers in honor of K. Kodaira), Tokyo: Univ. Tokyo Press, 1969, p. 21–71.

CARTIER (P.), ILLUSIE (L.), KATZ (N.M.) et al., eds.

[1990] *The Grothendieck Festschrift*, vol. I, vol. 86 of *Progress in Mathematics*, Boston: Birkhäuser, 1990.

GROTHENDIECK (A.)

[1955] A general theory of fibre spaces with structure sheaf, University of Kansas, 1955.

[1957] Sur quelques points d’algèbre homologique, *Tôhoku Math. J.*, 9 (1957), p. 119–221.

[1963] Résidus et dualité, prenotes for a “Hartshorne seminar”, manuscript. See Hartshorne [1966].

[1969] Standard conjectures on algebraic cycles, in *Algebraic Geometry* (Colloquium Tata Inst., Bombay, 1968), Oxford Univ. Press, 1969, p. 193–199.

[1985] *Récoltes et semailles : Réflexions et témoignages sur un passé de mathématicien*, Montpellier: Univ. Sci. et Tech. Languedoc et CNRS, 1985.

[EGA] *Éléments de Géométrie algébrique*. Inst. Hautes Études Sci. Publ. Math, 4, 8, 11, 17, 20, 24, 28, 32. In collaboration with J. Dieudonné.

[FGA] *Fondements de la Géométrie algébrique*. Extracts from the Séminaire Bourbaki 1957–1962: seminars 149, 182, 190, 195, 212, 221, 232 and 236.

[SGA] *Séminaire de Géométrie algébrique du Bois Marie*. The seminar SGA 5: *Cohomologie ℓ -adique et Fonctions L* (1965–66), was finally published: vol. 589 of Lecture Notes in Math., Berlin–Heidelberg: Springer, 1977.

HARTSHORNE (R.)

[1966] *Residues and duality*, vol. 20 of Lecture Notes in Math., Berlin–Heidelberg: Springer, 1966.

ILLUSIE (L.)

[1990] Catégories dérivées et dualité, travaux de J.-L. Verdier, *Enseign. math.*, 36 (1990), p. 369–391.

JANNSEN (U.), KLEIMAN (S.) and SERRE (J.-P.), eds.

[1994] *Motives*, vol. 55 of the *Proceedings of Symposia in Pure Mathematics*, Providence: Amer. Math. Soc., 1994.

SAAVEDRA (N.)

[1972] *Catégories tannakiennes*, vol. 265 of Lecture Notes in Math., Berlin–Heidelberg: Springer, 1972.

SERRE (J.-P.)

[1960] Analogues kähleriens de certaines conjectures de Weil, *Ann. of Math.*, 71 (1960), p. 392–394; *Œuvres*, vol. II, Berlin–Heidelberg: Springer, 1986, p. 1–3.

The Cohomology Algebra of the Complement, in an Affine Space, of a Finite Family of Affine Subspaces

P. Deligne, M. Goresky, and R. MacPherson

This paper is dedicated to W. Fulton in admiration for his mathematical achievements and his broad influence on the field of algebraic geometry.

0. Introduction

Let X be the complement, in a real affine space V , of a locally finite set E of proper affine subspaces of V . The flats of E are V and the non-empty intersections of elements of E . In [GM], Part III, Goresky and MacPherson apply Morse theory to a function “distance to $v_0 \in V$ ” in order (a) to decompose the homology of X into factors indexed by the flats, and (b) to give a combinatorial description of each factor. In Section 1 we recover these results by sheaf-theoretic techniques. These techniques make clear what the decomposition of [GM] depends on, and they also apply to the determination of the ℓ -adic cohomology of the complement of a finite family of affine subspaces in an affine space over an algebraically closed field.

By duality, these homology results are transported to cohomology. For E finite, one has a decomposition indexed by the flats

$$H^*(X) = \bigoplus h^*(A). \quad (0.1)$$

Let $\text{coor}(A)$ be the free rank-one $\mathbb{Z}$ -module with bases $\pm e$ the coorientations of A in V , placed in degree $\text{codim}(A)$; then

$$\text{coor}(A) = H_A^*(V, \mathbb{Z}). \quad (0.2)$$

The direct factor $h^*(A)$ is the tensor product of $\text{coor}(A)$ with a group depending only on the combinatorics of the arrangement, that is, only on the set $\mathcal{A}^+$ of flats ordered by inclusion. The decomposition (0.1) depends on the choice, for each flat A , of a point

$$x_A \in A^0 := A - \bigcup \{\text{smaller flats}\}.$$

Only the connected component of A^0 containing x_A matters. In particular, if the following connectedness hypothesis (CONN) is satisfied, then the decomposition (0.1) depends on no choice:

$$\text{if a flat } B \text{ is contained in a flat } A, \quad A = B \text{ or } \dim(A) - \dim(B) \geq 2. \quad (\text{CONN})$$

In Section 4.2, under the hypothesis (CONN), we compute the cup product of $H^*(X)$ in terms of the decomposition (0.1). The essential point is that the component

$$h(A) \otimes h(B) \longrightarrow h(C)$$

of the cup product can be non-zero only for $C = A \cap B$, the intersection being transverse. We prove this by interpreting the cup product as the exterior product followed by restriction to the diagonal $X \rightarrow X \times X$, and by treating this restriction as a special case of a restriction morphism

$$H^*(X) \longrightarrow H^*(X'),$$

where $X' = V' \cap X$ is the trace of X on an affine subspace.

Under (CONN), it follows from this calculation that the cohomology algebra $H^*(X)$ depends, up to unique isomorphism, only on the following data:

- (a) the set $\mathcal{A}^+$ of flats, ordered by inclusion;
- (b) the function $A \mapsto \text{codim}(A)$;
- (c) for each flat A , the free rank-one $\mathbb{Z}$ -module $\text{coor}(A)$;
- (d) for $C = \inf(A, B)$ and $\text{codim}(C) = \text{codim}(A) + \text{codim}(B)$, that is, for a transverse intersection $C = A \cap B$, the natural isomorphism from $\text{coor}(A) \otimes \text{coor}(B)$ to $\text{coor}(C)$.

M. de Longueville and C. Schultz independently determined the integral cohomology algebra of the complement of a central arrangement satisfying (CONN) [dS]. See also [FZ].

Acknowledgments. Our interest in these questions was drawn by the preprint [Y] of Yuzvinsky. In the case of complex arrangements, Yuzvinsky determines, up to isomorphism and using the results of Morgan [M], the rational cohomology algebra of the complement of an arrangement.

1 Decomposition of Cohomology

1.1

We shall use the following weak sense of stratification. A stratification of a topological space T , indexed by a finite ordered set I , is a decomposition indexed by I of T into non-empty disjoint subsets S_i , the strata, such that for every $i \in I$ the union

$$T_i := \bigcup_{j \leq i} S_j$$

is closed. The T_i are the closed strata. Note that $i \leq j$ if and only if $T_i \subset T_j$, and that the T_i determine the S_i by

$$S_i = T_i - \bigcup_{j < i} T_j. \quad (1.1.1)$$

Let $\varphi : T \rightarrow I$ be the map equal to i on S_i . If I is given the topology for which the closure of a point i is the interval

$$[\cdot, i] := \{j \mid j \leq i\},$$

then saying that the T_i are closed is equivalent to saying that φ is continuous.

1.2

We shall also write I for the category whose objects are the elements of I , with a unique arrow from i to j for $i \leq j$.

Each point i of the topological space I has a smallest neighborhood, namely the interval

$$[i, \cdot] := \{j \mid i \leq j\}.$$

For a sheaf F on I , the fiber F_i of F at i coincides with $F([i, \cdot])$. For $i \leq j$, restriction from $[i, \cdot]$ to $[j, \cdot]$ is a morphism from F_i to F_j , and this construction identifies sheaves of abelian groups on I with functors from I to (Ab) .

If I has a least element 0, the functor $\Gamma(I, F) = F_0$ is exact and the higher H^i vanish. If I has a greatest element 1, every constant sheaf is flasque and its higher H^i therefore vanish: the space I is acyclic, i.e. its cohomology with constant coefficients is that of a point.

Another proof is the following. Computing cohomology by the canonical flasque resolution, one verifies that the cohomology of I with constant coefficients coincides with that of the classifying space $|I|$ of the category I . In particular, it is the same for I and for I with the opposite order, and we have already seen that every I having a least element is acyclic.

Lemma 1.1 (Devissage, 1.3). *Let $\mathcal{C}$ be a class of sheaves on I . Suppose that $\mathcal{C}$ contains the zero sheaf, contains every constant sheaf on a closed subset $[\cdot, i]$ extended by zero to I , and has the following two-out-of-three property: for every short exact sequence*

$$0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$$

of sheaves on I , if two of the F_i belong to $\mathcal{C}$, then the third does as well. Then $\mathcal{C}$ contains all sheaves on I .

Proof. Proceed by induction on the number of elements of I . For a closed subset $i : J \hookrightarrow I$ different from I , applying the induction hypothesis to the class of sheaves on J such that $i_!F$ is in $\mathcal{C}$, one sees that every sheaf with support not equal to I is in $\mathcal{C}$. If I is empty, the lemma is clear. Otherwise, let x be a maximal element of I and let j be the inclusion of $\{x\}$ in I . Taking F_1, F_2, F_3 to be respectively a constant sheaf on $\{x\}$, on $[\cdot, x]$, and on $[\cdot, x[$, each extended by zero, one sees that for every abelian group Λ , $j_!\Lambda$ belongs to $\mathcal{C}$. For a sheaf F on I , conclude by observing that the support of $F/j_!j^*F$ is contained in the closed set $I - \{x\}$. $\square$

Proposition 1.2 (1.4). *If the closed strata are acyclic, then for every sheaf F on I one has*

$$H^*(I, F) \xrightarrow{\sim} H^*(T, \varphi^*F). \quad (1.4.1)$$

Proof. If F is a constant sheaf on a closed set $[\cdot, i]$, extended by zero to I , then φ^*F is constant on T_i , extended by zero to T , and (1.4.1) follows from the acyclicity of $[\cdot, i]$ and of T_i .

The general case follows by devissage: for a short exact sequence $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$ of sheaves on I , the two sides of (1.4.1) give long exact cohomology sequences, and if (1.4.1) is an isomorphism for two of the F_i , it is so for the third. Conclude by 1.3. $\square$

Remark 1.3 (1.5). If $|I|$ is the classifying space of the category I , define a continuous map $|I| \rightarrow I$ by sending the interior of the simplex $[i_0, \dots, i_n]$ with $i_0 < i_1 < \dots < i_n$ to i_n . Under this map the inverse image of $[\cdot, i]$ is $||[\cdot, i]||$, a cone with vertex i . The hypothesis of 1.4 is therefore satisfied, and taking F to be a constant sheaf one recovers the fact that I and $|I|$ have the same cohomology.

1.6

Assume that T is locally compact, separated, and countable at infinity, and that the closed strata T_i are acyclic manifolds without boundary. Being acyclic, T_i is orientable and connected.

For every connected orientable manifold V , write $\text{or}(V)$ for the free rank-one graded $\mathbb{Z}$ -module, pure of degree $\dim(V)$, whose generators $\pm e$ correspond to the two orientations of V . For every abelian group Λ and every point $x \in V$,

$$\Lambda \otimes \text{or}(V) = H_{\{x\}}^*(V, \Lambda) \xrightarrow{\sim} H_c^{\dim(V)}(V, \Lambda), \quad (1.6.1)$$

and, if V is acyclic, Poincaré duality gives

$$\Lambda \otimes \text{or}(V) \xrightarrow{\sim} H_c^*(V, \Lambda). \quad (1.6.2)$$

Write simply or_i for $\text{or}(T_i)$.

Choose, in each stratum S_i , a point x_i . For every sheaf F on I there is an inverse-image map in cohomology with support

$$\varphi^* : H_{\{i\}}^*([i, \cdot], F) \longrightarrow H_{S_i}^* \left(\bigcup_{i \leq j} S_j, \varphi^* F \right). \quad (1.6.3)$$

Note that $U_i := \bigcup_{i \leq j} S_j$ is an open subset of T , and that S_i is closed in U_i . There is a cup product

$$H_{S_i}^*(U_i, \varphi^* F) \otimes H_{\{x_i\}}^*(S_i, \mathbb{Z}) \longrightarrow H_{\{x_i\}}^*(U_i, \varphi^* F). \quad (1.6.4)$$

In the language of derived categories, $H_{S_i}^*(U_i, \varphi^* F)$ is a Hom^* in $D(U_i)$ from $\mathbb{Z}_{S_i}$ to $\varphi^* F$, $H_{\{x_i\}}^*(S_i, \mathbb{Z})$ is a Hom^* from $\mathbb{Z}_{\{x_i\}}$ to $\mathbb{Z}_{S_i}$, and the cup product is simply composition. The composite

$$H_{S_i}^*(U_i, \varphi^* F) \otimes_{\text{or}_i} \xrightarrow{1.6.4} H_{\{x_i\}}^*(U_i, \varphi^* F) \longrightarrow H_c^*(U_i, \varphi^* F) \quad (1.6.5)$$

depends only on the connected component of S_i in which x_i lies. Indeed, it factors through the tensor product by $H_{S_i}^*(U_i, \varphi^* F)$ of the morphism

$$\text{or}_i = H_{\{x_i\}}^*(S_i, \mathbb{Z}) \longrightarrow H_c^*(S_i, \mathbb{Z}).$$

Composing (1.6.3) and (1.6.5), one obtains morphisms

$$H_{\{i\}}^*([i, \cdot], F) \otimes \text{or}_i \longrightarrow H_c^*(U_i, \varphi^* F), \quad (1.6.6)$$

which, composed with $H_c^*(U_i, \varphi^* F) \rightarrow H_c^*(T, \varphi^* F)$, give

$$\bigoplus_i H_{\{i\}}^*([i, \cdot], F) \otimes \text{or}_i \longrightarrow H_c^*(T, \varphi^* F). \quad (1.6.7)$$

Theorem 1.4 (1.7). *Under the hypotheses and with the notation of 1.6, the morphism (1.6.7) is an isomorphism.*

Proof. For a short exact sequence $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$, the two sides of (1.6.7) give long exact cohomology sequences. Applying 1.3, we reduce to verifying the assertion for $F = \Lambda_{[\cdot, i]}$, the constant sheaf Λ on a closed set $[\cdot, i]$, extended by zero to I .

Compute $H_{\{j\}}^*([j, \cdot], \Lambda_{[\cdot, i]})$. If $j \not\leq i$, then $\Lambda_{[\cdot, i]}$ is zero on $[j, \cdot]$, and one gets 0. Suppose that $j \leq i$. There is a long exact sequence

$$\cdots \rightarrow H_{\{j\}}^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow H^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow H^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow \cdots$$

which identifies with

$$\cdots \rightarrow H_{\{j\}}^*([j, i], \Lambda) \rightarrow H^*([j, i], \Lambda) \rightarrow H^*([j, i], \Lambda) \rightarrow \cdots.$$

For $j < i$, both $[j, i]$ and $]j, i]$ are acyclic, and $H_{\{j\}}^*([j, i], \Lambda) = 0$. For $i = j$, one has $]j, i] = \emptyset$, $[j, i] = \{j\}$, and $H_{\{j\}}^*([j, i], \Lambda) = \Lambda$ in degree 0. The map (1.6.7) is therefore reduced to the isomorphism (1.6.2) for T_i . $\square$

Corollary 1.5 (1.8). *With the hypotheses and notation of 1.6, suppose that I has a greatest element 1, and let S_1 be the corresponding stratum. Let j_1 be the inclusion of $\{1\}$ in I , and let $j_{1!}\Lambda$ be the constant sheaf Λ on the open set $\{1\}$, extended by zero to I . Then the choice of a connected component of each S_i defines an isomorphism*

$$\bigoplus_i H_{\{i\}}^*([i, 1], j_{1!}\Lambda) \otimes_{\text{or}_i} \xrightarrow{\sim} H_c^*(S_1, \Lambda). \quad (1.8.1)$$

Proof. Let j be the inclusion of S_1 in T . The inverse image of $j_{1!}\Lambda$ on T is $j_!\Lambda$, the constant sheaf Λ on S_1 extended by zero to T . We have

$$H_c^*(S_1, \Lambda) \xrightarrow{\sim} H_c^*(T, j_!\Lambda),$$

and 1.8 follows from 1.7 applied to $j_{1!}\Lambda$. $\square$

Construction 1.6 (1.9). Let I be a finite ordered set with greatest element 1, and let j_1 be the inclusion of the open set $\{1\}$ in I . Then

$$H_{\{i\}}^*([i, 1], j_{1!}\Lambda) = \begin{cases} \Lambda & \text{in degree 0, if } i = 1, \\ \tilde{H}^{*-2}([i, 1[, \Lambda) & \text{otherwise.} \end{cases}$$

If $]i, 1[$ is empty, the reduced cohomology $\tilde{H}^*$ is Λ in degree -1 .

Construction. The case $i = 1$ is clear. Suppose $i < 1$ and put $F := j_{1!}\Lambda$. The groups $H^p([i, 1], F)$ are then all zero: for $p > 0$ because $[i, 1]$ has a least element, and for $p = 0$ because $F([i, 1]) = F_i = 0$. The long exact sequence

$$\cdots \rightarrow H_{\{i\}}^*([i, 1], \cdot) \rightarrow H^*([i, 1], \cdot) \rightarrow H^*([i, 1], \cdot) \rightarrow \cdots$$

therefore reduces, for F , to an isomorphism

$$\partial : H^p([i, 1], F) \xrightarrow{\sim} H_{\{i\}}^{p+1}([i, 1], F). \quad (1.9.1)$$

Let $\Lambda_{]i, 1[}$ be the constant sheaf Λ on the closed subset $]i, 1[$ of $[i, 1]$, extended by zero to $]i, 1]$. Since $]i, 1]$ is acyclic, because it has a greatest element, the long exact cohomology sequence defined by

$$0 \rightarrow F \rightarrow \Lambda \rightarrow \Lambda_{]i, 1[} \rightarrow 0$$

reduces to an isomorphism

$$\partial : \tilde{H}^p([i, 1[, \Lambda) \xrightarrow{\sim} H^{p+1}([i, 1], F). \quad (1.9.2)$$

The isomorphism of 1.9 is the composite of (1.9.1) and (1.9.2). $\square$

1.10. Variants: homology and cohomology of S_1

If V is an oriented pure n -dimensional manifold, cap product with the fundamental class is an isomorphism from compactly supported cohomology of V to homology of V :

$$H_c^i(V, \Lambda) \xrightarrow{\sim} H_{n-i}(V, \Lambda).$$

Under the hypotheses of 1.8, and up to a renumbering and a twist by $\text{or}(T)$, (1.8.1) therefore also computes the homology of S_1 .

The results obtained in compactly supported cohomology, or equivalently here in homology, give by duality—between homology and cohomology, or by Poincaré duality between cohomology and compactly supported cohomology—analogous results in cohomology.

For cohomology with coefficients in a field k , one simply has

$$H^i = \text{Hom}_k(H_i, k)$$

respectively $H^i = \text{Hom}_k(H_c^{n-i}, k)$, and in the present case these vector spaces are finite-dimensional. The same statement holds for $k = \mathbb{Z}/N\mathbb{Z}$. For Λ finite and Λ' its $\mathbb{Q}/\mathbb{Z}$ -dual, one similarly has

$$H^i(\cdot, \Lambda') = \text{Hom}(H_i(\cdot, \Lambda), \mathbb{Q}/\mathbb{Z}),$$

respectively

$$H^i(\cdot, \Lambda') = \text{Hom}(H_c^{n-i}(\cdot, \Lambda), \mathbb{Q}/\mathbb{Z}).$$

More generally, one may use Pontryagin duality, with values in $\mathbb{R}/\mathbb{Z}$.

Another method, for Λ finitely generated over $\mathbb{Z}$, is to work systematically in derived categories, taking as coefficient Λ an object of $D^b(\mathbb{Z})$ with finitely generated cohomology over $\mathbb{Z}$. If Λ' is the dual $R\text{Hom}(\Lambda, \mathbb{Z})$ of Λ , then one still has

$$R\Gamma(S_1, \Lambda') = R\text{Hom}(R\Gamma_c(S_1, \Lambda), \mathbb{Z}) \otimes \text{or}(T).$$

Notation. We shall write $h_c^*(i)$ for the groups

$$H_{\{i\}}^*([i, 1], F) \otimes \text{or}_i$$

which occur in the decomposition (1.8.1) of $H_c^*(S_1, \Lambda)$. The factors in the corresponding decompositions of $H_*(S_1, \Lambda)$ and $H^*(S_1, \Lambda)$ will be denoted by $h_*(i)$ and $h^*(i)$.

The free rank-one graded $\mathbb{Z}$ -module of coorientations of T_i in T , denoted $\text{coor}(T_i)$, or simply coor_i , is defined by

$$\text{or}(T_i) \otimes \text{coor}(T_i) \xrightarrow{\sim} \text{or}(T).$$

By duality, one deduces from 1.8 and 1.9 that

$$h^*(i) = \begin{cases} \Lambda & \text{in degree zero for } i = 1, \\ \tilde{H}_*([i, 1], \Lambda)[2] \otimes \text{coor}_i & \text{otherwise.} \end{cases} \quad (1.10.1)$$

In the second line, homological degree p is identified with cohomological degree $-p$: for $i \neq 1$,

$$h^p(i) = \tilde{H}_{\text{codim}(T_i)-p-2}([i, 1], \Lambda) \otimes \text{coor}_i.$$

Example 1.7 (1.11). Let V be a real affine space and let $\mathcal{A}$ be a finite set of affine subspaces $A \neq \emptyset, V$ of V , such that if the intersection of two elements of $\mathcal{A}$ is non-empty, then that intersection belongs to $\mathcal{A}$. Put

$$\mathcal{A}^+ := \mathcal{A} \cup \{V\}, \quad X := V - \bigcup \mathcal{A}.$$

We call $\mathcal{A}^+$, respectively $\mathcal{A}$, the set of flats, respectively of proper flats, of the arrangement $\mathcal{A}$. The flats are the closed strata of a stratification of V , indexed by the set $\mathcal{A}^+$ ordered by inclusion. Let $\varphi : V \rightarrow \mathcal{A}^+$ be the corresponding map. We shall sometimes write A^0 for the stratum corresponding to the flat A :

$$A^0 := \varphi^{-1}(A) = A - \bigcup \{\text{smaller flats}\}. \quad (1.11.1)$$

The flats are acyclic varieties. Applying 1.8, the choice in each stratum A^0 of a point x_A defines an isomorphism

$$H_c^*(X, \Lambda) = \bigoplus_A h_c^*(A) \quad (\text{sum over the flats}), \quad (1.11.2)$$

where

$$h_c^*(A) = H_{\{A\}}^*([A, V], j_{V!}\Lambda) \otimes \text{or}(A),$$

and j_V is the inclusion of $\{V\}$ in $\mathcal{A}^+$.

In [GM], Goresky and MacPherson obtain such a decomposition of the homology of X by applying Morse theory to a function “distance to $v_0 \in V$ ”. The Euclidean structure on V and the point v_0 are chosen generically, in the sense that:

- (a) for each flat A , the critical point on A of the distance function $d(v_0, \cdot)$, that is, the foot x_A of the perpendicular dropped from v_0 to A , lies in A^0 ;
- (b) the critical values $d(v_0, x_A)$ are distinct.

For the flat V , one has $x_V = v_0$.

Let $B(r)$ be the open ball with center v_0 and radius r . Endow it with the stratification induced by that of V , whose closed strata are the traces on $B(r)$ of the flats which meet $B(r)$. The hypotheses of 1.8 are again satisfied. If a flat A meets $B(r)$, then every flat $B \supset A$ also meets $B(r)$, and the group $h_c^*(A)$ is the same for $B(r)$ and for V . The diagram

$$\begin{array}{ccc}
\bigoplus_{A \cap B(r) \neq \emptyset} h_c(A) & \longrightarrow & \bigoplus_A h_c(A) \\
\sim \downarrow & & \downarrow \sim \\
H_c(X \cap B(r), \Lambda) & \longrightarrow & H_c(X, \Lambda)
\end{array} \tag{1.11.3}$$

is commutative. Orient V , and identify homology and compactly supported cohomology as in 1.10. When r , increasing, passes a critical value $r_A := d(v_0, x_A)$, the Morse group, in homology, which measures the change from $H_*(B(r_A - \epsilon), \Lambda)$ to $H_*(B(r_A + \epsilon), \Lambda)$ is, by (1.11.3), the group $h(A)$. Our definition of the map

$$h(A) \longrightarrow H_c^*(X \cap B(r_A + \epsilon), \Lambda) \longrightarrow H_c^*(X, \Lambda)$$

corresponding to the choice of x_A as marked point in A^0 is inspired by the proof in [GM] that the Morse function $d(v_0, \cdot)$ is perfect. The decomposition by flats of $H_*(X, \Lambda)$ implicit in [GM] therefore coincides with the decomposition (1.11.2) of $H_c^*(X, \Lambda)$, for the choice of the x_A as marked points.

Example 1.8 (1.12). If T is a convex open subset of $\mathbb{R}^n$ and E is a finite set of traces on T of affine subspaces, then the intersections $T \cap F$ for $F \in E$ which are non-empty are the closed strata of a stratification of T satisfying the hypotheses of 1.8. More generally, one may take T to be a convex subset of a space of negative curvature and E a set of totally geodesic subspaces.

Example 1.9 (1.13). In Examples 1.11 and 1.12, the ordered set I indexing the strata has the property that if i and j in I have a lower bound, then they have a greatest lower bound $\inf(i, j)$. This is not necessarily the case in applications of 1.8. For example, one may have a disk, two curves A and B , and the two points of intersection as closed strata; the two intersection points have the same two curve strata above them without a single greatest lower bound.

Example 1.10 (1.14). Our proof of 1.8 is sheaf-theoretic. It does not use Morse theory. It therefore applies in ℓ -adic cohomology. The only difference is that for V a connected smooth algebraic variety of dimension n over an algebraically closed field, the group $\text{or}(V)$ of 1.6 must be replaced by the Tate module $\mathbb{Z}_\ell(-n)$ placed in degree $2n$.

Let V be an affine space over k , let $\mathcal{A}$ be a finite set of affine subspaces, and let $\mathcal{A}^+$ and X be as in 1.11. One has

$$\bigoplus_A H_{\{A\}}^*([A, V], j_{V!} \mathbb{Z}_\ell)(-\dim A)[-2 \dim A] \xrightarrow{\sim} H_c^*(X, \mathbb{Z}_\ell). \tag{1.14.1}$$

Since the A^0 are connected here, (1.14.1) does not depend on the choice of marked points in the strata.

If $k = \mathbb{C}$, the same decomposition holds in mixed Hodge theory: in the decomposition (1.11.2) of $H_c^*(X, \mathbb{Z})$, the factor $h_c^*(A)$ is pure of Hodge type $(\dim_{\mathbb{C}} A, \dim_{\mathbb{C}} A)$. Dually, $h^*(A)$ is pure of Hodge type $(\text{codim}(A), \text{codim}(A))$.

In the complex case, cup product is compatible with the mixed Hodge structure. In algebraic geometry the arrangement $\mathcal{A}$ may be defined over a subfield k_0 of k of finite type over its prime subfield; the Galois group $\text{Gal}(\bar{k}_0/k_0)$ acts on cohomology, and cup product is compatible with the Galois action. Under the above hypotheses and in rational cohomology, respectively $\mathbb{Q}_\ell$ -cohomology, the component

$$h^*(A) \otimes h^*(B) \longrightarrow h^*(C)$$

of the cup product can therefore be non-zero only if

$$\text{codim}(C) = \text{codim}(A) + \text{codim}(B).$$

2 Functoriality in the Arrangement

2.1

Let $V, \mathcal{A}, \mathcal{A}^+ := \mathcal{A} \cup \{V\}$, $X := V - \bigcup \mathcal{A}$, and $\varphi : V \rightarrow \mathcal{A}^+$ be as in 1.11. Let $\mathcal{A}' \subset \mathcal{A}$ satisfy the same hypotheses as $\mathcal{A}$, and denote the corresponding objects by primes. The stratification of V defined by $\mathcal{A}$ refines that defined by $\mathcal{A}'$. If

$$\alpha : \mathcal{A}^+ \longrightarrow \mathcal{A}'^+$$

assigns to a flat of $\mathcal{A}$ the smallest flat of $\mathcal{A}'$ containing it, then the diagram

$$\begin{array}{ccc} & V & \\ \varphi \swarrow & & \searrow \varphi' \\ \mathcal{A}^+ & \xrightarrow{\alpha} & \mathcal{A}'^+ \end{array}$$

is commutative. Note that if the ordered sets $\mathcal{A}^+$ and $\mathcal{A}'^+$ are viewed as categories, then α is left adjoint to the inclusion incl of $\mathcal{A}'^+$ in $\mathcal{A}^+$. Indeed,

$$\alpha(A) \subset A' \iff A \subset \text{incl}(A').$$

Mark each flat A of $\mathcal{A}^+$ by a point $x_A \in A^0$, and mark A' in $\mathcal{A}'^+$ by $x_{A'} := x_{\text{incl}(A')}$. These points define decompositions (1.11.2) of $H_c^*(X, \Lambda)$ and $H_c^*(X', \Lambda)$.

Notation. Below, if a flat A of $\mathcal{A}^+$ lies in $\mathcal{A}'^+$, we shall write it as A' when it is to be considered as an element of $\mathcal{A}'^+$. Thus the interval $[A, V]$ in $\mathcal{A}'^+$ will be denoted $[A', V]$, and A'^0 is $A - \bigcup \{\text{smaller flats of } \mathcal{A}'\}$.

The inclusion $X \subset X'$ induces a morphism

$$H_c^*(X, \Lambda) \longrightarrow H_c^*(X', \Lambda). \quad (2.1.1)$$

Proposition 2.1 (2.2). *Let A be a flat of $\mathcal{A}^+$.*

- (i) *If A is not a flat of $\mathcal{A}'^+$, then (2.1.1) is zero on $h_c^*(A)$.*
- (ii) *If A is a flat of $\mathcal{A}'^+$, say $A = \text{incl}(A')$, then (2.1.1) sends $h_c^*(A)$ into $h_c^*(A')$. The morphism induced by (2.1.1) from $h_c^*(A)$ to $h_c^*(A')$ is the twist by $\text{or}(A)$ of*

$$\text{incl}^* : H_{\{A\}}^*([A, V], j_{V!}\Lambda) \longrightarrow H_{\{A'\}}^*([A', V], j_{V!}'\Lambda). \quad (2.2.1)$$

Note that $\text{incl}^{-1}(A)$ is reduced to $\{A'\}$, and that $j_{V!}'\Lambda$ is the inverse image on $[A', V]$ of $j_{V!}\Lambda$, so that (2.2.1) is defined. The inclusion of X , respectively X' , in V will be denoted by j_X , respectively $j_{X'}$.

Proof of (ii). In the definition of the morphism $h_c^*(A') \rightarrow H_c^*(X', \Lambda)$, we interpret $H_c^*(X', \Lambda)$ as $H_c^*(V, j_{X'}! \Lambda)$ and take the cup product of an element of

$$\text{or}_A = H_{\{x_A\}}^*(\varphi'^{-1}(A'), \mathbb{Z})$$

with a class in $H_{A'}^*(\varphi'^{-1}([A', V]), j_{X'}! \Lambda)$. Since we take the product with a class supported at x_A , it matters only through its restriction to a neighborhood of x_A . Take as neighborhood the open set $U := \varphi^{-1}([A, V])$. The morphism from $h_c^*(A')$ to $H_c^*(X', \Lambda)$ is deduced from

$$\varphi'^* : H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \rightarrow H_{A^0}(U, j_{X'}! \Lambda).$$

Again write α for the restriction $[A, V] \rightarrow [A', V]$ of α to $[A, V]$. The inverse image of A' in $[A, V]$ is reduced to A , and α induces

$$\alpha^* : H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \rightarrow H_{\{A\}}^*([A, V], \alpha^* j_{V'}! \Lambda). \quad (2.2.2)$$

The morphism α^* is an isomorphism. By the long exact sequence for cohomology with support, it is enough to verify that for every sheaf F on $[A', V]$ one has

$$H^*([A', V], F) \xrightarrow{\sim} H^*([A, V], \alpha^* F), \quad H^*(]A', V], F) \xrightarrow{\sim} H^*(]A, V], \alpha^* F).$$

This is an application of 1.4, with α viewed as a stratification of $[A, V]$ or $]A, V]$: the inverse image of $[\cdot, x]$ is $[\cdot, \text{incl}(x)]$, acyclic because it has a greatest element.

Consider the commutative diagram

$$\begin{array}{ccc} H_{\{A\}}^*([A, V], j_{V'}! \Lambda) & \rightarrow & H_{\{A\}}^*([A, V], \alpha^* j_{V'}! \Lambda) \xleftarrow{\sim} H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \\ \downarrow & & \swarrow \\ H_{\varphi^{-1}(A)}(U, j_X! \Lambda) & \longrightarrow & H_{\varphi^{-1}(A)}(U, j_{X'}! \Lambda) \end{array} \quad (2.2.3)$$

It defines a commutative diagram

$$\begin{array}{ccc} h_c^*(A) & \longrightarrow & h_c^*(A') \\ \downarrow & & \downarrow \\ H_c^*(X, \Lambda) & \longrightarrow & H_c^*(X', \Lambda), \end{array}$$

whose first line is the twist by $\text{or}(A)$ of the first line of (2.2.3). Since $\alpha \circ \text{incl}$ is the identity of $\mathcal{A}'^+$, the corresponding commutative diagram shows that this first line coincides with the inverse image by incl used in the statement of (ii). $\square$

Proof of (i) if $\bigcup \mathcal{A} = \bigcup \mathcal{A}'$. Under the hypothesis $\bigcup \mathcal{A} = \bigcup \mathcal{A}'$, one has $X = X'$. We prove more precisely that $h_c^*(A) = 0$ if the flat A of $\mathcal{A}^+$ is not in $\mathcal{A}'^+$. This is clear from the interpretation (1.10) of $h_c^*(A)$ as a Morse group: the “critical point” corresponding to A is superfluous, and its passage does not correspond to a change in the topology of $X' \cap B(r)$. Another proof is as follows. The inverse image of V by $\alpha : [A, V] \rightarrow [A', V]$ is reduced to V , and by (2.2.2), $h_c^*(A) \rightarrow h_c^*(A')$ is an isomorphism. From the commutative diagram

$$\begin{array}{ccc} \bigoplus_A h_c^*(A) & \longrightarrow & \bigoplus_{A'} h_c^*(A') \\ & \searrow \sim & \swarrow \sim \\ & H_c^*(X, \Lambda) & \end{array}$$

it follows that $h_c^*(A) = 0$ for A in $\mathcal{A}$ and not in $\mathcal{A}'$. $\square$

Proof of (i) if $A \neq V$ and $A \notin \bigcup \mathcal{A}'$. One has $\varphi^{-1}(A) \subset X'$. The diagram

$$\begin{array}{ccc} H_{\{A\}}^*([A, V], j_{V!}\Lambda) \otimes \text{or}(A) & \longrightarrow & H_{\{A\}}^*([A, V], \Lambda) \otimes \text{or}(A) \\ \downarrow & & \downarrow \\ H_c(X, \Lambda) & \longrightarrow & H_c^*(X', \Lambda) \end{array}$$

is commutative, and $H_{\{A\}}^*([A, V], \Lambda) = 0$, because $[A, V]$ and $]A, V]$ are both acyclic. $\square$

Proof of (i), general case. Let $\mathcal{A}''$ be the set of flats of $\mathcal{A}$ contained in $\bigcup \mathcal{A}'$. Then $\mathcal{A}' \subset \mathcal{A}'' \subset \mathcal{A}$, and it is enough to treat separately the inclusions of $\mathcal{A}'$ in $\mathcal{A}''$ and of $\mathcal{A}''$ in $\mathcal{A}$, both consequences of the cases already treated. $\square$

2.3

Assume that the arrangement $\mathcal{A}$ satisfies the connectedness hypothesis (CONN) of the introduction. Let A be a flat, let x_A be a point of the corresponding stratum A^0 , and let U be an open ball with center x_A , small enough to meet no flats other than those which contain A . As observed in 1.12, one may apply 1.8 to U , endowed with the stratification induced by $\mathcal{A}$, indexed by $[A, V]$. The inclusion of $U - \bigcup [A, V]$ in $V - \bigcup [A, V]$ is a homotopy equivalence, and by 2.2 one has a commutative diagram

$$\begin{array}{ccccc} \bigoplus_{B \in [A, V]} h_c(B) & \longrightarrow & \bigoplus h_c(B) & \longrightarrow & \bigoplus_{B \in [A, V]} h_c(B) \\ \downarrow & & \downarrow & & \downarrow \\ H_c^*(U - \bigcup [A, V], \Lambda) & \longrightarrow & H_c^*(X, \Lambda) & \longrightarrow & H_c^*(V - \bigcup [A, V], \Lambda). \end{array} \quad (2.3.1)$$

The composite in the second line is an isomorphism. It makes $H_c^*(U - \bigcup [A, V])$ a direct factor of $H_c^*(X, \Lambda)$, and the corresponding idempotent p_A is identified by 2.2 with the projection of $\bigoplus_{B \in \mathcal{A}^+} h(B)$ onto the partial sum $\bigoplus_{B \in [A, V]} h(B)$. The idempotents p_A for $A \in \mathcal{A}^+$ defined by the second line of (2.3.1) therefore commute with one another. They allow one to recover the decomposition of $H_c^*(X, \Lambda)$ indexed by the flats. The dual decomposition of $H^*(X, \Lambda)$ may similarly be reconstructed from the idempotents defined by

$$H^*(V - \bigcup [A, V], \Lambda) \longrightarrow H^*(X, \Lambda) \longrightarrow H^*(U - \bigcup [A, V], \Lambda),$$

whose composite is an isomorphism.

3 Restriction to a Linear Subspace

3.1

Resume the notation $V, \mathcal{A}, \mathcal{A}^+, X$, and $\varphi : V \rightarrow \mathcal{A}^+$ of 1.11. Let V' be an affine subspace of V not contained in $\bigcup \mathcal{A}$, let $\mathcal{A}'$ be the set of non-empty intersections of V' with a flat $A \in \mathcal{A}$, and let

$$X' := V' - \bigcup \mathcal{A}' = X \cap V'.$$

Choose a coorientation of V' in V . It defines a Gysin morphism

$$H_c^*(X', \Lambda) \longrightarrow H_c^*(X, \Lambda). \quad (3.1.1)$$

Theorem 3.1 (3.2). *Assume that $(V, \mathcal{A})$ and $(V', \mathcal{A}')$ satisfy the connectedness hypothesis (CONN). Then, for A in $\mathcal{A}$ and A' in $\mathcal{A}'$, the component*

$$h_c^*(A') \longrightarrow h_c^*(A) \quad (3.2.1)$$

of the morphism (3.1.1), in the decompositions (1.11.2), can be non-zero only if $A' = A \cap V'$ and the intersection of A and V' is transverse. In that case, the choice of the coorientation of V' in V defines an isomorphism $\epsilon : \text{or}(A') \xrightarrow{\sim} \text{or}(A)$; the map “trace on V' ”,

$$\alpha : [A, V] \longrightarrow [A', V'],$$

is injective, and (3.2.1) is the tensor product of ϵ with

$$\alpha^* : H_{\{A'\}}^*([A', V'], j_{V'!}\Lambda) \longrightarrow H_{\{A\}}^*([A, V], j_{V!}\Lambda). \quad (3.2.2)$$

Let $A \in \mathcal{A}$, let $\mathcal{A}_A := [A, V] \subset \mathcal{A}$, let $X_A := V - \bigcup \mathcal{A}_A$, let $\mathcal{A}'_A$ be the set of non-empty intersections of V' with a flat $B \in \mathcal{A}_A$, and let

$$X'_A = V' - \bigcup \mathcal{A}'_A = V' \cap X_A.$$

The group $h_c^*(A)$ is the same for $(V, \mathcal{A})$ and for $(V, \mathcal{A}_A)$. The diagram

$$\begin{array}{ccccc} H_c^*(X', \Lambda) & \longrightarrow & H_c^*(X, \Lambda) & \longrightarrow & h_c^*(A) \\ \downarrow & & \downarrow & & \parallel \\ H_c^*(X'_A, \Lambda) & \longrightarrow & H_c^*(X_A, \Lambda) & \longrightarrow & h_c^*(A) \end{array}$$

is commutative. For the square on the right, apply 2.2. The first vertical arrow was computed in 2.2. It remains to compute the composite in the second line. In other words, in order to compute $H_c^*(X', \Lambda) \rightarrow h_c^*(A)$, we have reduced to the case in which A is the least element of $\mathcal{A}$.

Lemma 3.2 (3.3). *If A is the least element of $\mathcal{A}$ and the intersection of A and V' is empty or is not transverse, then the composite*

$$H_c^*(X') \longrightarrow H_c^*(X) \longrightarrow h(A)$$

is zero.

Proof. Choose a Euclidean structure. If A and V' are disjoint, then for ϵ small enough the set U of points of V at distance $< \epsilon$ from V' is disjoint from A . The Gysin morphism factors through $H_c^*(U \cap X', \Lambda)$. Applying (1.8) to U , with the stratification induced by that of V , one sees that $H_c^*(U \cap X, \Lambda)$ maps into

$$\bigoplus_{B \cap U \neq \emptyset} h(B) \subset H_c^*(X, \Lambda),$$

and therefore has zero projection to $h(A)$.

If A meets V' and the intersection is non-transverse, there exists a hyperplane H containing A and V' . Let U be one of the two half-spaces defined by H . The map from $H_c^*(X', \Lambda)$ to $H_c^*(X, \Lambda)$ then factors through $H_c^*(U \cap X)$. This is clear if, applying 1.10, one replaces H_c^* by homology H_* , the Gysin morphism becoming a direct image in homology. Indeed, X' lies in the boundary of U , in a part of the boundary of U , namely $H \cap X$, where the closure $\overline{U}$ of U is a manifold with boundary. This permits any cycle traced on X' to be pushed into the interior of U .

As before, the image of $H_c^*(U \cap X)$ in $H_c^*(X)$ is contained in the sum of the $h_c^*(B)$ for $B \cap U \neq \emptyset$, and hence has zero projection to $h(A)$. $\square$

3.4

It remains to treat the case in which A is the least element of V , $A \cap V' \neq \emptyset$, and the intersection of A and V' is transverse. For affine spaces V_1 and V_2 , for a system $\mathcal{A}_1$ of flats of V_1 , and for a point 0 of V_2 , the triple $(V, V', \mathcal{A})$ is isomorphic to

$$(V_1 \times V_2, V_1 \times \{0\}, \{B \times V_2 \mid B \in \mathcal{A}_1\}).$$

If $X_1 = V_1 - \bigcup \mathcal{A}_1$, then $X = X_1 \times V_2$ and $X' = X_1 \times \{0\}$; the Gysin morphism is the Kunneth isomorphism

$$H_c^*(X_1, \Lambda) \otimes H_c^{\dim V_2}(V_2, \mathbb{Z}) \xrightarrow{\sim} H_c^*(X, \Lambda).$$

It sends each $h_c^*(B_1)$ isomorphically to $h_c^*(B_1 \times V_2)$, by the map indicated in 3.2.

3.5

Dually, consider the restriction morphism

$$H^*(X, \Lambda) \longrightarrow H^*(X', \Lambda). \quad (3.5.1)$$

In terms of the decompositions of 1.10 indexed by $\mathcal{A}$ and $\mathcal{A}'$, respectively

$$H^*(X, \Lambda) = \bigoplus h^*(A), \quad H^*(X', \Lambda) = \bigoplus h^*(A'),$$

3.2 implies by duality that the only non-zero components of the restriction morphism (3.5.1) are the

$$h_A \longrightarrow h_{A'}$$

for $A' = A \cap X'$, the intersection being transverse, and they are deduced by duality from (3.2.2).

4 Application to the Cup Product

4.1

Let $(V_1, \mathcal{A}_1)$ and $(V_2, \mathcal{A}_2)$ satisfy the hypotheses of (1.11), and let V be the affine space $V_1 \times V_2$. Use the notation of 1.11 for V_1 and V_2 , and also for V endowed with the product stratification, whose strata, respectively closed strata, are products of strata, respectively closed strata, of V_1 and V_2 . The notation for V_1 , respectively V_2 , will carry subscript 1, respectively 2. The ordered set $\mathcal{A}^+$ is naturally identified with the product of the ordered sets $\mathcal{A}_1^+$ and $\mathcal{A}_2^+$, and the diagram

$$\begin{array}{ccc} V & \xlongequal{\quad} & V_1 \times V_2 \\ \varphi \downarrow & & \downarrow \varphi_1 \times \varphi_2 \\ \mathcal{A}^+ & \xlongequal{\quad} & \mathcal{A}_1^+ \times \mathcal{A}_2^+ \end{array}$$

is commutative. Suppose that each stratum A^0 of V_i is equipped with a marked point x_A , and mark the stratum $A^0 \times B^0$ of $V_1 \times V_2$ by the point (x_A, x_B) .

Take as coefficients Λ a commutative ring. The cohomology of $X = X_1 \times X_2$ is given by the Kunneth formula, which it is convenient for us to write in the derived category. Only the case $\Lambda = \mathbb{Z}$ matters; the general case follows by extension of scalars $\otimes_{\mathbb{Z}}^L \Lambda$. With the notation and hypotheses of 1.6, the isomorphism (1.6.7) refines to an isomorphism in the derived category

$$\bigoplus_i R\Gamma_{\{i\}}([i, \cdot], F) \otimes_{\text{or}_i} \xrightarrow{\sim} R\Gamma_c(T, \varphi^* F). \quad (4.1.1)$$

The definition of the map is the same as in 1.6, and the fact that it is an isomorphism follows from 1.7.

Special case: for V_i with $i = 1, 2$,

$$\bigoplus_A R\Gamma_{\{A\}}([A, V_i], j_{V_i!}\Lambda) \otimes \text{or}(A) \xrightarrow{\sim} R\Gamma_c(X_i, \Lambda), \quad (4.1.2)$$

and similarly for V . One advantage of the derived category is that the Kunneth formulas take the simple form that the cohomology of a product is the derived tensor product of the cohomologies of the factors. The morphisms (4.1.2) are compatible with the following three Kunneth formulas: first, for the intervals $[A_i, V_i]$,

$$R\Gamma_{\{A_1\}}([A_1, V_1], j_{V_1!}\Lambda) \otimes^L R\Gamma_{\{A_2\}}([A_2, V_2], j_{V_2!}\Lambda) \xrightarrow{\sim} R\Gamma_{\{A_1 \times A_2\}}([A_1 \times A_2, V], j_{V!}\Lambda); \quad (4.1.3)$$

second, for the A_i , $\text{or}(A_1) \otimes \text{or}(A_2) \xrightarrow{\sim} \text{or}(A_1 \times A_2)$; and third, for $X_1 \times X_2$. The diagram

$$\begin{array}{ccc} (R\Gamma_{\{A_1\}}([A_1, V_1], j_{V_1!}\Lambda) \otimes \text{or}(A_1)) \otimes_{\Lambda}^L (R\Gamma_{\{A_2\}}([A_2, V_2], j_{V_2!}\Lambda) \otimes \text{or}(A_2)) & \longrightarrow & R\Gamma_c(X_1, \Lambda) \otimes_{\Lambda}^L R\Gamma_c(X_2, \Lambda) \\ \sim \downarrow & & \downarrow \sim \\ R\Gamma_{\{A_1 \times A_2\}}([A_1 \times A_2, V], j_{V!}\Lambda) \otimes \text{or}(A_1 \times A_2) & \longrightarrow & R\Gamma_c(X, \Lambda) \end{array} \quad (4.1.4)$$

is commutative. Thus the decomposition (4.1.2) of $R\Gamma_c(X, \Lambda)$ is simply the derived tensor product of the decompositions (4.1.2) of the factors $R\Gamma_c(X_i, \Lambda)$ for $i = 1, 2$.

Pass to the dual, denoted

$$D(\cdots) := R\text{Hom}_{\Lambda}(\cdots, \Lambda).$$

Under the finiteness hypotheses verified here, this is an involutive operation in the derived category and it commutes with tensor product. Poincare duality says that $R\Gamma(X_i, \Lambda)$ is the dual of $R\Gamma_c(X_i, \Lambda)$, twisted by $\text{or}(V_i)$:

$$R\Gamma(X_i, \Lambda) = D(R\Gamma_c(X_i, \Lambda)) \otimes \text{or}(V_i).$$

The same holds for X .

The isomorphism (4.1.2) therefore dualizes to an isomorphism

$$R\Gamma(X_i, \Lambda) \xrightarrow{\sim} \bigoplus_A D(R\Gamma_{\{A\}}([A_i, V_i], j_{V_i!}\Lambda)) \otimes \text{coor}(A),$$

and, by compatibility between the Kunneth formula and Poincare duality, the corresponding diagram for $R\Gamma(X_1, \Lambda) \otimes_{\Lambda}^L R\Gamma(X_2, \Lambda)$ and $R\Gamma(X, \Lambda)$ is commutative. The isomorphism in the second column is the tensor product of an isomorphism

$$\text{coor}(A_1 \times A_2) = \text{coor}(A_1) \otimes \text{coor}(A_2)$$

with the isomorphism deduced by duality from (4.1.3).

4.2

Now let $(V, \mathcal{A})$ be as in 1.11, and suppose that the connectedness hypothesis (CONN) of the introduction is satisfied. We apply 4.1 to $V \times V$. The cup product in cohomology is the composite of the exterior product deduced from the isomorphism studied in 4.1,

$$R\Gamma(X, \Lambda) \otimes^L R\Gamma(X, \Lambda) \xrightarrow{\sim} R\Gamma(X \times X, \Lambda),$$

and the restriction to the diagonal

$$\Delta^* : H^*(X \times X, \Lambda) \longrightarrow H^*(X, \Lambda).$$

This is the restriction to the diagonal subspace V of $V \times V$; it is a special case of (3.5.1). By (4.1.4), the exterior product sends $h_c^*(A_1) \otimes h_c^*(A_2)$ into $h_c^*(A_1 \times A_2)$. The condition that $A_1 \times A_2$ meet the diagonal transversely is equivalent to the condition that A_1 and A_2 meet transversely in V . By 3.5, the only non-zero components of the cup product are therefore the components

$$h^*(A) \otimes h^*(B) \longrightarrow h^*(C) \quad (4.2.1)$$

for $C = A \cap B$, this intersection being transverse. In that case, (4.2.1) is deduced from the dual of the isomorphism (4.1.3) and from the dual of

$$\alpha^* : R\Gamma_{\{C\}}([C, V], j_{V!}\Lambda) \rightarrow R\Gamma_{\{(A, B)\}}([A, V] \times [B, V], j_{(V, V)!}\Lambda),$$

where $\alpha : [A, V] \times [B, V] \rightarrow [C, V]$ is intersection.

Remark 4.1 (4.3). Let V be an affine space over an algebraically closed field k , and let $\mathcal{A}$, $\mathcal{A}^+$, and X be as in 1.14. The arguments used in 2.2 carry over unchanged to the case of ℓ -adic cohomology. By contrast, 3.3 creates a difficulty. Consequently, our arguments do not determine the cup product in ℓ -adic cohomology. We conjecture that it remains given by the same formulas as in 4.2.

References

- [dS] M. de Longueville and C. Schultz, *The cohomology rings of complements of subspace arrangements*, preprint, Technische Universität Berlin, 2000.
- [FZ] E. Feichtner and G. Ziegler, *On cohomology algebras of complex subspace arrangements*, Trans. Amer. Math. Soc. 352 (2000), 3523–3555.
- [GM] M. Goresky and R. MacPherson, *Stratified Morse theory*, Ergeb. Math. Grenzgeb. (3), 14, Springer-Verlag, Berlin, 1988.
- [M] J. W. Morgan, *The algebraic topology of smooth algebraic varieties*, Inst. Hautes Études Sci. Publ. Math. 48 (1978), 137–204.
- [Y] S. Yuzvinsky, *Small rational model of subspace complement*, preprint.

The Nagata Embedding Theorem

Pierre Deligne

Abstract

Except for a few corrections added in 2010, this article reproduces notes sent to Nagata in the 1970s, translating into the language of schemes his papers [2] and [3]. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to avoid the use of valuations. A more detailed exposition and additional results are given in Conrad [1]. Unless the contrary is expressly stated, all schemes considered below are assumed noetherian.

1 Recalls on blow-ups

0.1

Let $\mathfrak{a}$ be an ideal of the ring A . The scheme obtained from $X = \text{Spec}(A)$ by blowing up $\mathfrak{a}$ is the union of the open subschemes $D(x)$, where x runs through a system of generators of $\mathfrak{a}$, and $D(x)$ is the spectrum of the subring $A[\mathfrak{a}x^{-1}]$ of $A[x^{-1}]$. Moreover,

$$D(x) \cap (X - V(\mathfrak{a})) = \text{Spec}(A[x^{-1}]).$$

If $\mathfrak{c}$ is a second ideal of A , defining a closed subscheme $V(\mathfrak{c})$ of X , the trace on $D(x)$ of the strict transform of $V(\mathfrak{c})$ is defined by the ideal of $A[\mathfrak{a}x^{-1}]$ which is the trace of the ideal of $A[x^{-1}]$ generated by $\mathfrak{c}$.

Lemma (0.2). *Let X be a scheme and let $\mathfrak{a}$ and $\mathfrak{b}$ be two sheaves of ideals on X . Let $\widehat{X}$ be the blow-up of X along $\mathfrak{a} + \mathfrak{b}$. Then the open subset of $\widehat{X}$ which is the union of the $D(x)$ for $x \in \mathfrak{b}$ contains the strict transform of $V(\mathfrak{a})$.*

Proof. The scheme $\widehat{X}$ is the union of the $D(x)$ for $x \in \mathfrak{a}$ or $x \in \mathfrak{b}$, and, for $x \in \mathfrak{a}$, the open subset $D(x)$ is disjoint from the strict transform of $V(\mathfrak{a})$. $\square$

The lemma 0.3 of the original text was false. The corrected version of the lemma which follows dates from March 1996. It forced a change of notation. The original lemma used the following notation, to which the rest of the text refers: instead of F and G , one considered their complements U and V , which, with $Y' := U \cap p^{-1}(Y)$, formed a commutative diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \longrightarrow & X, \end{array}$$

where Y' is closed in $p^{-1}(V)$.

Lemma (0.3). *Let $p : Z \rightarrow X$ be a morphism, let $F \supset G$ be closed subschemes of X , and let Y be a closed subscheme of Z . Suppose that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G)$$

scheme-theoretically. Blow up X along G , obtaining $\widehat{X}$, and let $\tilde{p} : \widehat{Z} \rightarrow \widehat{X}$ be obtained from $p : Z \rightarrow X$ by base change. Then the closure of $Y - p^{-1}(G)$ in $\widehat{Z}$ and $\tilde{p}^{-1}(F - G)$, taken in $\widehat{X}$, are disjoint closed subschemes of $\widehat{Z}$.

Proof. We may suppose that Z and X are affine. Let $\mathfrak{G}$ be the ideal of G . Cover $\widehat{X}$ by the open subsets $D(g)$, for $g \in \mathfrak{G}$ (a system of generators of $\mathfrak{G}$ suffices), and cover $X - G$ by

$$U(g) = D(g) - (\text{inverse image of } G).$$

Thus

$$\begin{array}{ccc} X - G & \hookrightarrow & \widehat{X} \\ \uparrow & & \uparrow \\ U(g) & \hookrightarrow & D(g). \end{array}$$

It is enough to see that, for each g , the traces on $\tilde{p}^{-1}D(g)$ of $\tilde{p}^{-1}(F - G)$ and of the closure of $Y - p^{-1}G$ are disjoint. They are respectively the inverse image of the closure of $F \cap U(g)$ in $D(g)$, and the closure of $Y \cap p^{-1}(U(g))$ in $\tilde{p}^{-1}D(g)$.

Let a tilde denote inverse image, either from X to Z , or from $D(g)$ to $\tilde{p}^{-1}D(g)$. Since $p^{-1}G \supset Y \cap p^{-1}(F)$, the inverse image $\tilde{g}$ of g on Z can be written

$$\tilde{g} = y + f,$$

where y vanishes on Y and f vanishes on $p^{-1}(F)$. There are functions a_i on Z and elements f_i in the ideal of F such that

$$f = \sum_i a_i \tilde{f}_i.$$

Since $F \supset G$, the f_i are in the ideal of G , and the functions f_i/g are regular on $D(g)$. They vanish on $F \cap U(g)$, hence on the closure of $F \cap U(g)$ in $D(g)$, and therefore

$$\sum_i a_i (f_i/g)^\sim$$

vanishes on $\tilde{p}^{-1}\overline{F \cap U(g)}$. On $p^{-1}U(g)$ one has

$$1 = y/g + f/g = y/g + \sum_i a_i (f_i/g)^\sim,$$

and the same sum is therefore equal to 1 on $Y \cap p^{-1}U(g)$, hence on its closure in $\tilde{p}^{-1}D(g)$. This proves the required disjointness. $\square$

Remark. Suppose instead that one is given closed subschemes $F \supset G_0$ of X , with $Y \cap p^{-1}(F)$ contained set-theoretically in $p^{-1}(G_0)$. Then there exists an r such that the r -th infinitesimal neighbourhood G of G_0 in F satisfies the hypotheses of the lemma. If J is the ideal defining $(Y \cap p^{-1}(F))_{\text{red}}$ in $Y \cap p^{-1}(F)$, it is enough to take r such that $J^{r+1} = 0$. Thus the lemma permits, by a blow-up which does not touch the complement V of G_0 , the separation of the strict transform of F from the closure of $Y - p^{-1}(G_0)$ in $\widehat{Z}$.

Lemma (0.4). Let

$$\begin{array}{ccc} X' & \xrightarrow{p} & X' \\ q \downarrow & & \downarrow \\ U & \hookrightarrow & X \end{array}$$

be a commutative diagram of schemes, with U a dense open subset of X and of X' , with p and q separated of finite type, and with $q^{-1}(X) = X'$. Let F be a closed subscheme of X , and let G be a closed subscheme of X' containing $p^{-1}(F)$. Then there exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset F - \overline{F}$, such that, after base change by the blow-up morphism defined by $\mathfrak{a}$, $\widehat{X} \rightarrow X$, and after replacing the new X' by the closure of U , one has

$$q^{-1}(F) \subset G.$$

Proof. In Nagata's language, to take the closure of F in the Zariski–Riemann space of X , it is enough to take the “intersection” of the closures of F in the blow-ups of X whose centres are contained in $F - \overline{F}$.

Apply Lemma 0.3 to

$$\begin{array}{ccc} X' - q^*F & \longrightarrow & X' - G \\ \uparrow & & \uparrow \\ X - F & \longrightarrow & X - (F - \overline{F}). \end{array}$$

One obtains $\mathfrak{a}$, and, after the base change $\widehat{X} \rightarrow X$, the new closure of F is disjoint from the image of the closure of the old $X' - q^*F$ in $X' - G$. In particular, after replacing $\widehat{X}$ by U , this new closure is disjoint from the image of $X' - G$, which is what was required. $\square$

Lemma (0.5). *Let*

$$\begin{array}{ccc} X_i & \longrightarrow & X_i \quad (1 \leq i \leq n) \\ \uparrow & & \uparrow \\ U & \longrightarrow & X \end{array}$$

be a diagram of separated S -schemes of finite type, with X_i proper over S , and with U a dense open subscheme of X and of the X_i . Let X^ be the closure of U in the product of the X_i . Finally, let F_i be a closed subscheme of X_i ; suppose that the intersection of the inverse images of the F_i in X^* is empty. Then, after replacing the X_i by blow-ups whose centres lie in $F_i - \overline{F_i}$, one obtains an analogous situation in which the intersection of the inverse images of the $\overline{F_i}$ is empty.*

Proof. Let X^{**} be a blow-up of X^* , with projections $p_i : X^{**} \rightarrow X_i$, such that in X^{**} one has

$$\bigcap_i p_i^{-1}(F_i) = \emptyset;$$

for example, blow up $\bigcap_i p_i^{-1}(F_i)$. Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccc} p_i^{-1}(X_i) & \longrightarrow & X^{**} \\ \uparrow & & \uparrow \\ U & \longrightarrow & X_i, \end{array}$$

and to $F_i \subset X_i$, $G_i = p_i^{-1}(F_i)$. This provides the desired blow-ups of the X_i . Indeed, in the new situation, in the closure of U in X^{**} , one has

$$\bigcap_i p_i^{-1}(\overline{F_i}) \subset \bigcap_i G_i = \emptyset.$$

Since $X^{**} \rightarrow X^*$ is surjective, this remains true in X^* . $\square$

2 The theorems

Definition (1.1). *Let X and Y be two S -schemes, with Y separated over S . A quasi-domination $f : X \dashrightarrow Y$ is a pair consisting of a dense open subscheme U of X and a morphism of S -schemes $f : U \rightarrow Y$ whose graph is closed in $X \times_S Y$.*

Let X and Y be two S -schemes, with Y separated over S , let U be a quasi-compact open subscheme of X , and let $f : U \rightarrow Y$. We say that X is quasi-dominant over Y , relative to U and f , if there exists a quasi-domination of the schematic closure $\overline{U}$ of U , with values in Y , which extends f . There is at most one such quasi-domination; its graph is the closure of the graph of f .

A quasi-domination f is called proper or complete if the morphism $f : U \rightarrow Y$ is proper.

Theorem (1.2). *Let X and Y be two S -schemes, and let $U \subset V \subset X$ be two dense open subschemes of X . Suppose that Y is separated of finite type over S . Let $f : U \rightarrow Y$ be a quasi-dominion of V in Y . Then there exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset X - V$, such that the blow-up $\widehat{X}$ of X with centre $V(\mathfrak{a})$ quasi-dominates Y .*

Proof. By the base change $X \rightarrow S$ one is reduced to the case $X = S$. Then Y is a scheme over X , f is a section of Y over U , and by hypothesis $f(U)$ is closed over V .

Case 1: $U = V$, $p : Y \rightarrow X$ quasi-affine, and X affine. By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & E_X \\ & \searrow & \downarrow \\ & & X, \end{array}$$

where E_X is an affine space over X . If $\widehat{X}$ quasi-dominates E_X relative to f , then a fortiori $\widehat{X}$ quasi-dominates Y . We may therefore suppose $Y = E_X$. A preliminary blow-up allows us to suppose that $X - U$ is a divisor D . Locally on X , the section f has coordinates $(f_i u^{-n})$, where u is a local equation of D , where the f_i are regular, and where n is chosen once and for all sufficiently large. Let $\mathfrak{a}$ be the ideal generated by u^n and the f_i . After blowing up $V(\mathfrak{a})$, the section f extends to a section of the projective completion of E_X , and we are done.

Case 2: $U = V$ and p quasi-affine. By Case 1, if $(U_i)_{0 \leq i \leq n}$ is an affine open covering of X , there exists on U_i an ideal $\mathfrak{a}_i$ which works. This ideal admits an extension $\tilde{\mathfrak{a}}_i$ to X , with $V(\tilde{\mathfrak{a}}_i) \subset X - V$, and it is enough to blow up the product of the $\tilde{\mathfrak{a}}_i$.

Case 3: p quasi-affine. By Lemma 0.3, applied to

$$\begin{array}{ccc} f(U) & \longrightarrow & Y \\ & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X, \end{array}$$

a preliminary blow-up allows us to suppose that

$$p(f(U)) \cap (V - U) = \emptyset.$$

There is therefore a quasi-compact open subscheme X' of X with $X' \cap V = U$ and $p(f(U)) \subset X'$. By Case 2, there exists an ideal $\mathfrak{a}$ on X' , with $V(\mathfrak{a}) \subset X' - U$, such that the blow-up $\widehat{X'}$ quasi-dominates Y , that is, $f(U)$. It remains only to extend $\mathfrak{a}$ to an ideal on X , with $V(\mathfrak{a}) \subset X - V$.

General case. Let (Y_i) be a finite covering of Y by affine open subschemes, and put

$$U_i = f^{-1}(Y_i), \quad X_i = \overline{U_i}, \quad V_i = X_i \cap V.$$

Case 3 applies to

$$Y_i \cap p^{-1}(X_i), \quad U_i \subset V_i \subset X_i,$$

and gives an ideal $\mathfrak{a}_i$ on X_i . This ideal defines an ideal on X , still denoted $\mathfrak{a}_i$, with $V(\mathfrak{a}_i) \subset X - V$. Blow up the product of the $\mathfrak{a}_i$. The blow-up $q : \widehat{X} \rightarrow X$ thus obtained is such that $q^{-1}(X_i)$ dominates the blow-up $\widehat{X}_i$. If $X'_i = q^{-1}(X_i)$, there exists an open U'_i of X'_i , with $U'_i \supset q^{-1}(U_i)$, such that, above X_i , $f(U_i)$ is a section of $U'_i \rightarrow Y$. One concludes that, after this blow-up,

$$f(U) = \bigcup_i f_i(U'_i).$$

For the proof of Theorem 1.2, we may replace Y by $f(U)$, and conclude by Case 3 and the following lemma. $\square$

Lemma (1.3). *The scheme $f(U)$ is quasi-affine over $\widehat{X}$.*

Proof. Indeed, $f(U)$ is quasi-finite over $\widehat{X}$. □

Corollary (1.4, a variant of Chow's lemma). *Let $f : X \rightarrow S$ be a separated morphism of finite type. Then for every open subscheme U which is quasi-projective over S and dense in X , there exists a diagram of S -schemes*

$$\begin{array}{ccc} & X' & \\ q \swarrow & & \searrow j \\ X & & \overline{X}, \end{array}$$

where q is the blow-up morphism defined by an ideal $\mathfrak{a}$ satisfying $V(\mathfrak{a}) \subset X - U$, and where $\overline{X}$ is projective over S . In particular, q is projective and surjective and $q^{-1}(U) \simeq U$.

Proof. Let U^* be a projective S -scheme containing U as a schematically dense open subscheme. By Theorem 1.2, applied to the injection of U into X , one may choose U^* quasi-dominant over X . Thus one has a diagram

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ U^* & \longrightarrow & X, \end{array} \quad (1.4.1)$$

and here, since U^* is proper over S , φ is proper. □

Lemma (1.5). *Let*

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ Y & \longrightarrow & X \end{array}$$

be a diagram of separated S -schemes of finite type, with U dense in X , dense in Y , and schematically dense in V , and with φ a quasi-domination of Y in X . If $\mathfrak{a}$ is an ideal of X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\widehat{X}$ of X along $\mathfrak{a}$ quasi-dominates V , relative to U , and such that U is schematically dense in $\widehat{X}$, giving a diagram

$$\begin{array}{ccc} U & \hookrightarrow & W \\ \downarrow & & \downarrow \psi \\ \widehat{X} & \longrightarrow & V, \end{array}$$

then:

(a) ψ is the blow-up morphism of V along $\varphi^*\mathfrak{a}$;

(b) the graph of ψ is closed in $\widehat{X} \times_S Y$.

Proof. Let $\widehat{V}$ be the blow-up of V along $\varphi^*\mathfrak{a}$. By the universal property of $\widehat{X}$, there is a unique X -morphism $\alpha : \widehat{V} \rightarrow \widehat{X}$. By the universal property of $\widehat{V}$, there is a unique V -morphism $\beta : W \rightarrow \widehat{V}$. The relevant diagram is

$$\begin{array}{ccccc} U & \hookrightarrow & W & \xrightarrow{\beta} & \widehat{V} \\ \downarrow & & \downarrow \psi & & \downarrow \alpha \\ V & \xleftarrow{\varphi} & Y & \longrightarrow & \widehat{X} \end{array}$$

The graph of φ is, trivially, proper over V ; the same is true of its inverse image on $V \times_S \widehat{X}$, and it follows that the graph of ψ is proper over V , hence that ψ is proper. Thus β is proper,

and its image is schematically dense, because already U is schematically dense in $\widehat{V}$. On the other hand, β is locally a section of α on W (because $\beta|_U$ is such a section over U); hence β is an isomorphism. By hypothesis the graph of ψ is closed in $V \times_S \widehat{X}$, and is contained in the inverse image of the graph of φ , which is closed in $Y \times X$. One concludes that $\Gamma(\psi)$ is closed in $Y \times \widehat{X}$. $\square$

Lemma (1.7). *Let*

$$X_2 \leftarrow U \rightarrow X_1$$

be a diagram of separated S -schemes of finite type, where U is a dense open subscheme of each X_i . Then there exists $j : U \hookrightarrow X$, with X separated of finite type over S and U schematically dense in X , and proper quasi-dominations

$$p_i : X \dashrightarrow X_i.$$

Proof. By Theorem 1.2, after blowing up X_1 without touching U , one reduces to the case where X_1 quasi-dominates X_2 :

$$\begin{array}{ccc} & X_1 & \\ \swarrow & & \searrow \\ U & \xhookrightarrow{\quad} & X_2. \end{array}$$

Apply Lemma 1.5 to this diagram, in order to obtain

$$\begin{array}{ccc} W & \xrightarrow{\psi} & X'_2 \\ \uparrow & & \uparrow \\ U & \xhookrightarrow{\quad} & V, \\ & & \uparrow \\ & & X_1 \end{array}$$

where

- (a) ψ is the blow-up morphism of an ideal $\mathfrak{a}$ of V , with $V(\mathfrak{a}) \subset V - U$;
- (b) the graph of ψ is closed in $X_1 \times X'_2$;
- (c) X'_2 is a blow-up of X_2 .

Let $\mathfrak{a}'$ be an ideal of X_1 extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset X_1 - U$, and let $q : X'_1 \rightarrow X_1$ be the blow-up of X_1 along $\mathfrak{a}'$. Then ψ induces an isomorphism between W and $q^{-1}(V)$, and the graph of this isomorphism is closed in $X'_1 \times X'_2$. Define X by gluing X'_1 and X'_2 along this isomorphism. $\square$

Remark. *The proof shows that one may take for $p_i : U_i \rightarrow X_i$, with $U_i \subset X$, a blow-up morphism of an ideal outside U .*

Theorem (1.6). *Let $f : X \rightarrow S$ be separated of finite type. There exists an immersion of X into an S -scheme proper over S .*

Proof. Let (U_i) be a finite open covering of X by dense quasi-projective open subschemes. For each U_i , there is, by Corollary 1.4, a commutative diagram of S -schemes

$$\begin{array}{ccc} X'_i & \xrightarrow{q_i} & X_i \\ j_i \downarrow & & \\ U_i & \xhookrightarrow{\quad} & \overline{X}_i, \end{array}$$

where q_i is proper, $\overline{X}_i$ is proper over S , and U_i is dense in $\overline{X}_i$. Let $F_i = \overline{X}_i - U_i$ and $F'_i = q_i^{-1}(F_i)$. Finally, let X^* be the closure of U in the product of the $\overline{X}_i$, with projections

$$p_i : X^* \rightarrow \overline{X}_i.$$

By Lemma 0.5, a preliminary blow-up of the $\overline{X}_i$, not touching X'_i , allows one to suppose that

$$\bigcap_i p_i^{-1}(F'_i) = \emptyset.$$

Let M_i be the S -scheme, separated over S , obtained by gluing X and $\overline{X}_i - F'_i$ along U_i :

$$\begin{array}{ccc} \overline{X}_i - F'_i & & X \\ & \swarrow \quad \searrow & \\ & M_i & \end{array}$$

Apply Lemma 1.7 to X and to the M_i , to obtain $X \hookrightarrow M$, where M properly quasi-dominates each M_i . It remains to prove that M is proper over S .

Indeed, one has morphisms

$$X^* - p_i^{-1}(F'_i) \longrightarrow M_i$$

which glue on $U = \bigcap_i U_i$. Replacing X^* by a blow-up X^{**} , one obtains morphisms

$$X^{**} - p_i^{-1}(F_i) \longrightarrow M$$

which glue, and hence a morphism from the schematic closure of U in X^{**} to M . The scheme M , as the image of this morphism, is proper over S . $\square$

Comments. *I hope that, by being still more constructive, one can prove Theorem 1.6 for $f : X \rightarrow S$ separated of finite type, with S quasi-compact and quasi-separated. The rules to follow are:*

- (a) *consider only quasi-compact open subsets and closed subsets defined by ideals of finite type;*
- (b) *never take schematic closure (one would leave (a)), but take only a closed subset containing the given part and sufficiently small for the later constructions;*
- (c) *do not speak of a dense quasi-projective open subset, cf. (b). Remember that, to construct such open subsets, one could start from an affine covering (U_i) , and take*

$$U = \bigcup_i \left(U_i - \bigcup_{j < i} U_j \right).$$

Translate the arguments in terms of the original U_i , and apply (a) and (b).

On the other hand, I have no idea in the case of algebraic spaces.

Acknowledgements. The original notes were handwritten. They were typed by V. Maillot.

References

- [1] B. Conrad, Deligne's notes on Nagata compactifications, *J. Ramanujan Math. Soc.* **22** (2007), 205–257.
- [2] M. Nagata, Imbedding of an abstract variety in a complete variety, *J. Math. Kyoto Univ.* **2** (1962), 1–10.
- [3] M. Nagata, A generalization of the imbedding problem of an abstract variety in a complete variety, *J. Math. Kyoto Univ.* **3** (1963), 89–102.

Semisimplicity of Tensor Products in Characteristic p

Pierre Deligne

English translation, literal mathematical layer

Abstract. Let G be an affine group scheme over a field k of characteristic $p \neq 0$, and let (V_i) be a finite family of semisimple representations of G . We show that, if

$$\sum_i (\dim V_i - 1) < p,$$

then the tensor-product representation of G on $\bigotimes_i V_i$ is again semisimple. Under the additional hypothesis that G is smooth, this theorem was proved by J.-P. Serre in 1994. We reduce to that case. More generally, the V_i may be taken to be objects of a Tannakian category over k .

Contents

1	Introduction	1
2	Proof of Theorem 0.7	3
3	Saturation (cf. [4] §4)	11
4	The case in which k is algebraically closed	12
5	Extension of scalars in Tannakian categories and proof of 0.1	13
6	Tensor products of exterior powers	17

1 Introduction

Let T be a Tannakian category over a field k of characteristic $p \neq 0$. For the definition of Tannakian categories we refer to [3], p. 193, or to [1], 2.1 and 2.8. Recall that among the axioms are the existence of duals, and the existence of a fiber functor ω with values in the vector spaces over a suitable extension K of k . The dimension $\dim(V)$ of an object V of T is $\dim_K \omega(V)$. Since two fiber functors become isomorphic over a suitable larger extension ([1] 1.12, 1.13), this dimension does not depend on the choice of the fiber functor ω .

Theorem 0.1. Let $(V_i)_{i \in I}$ be a finite family of objects of T . If the V_i are semisimple and

$$\sum_{i \in I} (\dim(V_i) - 1) < p,$$

then the tensor product $\bigotimes_{i \in I} V_i$ is a semisimple object of T .

Example 0.2. If T is the category of finite-dimensional linear representations of an affine group scheme over k , one obtains the theorem announced in the abstract.

Other examples are the category of finite-dimensional representations of a Lie algebra over k , or that of finite-dimensional p -representations of a p -Lie algebra over k . By passage to an algebraic envelope, these cases also reduce to the preceding one.

Example 0.3. Suppose that a group Γ acts on a commutative field K of characteristic p , and let $k := K^\Gamma$ be the subfield of invariants. Let T be the category of finite-dimensional K -vector spaces, endowed with a semilinear action of Γ . With the tensor product over K , this category is Tannakian over k : it admits the fiber functor over K “underlying vector space.” If semilinear representations V_i of Γ on K are semisimple and

$$\sum_i (\dim_K(V_i) - 1) < p,$$

then the semilinear tensor-product representation of the V_i is therefore semisimple.

0.4. Let V be a finite-dimensional vector space over k of characteristic p . As explained in [4], §4, an automorphism g of V such that $g^p = 1$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto g^t,$$

where g^t is defined by the binomial formula

$$g^t := \sum_{n < p} \binom{t}{n} (g - 1)^n. \quad (0.4.1)$$

Similarly, an endomorphism N of V such that $N^p = 0$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto \exp(tN),$$

where $\exp(tN)$ is defined by

$$\exp(tN) := \sum_{n < p} \frac{(tN)^n}{n!}. \quad (0.4.2)$$

Indeed, after any extension of scalars to a commutative k -algebra Λ , if for $t \in \Lambda$ one defines $\exp(tN)$ by (0.4.2), the hypothesis $N^p = 0$ ensures that

$$\exp((t' + t'')N) = \exp(t'N) \exp(t''N).$$

Definition 0.5. Suppose that k is algebraically closed of characteristic p , and let G be a subgroup scheme of $\mathrm{GL}(V)$.

- (i) G is *saturated* if, for every $g \in G(k)$ such that $g^p = 1$, the morphism $t \mapsto g^t$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .
- (ii) G is *infinitesimally saturated* if, for every $N \in \mathrm{Lie}(G)$ such that $N^p = 0$ (in the sense of the p -Lie-algebra structure of $\mathrm{Lie}(G)$, or as an endomorphism of V , which amounts to the same thing), the morphism $t \mapsto \exp(tN)$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .

We shall say that G is *doubly saturated* if it is saturated and infinitesimally saturated.

0.6. Let G be an affine group scheme of finite type over an algebraically closed field k of characteristic p . In the statement 0.1, take T to be the category of finite-dimensional representations of G , suppose that the family (V_i) of representations is faithful, and put $V := \bigoplus V_i$. To prove 0.1 when G is reduced, Serre [4] begins by reducing to the case in which G is saturated in $\mathrm{GL}(V)$. If that is the case, the finite group $\pi_0(G)$ of connected components of G has order prime to p , and this permits a reduction to the case in which G is connected. For G not necessarily reduced, we use an analogous argument to reduce to the case in which G is infinitesimally saturated in $\mathrm{GL}(V)$, and our key result in that case is the following structure theorem. This theorem will provide a reduction to the case in which G is smooth.

Let G^0 be the identity component of G , let G_{red}^0 be the reduced subgroup scheme of G^0 , and let $R_{\mathrm{u}} G_{\mathrm{red}}^0$ be the unipotent radical of G_{red}^0 . Both G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$ are smooth, i.e. linear algebraic groups in the sense of the Chevalley seminar of 1956/58.

Theorem 0.7. Let $(V_i)_{i \in I}$ be a finite faithful family of representations of G . Let V be the sum of the V_i . Suppose that each representation V_i has dimension $< p$ and that, as a subgroup of $\mathrm{GL}(V)$, G is infinitesimally saturated. Under these hypotheses:

- (i) G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$ are invariant subgroup schemes of G , and G^0/G_{red}^0 is of multiplicative type.
- (ii) If the representations V_i are semisimple, G_{red}^0 is reductive.
- (iii) If G_{red}^0 is reductive, there exists in G^0 a central connected subgroup scheme M of multiplicative type such that the morphism

$$M \times G_{\mathrm{red}}^0 \longrightarrow G^0$$

realizes G^0 as a quotient of $M \times G_{\mathrm{red}}^0$.

2 Proof of Theorem 0.7

Proposition 1.1. In an affine group scheme G of finite type over a field, every connected subgroup scheme of multiplicative type is contained in a maximal connected subgroup scheme of multiplicative type.

Proof, after a letter from M. Raynaud to the author. It is enough to show that, in the set of connected subgroup schemes of multiplicative type of G , ordered by inclusion, every totally ordered subset $\mathcal{M}$ is bounded above (Bourbaki, Ens. III, §2, no. 4, Theorem 2). For M in $\mathcal{M}$, the centralizer $Z_G(M)$ of M in G is a subgroup scheme, automatically closed, and the map $M \mapsto Z_G(M)$ is decreasing. The ordered set of closed subgroup schemes of G is noetherian. The set of the $Z_G(M)$ for M in $\mathcal{M}$ therefore has a least element C . Every M in $\mathcal{M}$ is contained in the center $Z(C)$ of C . This center is affine and commutative. The identity component of its largest subgroup scheme of multiplicative type (SGA3 XVII 7.2.1) bounds $\mathcal{M}$ from above. This is the required upper bound.

1.2. Fix k , G , and the finite family of representations $(V_i)_{i \in I}$, with sum V , satisfying the hypotheses of 0.7: k is algebraically closed of characteristic $p \neq 0$, $\dim(V_i) < p$, and, as a subgroup scheme of $\mathrm{GL}(V)$, G is infinitesimally saturated.

Lemma 1.3.

- (i) The identity component G^0 of G is infinitesimally saturated in $\mathrm{GL}(V)$.
- (ii) If a representation W of G is semisimple, its restriction to G^0 is also semisimple.
- (iii) If the reduced subgroup G_{red}^0 of G^0 (respectively its unipotent radical $R_{\mathrm{u}} G_{\mathrm{red}}^0$) is an invariant subgroup scheme of G^0 , then it is one of G .

Proof. Assertion (i) follows from $\mathrm{Lie} G = \mathrm{Lie} G^0$ and from the connectedness of $\mathbb{G}_a$.

It is enough to prove (ii) for W irreducible. Let $S \subset W$ be the sum of the irreducible G^0 -subrepresentations of W . We must show that $S = W$. Each $g \in G(k)$ normalizes G^0 . By transport of structure, S is therefore stable under g . Since k is algebraically closed, G is the union of the gG^0 for $g \in G(k)$, and S is stable under G . Since W is irreducible and $S \neq 0$, S is all of W .

Similarly, by transport of structure, each $g \in G(k)$ normalizes G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$, and (iii) follows because G is the union of the gG^0 .

1.4. Lemma 1.3 shows that, to prove 0.7 for G , it suffices to prove it for G^0 . Moreover, if $p = 2$, each V_i has dimension one, G is of multiplicative type, and 0.7 is trivial.

In the rest of this section, besides the hypotheses of 1.2, we shall suppose that $p \neq 2$ and that G is connected: $G = G^0$. We shall prove that G_{red} is an invariant subgroup scheme of G in 1.18, that G/G_{red} is of multiplicative type in 1.19, the invariance in G of $R_u G_{\text{red}}$ in 1.20–1.23, 0.7(ii) in 1.24, and 0.7(iii) in 1.25.

1.5. Let H be a maximal connected subgroup of multiplicative type of G (1.1), and let $Z_G^0(H)$ be the identity component of its centralizer $Z_G(H)$. The quotient

$$U := Z_G^0(H)/H$$

contains no subgroup μ_p , since the inverse image in $Z_G^0(H)$ of such a subgroup would be an extension of μ_p by H , hence of multiplicative type (SGA3 XVII 7.1.1 b)). Thus U is unipotent (SGA3 XVII 4.3.1(v)).

1.6. Let X be the group of characters of H . The connectedness of H is equivalent to X having no torsion prime to p . Giving an action of H on a vector space W is equivalent to giving an X -grading

$$W = \bigoplus_{\alpha \in X} W^\alpha$$

of W , and this construction is functorial and compatible with tensor product. In particular, V and the V_i are X -graded, and the action by inner automorphisms of H on G defines an X -grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha$$

of $\text{Lie}(G)$.

Lemma 1.7. The central extension $Z_G^0(H)$ of U by H is split.

Since U is unipotent and H is of multiplicative type, there is no nontrivial morphism from U to H (SGA3 XVII 2.4(ii)). The splitting guaranteed by Lemma 1.7 is therefore unique. We shall use the corresponding isomorphism of $H \times U$ with $Z_G^0(H)$ to regard U as a subgroup of $Z_G^0(H)$.

Proof. Let V_i^α ($\alpha \in X$) be the homogeneous components of the V_i . The group $Z_G^0(H)$ acts on V_i^α . Let χ_i^α be the determinant character of this representation. Its restriction to H is $\dim(V_i^\alpha)$ times the character α of H .

Let $J \subset I \times X$ be the set of pairs (i, α) such that $V_i^\alpha \neq 0$, and for $j = (i, \alpha)$ put

$$V^j := V_i^\alpha, \quad \chi_j := \chi_i^\alpha.$$

The family (V_i) of representations of H is faithful: the α for $j = (i, \alpha)$ in J generate X . If $j = (i, \alpha)$ is in J , $\dim V_i^\alpha \leq \dim V_i < p$ is prime to p . The restrictions to H of the χ_j ($j \in J$) therefore generate a subgroup of X of finite index prime to p . Let F be the kernel of

$$(\chi_j) : H \longrightarrow \mathbb{G}_m^J$$

and let Q be the cokernel. The group F is finite, reduced, and of order prime to p . In the commutative diagram with exact rows

$$\begin{array}{ccccccc} 1 & \longrightarrow & H/F & \longrightarrow & Z_G^0(H)/F & \longrightarrow & U \longrightarrow 1 \\ & & \parallel & & \downarrow & & \downarrow \\ 1 & \longrightarrow & H/F & \longrightarrow & \mathbb{G}_m^J & \longrightarrow & Q \longrightarrow 1, \end{array}$$

the morphism from the unipotent group U to the group of multiplicative type Q is trivial. The kernel of (χ_j) therefore lifts U into $Z_G^0(H)/F$. Its inverse image in $Z_G^0(H)$ is a central extension E of U by F . By Lemma 1.8 below, this extension is trivial. A lifting of U into E splits the central extension $Z_G^0(H)$ of U by H .

Lemma 1.8. A central extension of a unipotent group by a finite reduced group of order prime to p is trivial.

We give two proofs of the lemma, based on different ideas. The second assumes that U is connected, which is sufficient for the application above.

First proof. Let

$$1 \longrightarrow F \longrightarrow E \longrightarrow U \longrightarrow 1$$

be a central extension as in the lemma, and let a be an integer prime to p that kills F . View E as a central extension in the category of fppf sheaves of groups on $\text{Spec}(k)$. For any S/k , $U(S)$ admits a finite central filtration with successive quotients killed by p (apply SGA3 XV 3.5(v)). Hence every finite subset X of $U(S)$ generates a finite p -group P_X . Let Q_X be its intersection with the image of $E(S)$. The inverse image of Q_X in $E(S)$ is a central extension E_X of Q_X by $F(S) = F^{\pi_0(S)}$. The group $H^2(Q_X, F(S))$ classifying these extensions is trivial, since it is killed both by a and by the power $|Q_X|$ of p . The extension E_X is therefore trivial. Moreover, the lifting of Q_X into E_X is unique: it is the set of the a -th powers of elements of E_X .

Passing to the inductive limit over X and sheafifying, one deduces that the subsheaf of sets of E which is the image of

$$x \longmapsto x^a : E \longrightarrow E$$

is a subsheaf in groups lifting U into E .

Second proof. The action by right translations of F on E makes E an F -torsor over U . It is trivialized over the identity element, and therefore corresponds to a morphism

$$\pi_1(U, e) \longrightarrow F.$$

The reduced scheme U_{red} is isomorphic to an affine space $\mathbb{A}^N$. Since

$$\pi_1(U, e) = \pi_1(U_{\text{red}}, e) \simeq \pi_1(\mathbb{A}^N, 0)$$

has no nontrivial quotient of order prime to p (SGA4 XV 2.2), the F -torsor E is trivial: the identity component of E lifts U into E .

Lemma 1.9. Let $f : V \rightarrow W$ be a morphism of schemes of finite type over k , let $v \in V(k)$, and let $(df)_v$ be the morphism induced by f from the Zariski tangent space of V at v to the Zariski tangent space of W at $f(v)$. If V is smooth over k and $(df)_v$ is an isomorphism, then f is etale at v and W is smooth over k in a neighborhood of $f(v)$.

Proof. Let A (respectively B) be the completion of the local ring of V at v (respectively of W at $f(v)$). Let m be the maximal ideal of A and n that of B . The injectivity of $(df)_v$ ensures that A is a quotient of B . We must show that $B \xrightarrow{\sim} A$ (SGA1 I 4.4). Let $t_1, \dots, t_N$ in n lift a basis of n/n^2 . The bijectivity of $(df)_v$ ensures that the images of the t_i in m/m^2 form a basis. The t_i define an epimorphism

$$g : T_i \mapsto t_i : k[[T_1, \dots, T_N]] \longrightarrow B.$$

In the commutative diagram

$$\begin{array}{ccc} & k[[T_1, \dots, T_N]] & \\ \swarrow & & \searrow g \\ A & \xleftarrow{\hspace{1.5cm}} & B \end{array}$$

the smoothness of V at v ensures that g is an isomorphism. It follows that $B \xrightarrow{\sim} A$.

In characteristic p , the Lie algebra of a group scheme is a p -Lie algebra. We shall write $\gamma \mapsto \gamma^p$ for the p -th power operation, often denoted $\gamma^{[p]}$. For every linear representation ρ of the group, one has

$$\rho(\gamma^p) = \rho(\gamma)^p.$$

Lemma 1.10. For every $\gamma \in \text{Lie } U$, one has $\gamma^p = 0$.

As explained under 1.7, we regard here U as a subgroup of $Z_G^0(H) \subset G$.

Proof. Since U is a unipotent group scheme, for each i the image γ_i of γ in $\text{End}(V_i)$ is nilpotent. Since $\dim(V_i) < p$, one has $\gamma_i^p = 0$. Since the family (V_i) of representations is faithful, $\gamma^p = 0$.

Corollary 1.11. The unipotent group U is smooth.

Proof. Let $(\gamma_a)_{1 \leq a \leq A}$ be a basis of $\text{Lie } U$. By 1.10, for each a ,

$$\rho_a : \lambda \mapsto \exp(\lambda \gamma_a) : \mathbb{G}_a \longrightarrow \text{GL}(V)$$

is defined by 0.4. The hypothesis that G is infinitesimally saturated ensures that this morphism factors through G . Since U centralizes H , the action of H by inner automorphisms fixes $\text{Lie } U$, hence each ρ_a , and ρ_a factors through $Z_G(H)$, and even through $Z_G^0(H) = H \times U$ because $\mathbb{G}_a$ is connected. Every morphism from $\mathbb{G}_a$ to H being trivial, ρ_a factors through U . Let $\mathbb{E}^A$ be affine space of dimension A and set

$$f : \mathbb{E}^A \longrightarrow U, (t_a) \mapsto \prod \rho_a(t_a),$$

the product being taken with respect to the group law of U . By construction, df is an isomorphism at 0. Applying 1.9, one deduces that U is smooth in a neighborhood of the identity element, hence everywhere by homogeneity.

Recall that the action by inner automorphisms of H on $\text{Lie } G$ defines a grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha.$$

Lemma 1.12. If $\alpha \in X$ is nonzero and $\gamma \in \text{Lie}(G)^\alpha$, then $\gamma^p = 0$.

Proof. Regard γ as an endomorphism of V . If γ^p were nonzero, in the sense of the p -Lie algebra structure of $\text{Lie } G$ or as an endomorphism of V (which amounts to the same thing), there would exist $i, \beta \in X$, and $v \in V_i^\beta$ such that $\gamma^p v \neq 0$. This implies the nonvanishing of the $\gamma^l v$ for $0 \leq l \leq p$. The element $\gamma^l v$ of V_i lies in $V_i^{\beta + l\alpha}$. For $0 \leq l < p$, the $\beta + l\alpha$ are distinct, because X has no torsion prime to p . The $\gamma^l v$ for $0 \leq l < p$ are therefore linearly independent. This contradicts the hypothesis $\dim V_i < p$.

1.13. Let $(\gamma_b)_{1 \leq b \leq B}$ be a family of elements of $\text{Lie } G$, the disjoint union of bases of the $(\text{Lie } G)^\alpha$ for $\alpha \neq 0$. Let α_b be the character α of H such that $\gamma_b \in (\text{Lie } G)^\alpha$. Since G is infinitesimally saturated, the morphism $\lambda \mapsto \exp(\lambda \gamma_b) : \mathbb{G}_a \rightarrow \text{GL}(V)$ provided by 0.4 and 1.12 factors through a morphism

$$\rho_b : \mathbb{G}_a \longrightarrow G.$$

This morphism is equivariant for the action of H on $\mathbb{G}_a$ through α_b and on G by inner automorphisms, because the same equivariance holds after composition with the inclusion of G in $\text{GL}(V)$.

Let q (respectively q_0) be the projection of G onto $G/Z_G(H)$ (respectively $G/Z_G^0(H)$). Consider the morphism

$$f : \mathbb{E}^B \longrightarrow G, \quad (t_b) \longmapsto \prod_{b=1}^B \rho_b(t_b),$$

the product being taken with respect to the group law of G .

Lemma 1.14. The composite morphism $qf : \mathbb{E}^B \rightarrow G/Z_G(H)$ is étale at 0.

Since $G/Z_G^0(H)$ is étale over $G/Z_G(H)$, the same statement holds for q_0f .

Proof. Let e be the identity element of G , and let $T_{q(e)}$ denote the tangent space of $G/Z_G(H)$ at $q(e)$. By 1.9, it is enough to check that qf induces an isomorphism from the tangent space at 0 onto $T_{q(e)}$, and for that it is enough to check that

$$dq : \text{Lie}(G) \longrightarrow T_{q(e)}$$

induces an isomorphism

$$\bigoplus_{\alpha \neq 0} \text{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

If G were smooth, $Z_G(H)$ would be smooth as the fixed locus of an action on G of the group of multiplicative type H ; one would deduce an exact sequence

$$0 \longrightarrow \text{Lie } Z_G^0(H) \longrightarrow \text{Lie } G \longrightarrow T_{q(e)} \longrightarrow 0,$$

and one would conclude by noting that $\text{Lie } Z_G^0(H) = \text{Lie}(G)^H$.

We shall deduce 1.14 from the following lemma.

Lemma 1.15. Suppose that a group H of multiplicative type, with character group X , acts on an affine k -scheme of finite type $S = \text{Spec}(A)$. Let S^H be the fixed locus and let $s \in S^H(k)$. Let I be the ideal defining S^H in S . View I/I^2 as a coherent sheaf on S^H , and let

$$(I/I^2)_s = s^*(I/I^2)$$

be its fiber at s . Let m be the maximal ideal at s . The group H acts on m/m^2 , hence m/m^2 is X -graded. The inclusion of I in m induces an isomorphism

$$(I/I^2)_s \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha. \quad (1.15.1)$$

The hypothesis “ S affine” is probably unnecessary.

Proof. The action of H gives an X -grading of A . The quotient A/I is the largest graded quotient of A purely of degree 0: I is the ideal generated by the A^α for $\alpha \neq 0$. The two sides of (1.15.1) are unchanged when A is replaced by A/I^2 . We may therefore suppose that $I^2 = 0$. Then $A^\alpha A^\beta = 0$ for $\alpha, \beta \neq 0$, I is the sum of the A^α for $\alpha \neq 0$, and $A^0 \xrightarrow{\sim} A/I$. Let m_0 be the maximal ideal of s in $A/I \xleftarrow{\sim} A^0$. We have

$$m = m_0 \oplus \bigoplus_{\alpha \neq 0} A^\alpha.$$

Thus

$$m/m^2 = m_0/m_0^2 \oplus \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

and

$$I/I^2 \otimes_{A^0} A^0/m_0 = \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

which gives (1.15.1).

End of the proof of 1.14. The morphism $G \rightarrow G/Z_G(H)$ is flat. If I is the ideal defining $Z_G(H)$ in G , and n is the maximal ideal at the point $q(e)$ of $G/Z_G(H)$, then the vector bundle I/I^2 on $Z_G(H)$ is the inverse image of n/n^2 on $q(e)$. Its fiber at the identity element e of G is n/n^2 , and therefore by (1.15.1)

$$n/n^2 \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha.$$

Passing to duals gives

$$\bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

Corollary 1.16. The morphism

$$F : \mathbb{E}^B \times Z_G^0(H) \longrightarrow G, \quad (t_b), z \longmapsto \prod \rho_b(t_b) \cdot z$$

is etale at $(0, e)$.

Proof. The square

$$\begin{array}{ccc} \mathbb{E}^B \times Z_G(H) & \xrightarrow{F} & G \\ \downarrow & & \downarrow \\ \mathbb{E}^B & \xrightarrow{q_0 f} & G/Z_G^0(H) \end{array}$$

is cartesian. Hence F is etale at every point lying above $0 \in \mathbb{E}^B$.

1.17. If $f : T \rightarrow U$ is a quasi-compact morphism of schemes, the closed scheme-theoretic image $\overline{f(T)}$ of f is the smallest closed subscheme of U through which f factors. If $f : \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is affine, it is defined by the ideal kernel of $f^* : B \rightarrow A$. Its formation is compatible with every flat base change $U' \rightarrow U$. If f is a morphism of schemes over S , if a group scheme H flat over S acts on T and U , and if f is equivariant, this compatibility with base change implies that the action of H on U induces an action of H on the closed scheme-theoretic image $\overline{f(T)}$.

1.18. Proof of the invariance of G_{red} in G . The morphism F of 1.16 is equivariant for the action of H on $\mathbb{E}^B$ by the α_b , and on G by inner automorphisms. Since U is smooth (1.11), the reduced subscheme of

$$Z_G^0(H) = H \times U$$

is $H_{\text{red}} \times U$. An étale morphism induces an étale morphism on reduced subschemes. The morphism induced by F from $\mathbb{E}^B \times H_{\text{red}} \times U$ to G_{red} is therefore étale at 0, and its image contains a nonempty open of G_{red} . The closed scheme-theoretic image of the morphism

$$F_1 : \mathbb{E}^B \times H_{\text{red}} \times U \longrightarrow G$$

induced by F therefore coincides with G_{red} . Since this morphism is H -equivariant, 1.17 ensures that G_{red} is stable under the action of H on G by inner automorphisms.

Because F is étale at the origin, G is generated as an fppf sheaf by the image of F , hence by H and G_{red} , since $U \subset G_{\text{red}}$. Since H normalizes G_{red} , it follows that G_{red} is an invariant subgroup scheme of G .

Proposition 1.19. One has

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}.$$

Consequently G/G_{red} is of multiplicative type.

Proof. If Y_1 and Y_2 are two closed subschemes of Z , then for any morphism $u : Z' \rightarrow Z$ one has

$$u^{-1}(Y_1 \cap Y_2) = u^{-1}(Y_1) \cap u^{-1}(Y_2).$$

If u is étale, and $Y_2 = Z_{\text{red}}$, one obtains

$$u^{-1}(Y_1 \cap Z_{\text{red}}) = u^{-1}(Y_1) \cap Z'_{\text{red}}.$$

Apply this to F , which is étale in a neighborhood of $\{0\} \times H \subset \{0\} \times Z_G^0(H)$. One obtains that, in a neighborhood of $\{0\} \times H$, the inverse image of $H \cap G_{\text{red}}$ is H_{red} ; hence $H_{\text{red}} \rightarrow H \cap G_{\text{red}}$ is étale, therefore an isomorphism. If H , H_{red} , G , G_{red} and the quotients are interpreted as fppf sheaves, the isomorphism

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}$$

follows from H normalizing G_{red} , from $HG_{\text{red}} = G$, and from $H_{\text{red}} \xrightarrow{\sim} H \cap G_{\text{red}}$.

1.20. Proof of the invariance in G of $R_{\text{u}} G_{\text{red}}$. Let R denote the unipotent radical $R_{\text{u}} G_{\text{red}}$ of the smooth group G_{red} , and let

$$L := G_{\text{red}}/R.$$

Let T_L be a maximal torus of the reductive group L . It is a maximal connected subgroup of multiplicative type. The inverse image of T_L in G_{red} is an extension of the torus T_L by the smooth connected unipotent group R . By SGA3 XVII 5.1.1(i), such an extension is trivial: the torus T_L lifts to a torus T , automatically maximal, of G_{red} . We shall take H to be a maximal connected subgroup of multiplicative type of G containing T . Then

$$T = H_{\text{red}} = H \cap G_{\text{red}}.$$

Lemma 1.21. The connected centralizer $Z_{G_{\text{red}}}^0(T)$ of T in G_{red} is a product $T \times W$, with W unipotent. The group scheme H normalizes W .

Proof. Since H normalizes T , H normalizes

$$Z := Z_{G_{\text{red}}}^0(T).$$

As with G , the subgroup G_{red} of $\text{GL}(V)$ is infinitesimally saturated. Applying 1.7 to G_{red} and T (instead of G and H) gives the decomposition

$$Z = T \times W.$$

The proof of 1.7 describes $W \subset Z$ as the identity component of the intersection of the kernels of the determinant characters of the representations of Z in the V_i^β , for β a weight of T in V . This description shows that H normalizes W .

Lemma 1.22. The action of H on $\text{Lie}(G_{\text{red}})$ preserves the subspace $\text{Lie}(R)$.

Proof. Decompose

$$l = \text{Lie } L, \quad g = \text{Lie } G_{\text{red}}, \quad r = \text{Lie } R$$

according to the weights of T . Since H normalizes T , the decomposition obtained for g is respected by H . We shall show that each r^α , hence their sum r , is stable under H .

The connected centralizer of T_L in L is reduced to T_L . Hence we have a morphism of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & W & \longrightarrow & Z = T \times W & \longrightarrow & T_L \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & R & \longrightarrow & G_{\text{red}} & \longrightarrow & L. \end{array}$$

The part of weight 0 of g is $\text{Lie}(Z)$. Its intersection with $\text{Lie } R$ is $\text{Lie } W$, stable under H by 1.21.

If α is not a weight of T on l , then $r^\alpha = g^\alpha$, which is stable under H .

If α is a nonzero weight of T on l , then l^α and $l^{-\alpha}$ have dimension one. Under our hypothesis that $p \neq 2$, the bracket induces a nondegenerate pairing of l^α and $l^{-\alpha}$ with values in a line $h_\alpha \subset \text{Lie } T_L$. The bracket of g induces a pairing of g^α and $g^{-\alpha}$ with values in

$$g^0 = \text{Lie } Z.$$

Its composite with $g^0 \rightarrow g^0 / \text{Lie } W \simeq \text{Lie } T_L$ factors through $g^{\pm\alpha} \rightarrow l^{\pm\alpha}$ and the preceding pairing. It therefore has values in a line d of $g^0 / \text{Lie } W$, and

$$r^{\pm\alpha} = \text{Ker}(g^{\pm\alpha} \rightarrow \text{Hom}(g^{\mp\alpha}, d)) :$$

the $r^{\pm\alpha}$ are the kernels of the pairing of g^α and $g^{-\alpha}$ with values in d . The decomposition of g according to the weights of T , and $W \subset g^0$, are respected by H ; this description shows that the r^α , and hence r , are subrepresentations of H acting on g .

1.23. End of the proof of the invariance of R . Let Fil denote the finite increasing filtration of $\bigoplus V_i$ such that $\text{Fil}_0 = 0$ and $\text{Fil}_{n+1} / \text{Fil}_n$ is the space of invariants of R in $\bigoplus V_i / \text{Fil}_n$. It is the sum of filtrations of the V_i . It is respected by G_{red} , since R is an invariant subgroup of G_{red} .

The reduced subgroup of the largest subgroup of G_{red} which acts trivially on

$$\text{Gr}^{\text{Fil}}(\bigoplus V_i)$$

is a smooth invariant unipotent subgroup containing R . It therefore coincides with R .

If $\gamma \in \text{Lie } R$, then γ acts nilpotently on V_i . Since the family of representations V_i of dimension $< p$ is faithful, one has $\gamma^p = 0$. Since G_{red} is infinitesimally saturated, the corresponding morphism

$$\rho : \mathbb{G}_a \longrightarrow \prod \text{GL}(V_i)$$

factors through G_{red} . Since this representation of $\mathbb{G}_a$ is trivial on

$$\text{Gr}^{\text{Fil}}(\bigoplus V_i),$$

the morphism ρ even factors through R .

Choose a basis $(\gamma_c)_{1 \leq c \leq C}$ of $r = \text{Lie } R$, the disjoint union of bases of the r^β for β a weight of H acting on r . Let $\rho_c : \mathbb{G}_a \rightarrow R$ be the morphism defined by γ_c . As in 5.6, the morphism of schemes

$$f : \mathbb{E}^C \longrightarrow R, \quad (t_c) \longmapsto \prod_{c=1}^C \rho_c(t_c)$$

is etale at 0. Its composite with the inclusion of R in G_{red} is H -equivariant, for a suitable action of H on $\mathbb{E}^C$. By the same argument as in 1.17 and 1.18, it follows that R is stable under the action of H on G_{red} .

Corollary 1.24. If the representations V_i are semisimple, the group scheme G_{red} is reductive.

Proof. Replacing each V_i by a family of irreducible representations of which V_i is the sum, we reduce to supposing the V_i irreducible. We must prove that the unipotent radical $R = R_u G_{\text{red}}$ is trivial.

Since R is unipotent, the subspace V_i^R of R -invariants in V_i is nonzero. Because R is invariant in G , V_i^R is stable under G , hence, by irreducibility of V_i , coincides with V_i . Since the family of the V_i is faithful, R is trivial.

1.25. Proof of 0.7(iii). Suppose that G_{red} is reductive, let T be a maximal torus of G_{red} , and take H to be a maximal connected subgroup of multiplicative type of G containing T . It normalizes G_{red} and its action on G_{red} fixes T . The group scheme of automorphisms of the reductive group G_{red} which fix T is the image T^{ad} of T in the adjoint group. If Z is the subgroup of H which centralizes G_{red} , the morphism

$$Z \times T \longrightarrow H$$

is a quotient by T : the image T^{ad} of T in the adjoint group. View all these group schemes as fppf sheaves. Since H and G_{red} generate G , Z is central. Since the action of H on G_{red} factors through a quotient of T , Z and T generate H . Since Z is generated by Z^0 and Z_{red} and $Z_{\text{red}} \subset G_{\text{red}}$, Z^0 and G_{red} generate G : the morphism

$$Z^0 \times G_{\text{red}} \longrightarrow G$$

is an epimorphism, and this verifies 0.7(iii).

3 Saturation (cf. [4] §4)

2.1. Let V be a finite-dimensional vector space over an algebraically closed field k of characteristic p , and let G be a subgroup scheme of $\text{GL}(V)$. The intersection G^* of the doubly saturated subgroup schemes (0.5) of $\text{GL}(V)$ containing G is doubly saturated. We call G^* the double saturation of G .

Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k . For each i , let N_i be an endomorphism of V_i such that $N_i^p = 0$. It defines an action $t \mapsto \exp(tN_i)$ of $\mathbb{G}_a$ on V_i (0.4). For $a \in I$, again denote by N_a the endomorphism of $\bigotimes V_i$ which is the tensor product of $N_a \in \text{End}(V_a)$ and of the identities $1 \in \text{End}(V_j)$ for $j \neq a$. Let N be the endomorphism $\sum N_a$ of $\bigotimes V_i$.

Proposition 2.2. Suppose there exist integers v_i such that $N_i^{v_i} = 0$ and

$$\sum_i (v_i - 1) < p.$$

Then $N^p = 0$, and the action $t \mapsto \exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .

Proof. The endomorphisms N_a of $\bigotimes V_i$ commute with one another, and every monomial of degree p in the N_a has degree at least v_a in N_a for some $a \in I$, hence is zero. It follows that

$$N^p = \left(\sum N_a \right)^p = 0,$$

and also that the truncated exponentials satisfy

$$\exp(tN) = \exp\left(\sum tN_a\right) = \prod \exp(tN_a) = \bigotimes \exp(tN_a).$$

Corollary 2.3. If $\sum_i (\dim(V_i) - 1) < p$, the action $\exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .

Proof. Take $v_i = \dim(V_i)$ in 2.2.

By the same arguments as those of [4], §4, one deduces from [4], §4, Proposition 10, and from 2.3 the following proposition 2.4. Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k and let G be a subgroup scheme of $\prod \text{GL}(V_i)$. The subgroup scheme $\prod \text{GL}(V_i)$ of $\text{GL}(\bigoplus V_i)$ is saturated and infinitesimally saturated. The double saturation G^* of G , viewed as a subgroup of $\text{GL}(\bigoplus V_i)$, is therefore contained in $\prod \text{GL}(V_i)$.

Proposition 2.4. If $\sum_i (\dim(V_i) - 1) < p$, then a subspace W of $\bigotimes V_i$ is stable under the action of G if and only if it is stable under that of G^* . Thus the representation $\bigotimes V_i$ of G is semisimple if and only if it is semisimple as a representation of G^* .

4 The case in which k is algebraically closed

Let G be an affine group scheme over an algebraically closed field k of characteristic p , and let $(V_i)_{i \in I}$ be a finite family of representations of G .

Proposition 3.1. If the representations V_i are semisimple and $\sum_i (\dim V_i - 1) < p$, then the representation $\bigotimes V_i$ of G is again semisimple.

Proof. We leave to the reader the verification that this statement is implied by the special case in which one makes the additional hypotheses that each V_i is simple of dimension ≥ 2 , and that $|I| \geq 2$. These hypotheses imply that $p \neq 2$, and that each V_i has dimension $< p$.

Replacing G by its image in $\prod \text{GL}(V_i)$, one may suppose that G is a subgroup scheme of $\prod \text{GL}(V_i)$. By 2.4, one may moreover suppose that, as a subgroup of $\text{GL}(\bigoplus V_i)$, G is doubly saturated. In the rest of the proof we shall suppose all these properties satisfied. That G is

saturated is equivalent to G_{red} being so and, by [4], §4, Proposition 11, implies that the finite group

$$G/G^0 = G_{\text{red}}/G_{\text{red}}^0$$

has order prime to p .

By 0.7(ii)(iii), the group G_{red}^0 is reductive and G^0 contains a central connected group M of multiplicative type such that

$$M \times G_{\text{red}}^0 \longrightarrow G^0$$

is an epimorphism.

By Lemma 1.3(ii), the V_i are semisimple as representations of G^0 . If W is an irreducible representation of G^0 , then M , being of multiplicative type and central, acts on W by a character. The representation W is therefore still irreducible as a representation of G_{red}^0 . Thus the V_i are semisimple as representations of G_{red}^0 , and by [4] the tensor product of these representations is again semisimple as a representation of G_{red}^0 . The following lemma shows that it is semisimple as a representation of G .

Lemma 3.2. Suppose that a group scheme G is an extension of a linearly reductive group scheme A by a group scheme B . Then every representation V of G which is semisimple as a representation of B is semisimple.

That A is linearly reductive means that all its linear representations are semisimple. A group of multiplicative type, and a finite group of order prime to p , are linearly reductive.

Proof. Let S be the sum of the irreducible subrepresentations of V . We must show that $S = V$, and to verify this it is enough to show that S has a complement T in V stable under G . Consider the exact sequence

$$1 \longrightarrow \text{Hom}_B(V/S, S) \longrightarrow \text{Hom}_B(V, S) \longrightarrow \text{Hom}_B(S, S) \longrightarrow 1.$$

It is an exact sequence of representations of A . Since A is linearly reductive, the identity 1_S of S , fixed by A , lifts to a homomorphism $f : V \rightarrow S$ of representations of B which is fixed by A , i.e. is a morphism of representations of G . Its kernel is the required complement.

Corollary 3.3. The statement 0.1 is true when k is algebraically closed.

Proof. Let T' be the Tannakian subcategory of T generated from the V_i by tensor product, duality, and passage to a subquotient. By [3] 3.3.1.1, T' admits a fiber functor over k , hence is equivalent to the category of representations of an affine group scheme over k . Another reference is [1] 6.20.

Remark 3.4. The reduction from T to T' is in fact unnecessary. I have checked (unpublished) that every Tannakian category over an algebraically closed field k is equivalent to the Tannakian category of representations of an affine group scheme over k .

5 Extension of scalars in Tannakian categories and proof of 0.1

4.1. Let k be a commutative field and let T be a k -linear abelian category. Define the tensor product of an object X of T and a finite-dimensional k -vector space V to be the object of T which represents the functor

$$Y \longmapsto V \otimes \text{Hom}(Y, X).$$

The choice of a basis $(e_i)_{i \in I}$ of V identifies $V \otimes X$ with the sum of a family of copies of X indexed by I .

Let k' be a finite extension of k . We define as follows the k' -linear abelian category $T_{k'}$ obtained from T by extension of scalars to k' , the scalar-extension functor

$$e_{k'/k} : T_k \longrightarrow T_{k'},$$

and the scalar-restriction functor

$$r_{k \setminus k'} : T_{k'} \longrightarrow T_k.$$

The category $T_{k'}$ is the category of objects X of T endowed with a k' -module structure $k' \otimes X \rightarrow X$. For X in T ,

$$e_{k'/k}(X) := k' \otimes X,$$

endowed with its natural k' -module structure. The functor $r_{k \setminus k'}$ forgets the k' -module structure. The functors $e_{k'/k}$ and $r_{k \setminus k'}$ are exact, and $r_{k \setminus k'}$ is right adjoint to $e_{k'/k}$. Example: if T is the category of left modules over a k -algebra A , then $T_{k'}$ identifies with the category of left modules over the k' -algebra

$$A' := k' \otimes_k A,$$

and $e_{k'/k}$ and $r_{k \setminus k'}$ are the usual extension- and restriction-of-scalars functors.

Suppose that the objects of T have finite length and that the Hom groups are finite-dimensional over k . If X is a simple object of T , the division algebra $\text{End}(X)$, being finite-dimensional over k , is central simple over its center, which is a finite extension of k .

Lemma 4.2. Under these hypotheses:

- (i) Let X be a simple object of T , $D := \text{End}(X)$, and F the center of D . The object $e_{k'/k}(X)$ of $T_{k'}$ is semisimple if and only if $F \otimes_k k'$ is a product of fields.
- (ii) If Y is a semisimple object of $T_{k'}$, then $r_{k \setminus k'}(Y)$ is semisimple.
- (iii) If X is in T and $e_{k'/k}(X)$ is semisimple, then X is semisimple.

Proof. (i) The functor $Z \mapsto \text{Hom}(X, Z)$ is an equivalence from the category of objects of T which are sums of copies of X to the category of finite-dimensional right D -vector spaces. The category of sums of copies of X endowed with a k' -module structure identifies with the category of finite-type right $D \otimes_k k'$ -modules. The object $e_{k'/k}(X)$ of $T_{k'}$ thus corresponds to the $D \otimes_k k'$ -module $D \otimes_k k'$. It is known to be semisimple if and only if $F \otimes_k k'$ is a product of fields.

(ii) Reduce to the case in which Y is simple. The sum S of the simple subobjects of $r_{k \setminus k'}(Y)$ is nonzero and is a k' -submodule. By the simplicity of Y , it coincides with $r_{k \setminus k'}(Y)$.

(iii) The object X is a direct factor of $r_{k \setminus k'} e_{k'/k}(X)$, and one applies (ii).

4.3. Now suppose that T is a Tannakian category over k , and equip $T_{k'}$ with the tensor product over k' :

$$X \otimes_{k'} Y = \text{coker}(X \otimes k' \otimes Y \rightrightarrows X \otimes Y).$$

The scalar-extension functor $e_{k'/k}$ is compatible with tensor products and with their associativity and commutativity data. It sends the unit object to the unit object, and duals, in the sense of [1] 2.2, to duals. The category $T_{k'}$ admits internal Homs: if X and Y in T are endowed with k' -module structures, their internal Hom in $T_{k'}$ is the kernel of the two morphisms

$$\text{Hom}(X, Y) \rightrightarrows \text{Hom}(k' \otimes X, Y) = k'^{\vee} \otimes \text{Hom}(X, Y),$$

where $k'^{\vee}$ is the k -dual of the k -vector space k' . We shall denote it by $\text{Hom}_{k'}(X, Y)$.

Theorem 4.4. If T is a Tannakian category over k and k' is a finite extension of k , then the category $T_{k'}$, endowed with the tensor product over k' , is Tannakian over k' .

It is possible to deduce this theorem from Theorem 1.12 of [1], which characterizes Tannakian categories as categories of representations of affine groupoids G acting transitively on a k -scheme S ($G \rightarrow S \times_k S$ affine and faithfully flat): if T is such a category of representations, then $T_{k'}$ identifies with the category of representations of the groupoid obtained from it by extension of scalars to k' .

Conversely, and according to a strategy indicated to me by Grothendieck about thirty years ago, the more elementary proof of 4.4 that follows should make it possible to prove [1] 1.12 more simply.

If $k' \supset k'' \supset k$, then $T_{k'}$ identifies with $(T_{k''})_{k'}$. This observation permits one to reduce the proof of 4.4 to the following two special cases: k'/k separable, and k'/k inseparable of degree p . The delicate point is to show the existence of duals, in the sense of [1] 2.2, in $T_{k'}$. Once this is acquired, if ω is a fiber functor over the k -algebra K , then ω provides a fiber functor ω' of $T_{k'}$ over the k' -algebra

$$K' := k' \otimes_k K :$$

to X in T , endowed with a k' -module structure, attach $\omega(X)$, endowed with its k' -module structure. This structure makes $\omega(X)$ a $k' \otimes K$ -module.

Proof if k' is a finite separable extension of k . If X is in T and has $X^\vee$ as dual, in the sense of [1] 2.2, the object $e_{k'/k}(X)$ of $T_{k'}$ admits $e_{k'/k}(X^\vee)$ as dual. One concludes by observing that every object Y of $T_{k'}$ is a direct factor of an object of this form. More precisely, the adjunction morphism

$$e_{k'/k} r_{k \setminus k'}(Y) \longrightarrow Y$$

is split: if one regards Y as an object of T , $k' \otimes Y$ has two natural k' -module structures, one coming from k' (this is that of $e_{k'/k} r_{k \setminus k'}(Y)$), the other from Y . The adjunction morphism is the natural morphism

$$k' \otimes Y \longrightarrow (k' \otimes Y) \otimes_{k' \otimes k} k'.$$

It is split because the morphism $k' \otimes_k k' \rightarrow k'$ is the projection onto a direct factor.

Proof if $k' = k(a^{1/p})$. Let ω be a fiber functor over an extension K of k . The key point is the following lemma.

Lemma 4.5. If an object X of T is endowed with a k' -module structure, this structure makes $\omega(X)$ a $k' \otimes_k K$ -module. This module is free.

Proof. If $K' := k' \otimes_k K$ is a field, this is clear. Otherwise, a has a p -th root α in K , and the K -algebra K' is isomorphic to $K[\varepsilon]/(\varepsilon^p)$: take

$$\varepsilon = \sqrt[p]{a} - \alpha.$$

Let d be the dimension over K of $\omega(X)/\varepsilon\omega(X)$. The K' -module $\omega(X)$ is isomorphic to a sum of d monogenic modules $K[\varepsilon]/(\varepsilon^j)$, $1 \leq j \leq p$. One checks that it is free if and only if the exterior power over K'

$$\bigwedge \text{olimits}_{K'}^d \omega(X)$$

is free of rank one. By the exactness of ω , this exterior power over K' is the image under ω of the exterior power over k' of X , defined as the image of the antisymmetrization

$$a : \bigotimes \text{olimits}_{k'}^d X \longrightarrow \bigotimes \text{olimits}_{k'}^d X.$$

Replacing X by this exterior power, we are reduced to the case in which X is such that $\omega(X)$ is a nonzero monogenic K' -module.

The k' -module structure of X gives a morphism

$$k' \otimes \mathbf{1} \longrightarrow \text{Hom}(X, X). \quad (4.5.1)$$

One knows that the unit object $\mathbf{1}$ is simple ([2] 1.17). The kernel of (4.5.1) is therefore of the form $A \otimes \mathbf{1}$, for A a vector subspace of k' . This subspace is a proper ideal of k' , so $A = 0$ and (4.5.1) is a monomorphism. Applying ω , one deduces that $\omega(X)$ is a faithful K' -module, and one concludes by observing that a faithful monogenic K' -module is free.

4.6. End of the proof if $k' = k(a^{1/p})$. Let $\mathbf{1}'$ denote the unit object $e_{k'/k}(\mathbf{1})$ of $T_{k'}$. For X in $T_{k'}$, put

$$X^\vee := \text{Hom}_{k'}(X, \mathbf{1}').$$

We show that $X^\vee$ is a dual of X in the sense of [1] 2.2. In $T_{k'}$ there is an evaluation morphism

$$\text{ev} : X^\vee \otimes_{k'} X \longrightarrow \mathbf{1}' \quad (4.6.1)$$

and, for every Y in $T_{k'}$, a morphism

$$Y \otimes_{k'} X^\vee \longrightarrow \text{Hom}_{k'}(X, Y). \quad (4.6.2)$$

The functor ω' , with values in K' -modules, transforms this morphism into

$$\omega(Y) \otimes \omega(X)^\vee \longrightarrow \text{Hom}_{K'}(\omega(X), \omega(Y)). \quad (4.6.3)$$

Lemma 4.5 ensures that (4.6.3) is an isomorphism. Since ω' is exact and satisfies $\omega(Z) = 0 \Rightarrow Z = 0$, because ω has these properties, it is conservative and (4.6.2) is an isomorphism. Taking $Y = X$, and composing the evident morphism $\mathbf{1}' \rightarrow \text{Hom}_{k'}(X, X)$ with the inverse of (4.6.2), we obtain

$$\delta : \mathbf{1}' \longrightarrow X \otimes_{k'} X^\vee.$$

The morphisms ev and δ satisfy the identities [1] (2.1.2) making $X^\vee$ a dual of X , since these identities become true after application of ω' .

Corollary 4.7. Let X be an object of a Tannakian category T over k . If X admits a module structure over an extension k' of k , then $[k' : k]$ divides $\dim X$.

Proof. Let ω be a fiber functor of T over an extension K of k , put $K' := k' \otimes K$, and regard $\omega(X)$, with its k' -module structure deduced from that of X , as a K' -module. If X is regarded as an object of the Tannakian category $T_{k'}$, this K' -module is the image of X under the fiber functor ω' of $T_{k'}$ deduced from ω . As the image of the object X of the Tannakian category $T_{k'}$ under the fiber functor ω' , it is a locally free K' -module of constant rank, equal to the dimension of X viewed as an object of $T_{k'}$. Thus

$$\begin{aligned} \dim(X) &:= \dim_K \omega(X) \\ &= [K' : K] \cdot \text{rank}_{K'} \omega(X) \\ &= [k' : k] \cdot \dim(X \text{ viewed as an object of } T_{k'}). \end{aligned}$$

4.8. Warning. If D is a division algebra over k and a nonzero object X admits a D -module structure, one does not necessarily have $\dim X \geq [D : k]$. Example: take T to be the category of $\mathbb{Z}/2$ -graded complex vector spaces V endowed with a homogeneous antilinear automorphism j whose square is 1 on V^0 and -1 on V^1 . Tensor product is taken over $\mathbb{C}$. This category is Tannakian over $\mathbb{R}$, and over $\mathbb{C}$ admits the fiber functor “underlying complex vector space.” An odd object identifies with a vector space over the quaternion field $\mathbb{H}$. A simple odd object S has dimension 2 and $\text{End}(S)$ is isomorphic to $\mathbb{H}$.

Corollary 4.9. If X is a simple object of a Tannakian category T over k of characteristic p and $\dim X < p$, then the center of $\text{End}(X)$ is a separable extension of k .

Proof. By 4.7 it is an extension of degree $< p$.

4.10. Let k_1 be an algebraic extension of k . If T is a k -linear abelian category, the categories $T_{k'}$, for k' a finite extension of k in k_1 , form a 2-inductive system of abelian categories. I write “2-inductive” rather than “inductive,” because the scalar-extension functors

$$e_{k''/k'} : T_{k'} \longrightarrow (T_{k'})_{k''} = T_{k''}$$

for $k' \subset k''$ satisfy

$$e_{k'''/k''} e_{k''/k'} = e_{k'''/k'}$$

only up to a canonical isomorphism; these isomorphisms satisfy a cocycle condition for $k' \subset k'' \subset k''' \subset k''''$. Define T_{k_1} to be the 2-inductive limit of the $T_{k'}$, for k' a finite extension of k in k_1 . Since the functors $e_{k''/k'}$ are exact, the category T_{k_1} is again abelian.

If T is Tannakian, with fiber functor ω over K , the categories $T_{k'}$ for $k' \subset k_1$ are Tannakian, and ω provides a fiber functor of $T_{k'}$ over $k' \otimes K$. Passing to the inductive limit, one sees that T_{k_1} is Tannakian, that ω provides a fiber functor ω_1 of T_{k_1} over $k_1 \otimes K$, and that the $e_{k'/k}$ provide a scalar-extension functor

$$e_{k_1/k} : T \longrightarrow T_{k_1},$$

compatible with tensor product.

Corollary 4.11. Let X be an object of T .

- (i) If the object $e_{k_1/k}(X)$ of T_{k_1} is semisimple, then X is semisimple.
- (ii) If $\dim(X) < p$ and X is semisimple, then $e_{k_1/k}(X)$ is semisimple.

Proof. For k' a finite extension of k in k_1 , the length of $e_{k'/k}(X)$ increases with k' and is bounded by $\dim(X)$. There exists therefore a finite extension k' such that the length of $e_{k''/k}(X)$ is constant for $k'' \supset k'$.

Prove (i) for X . By 4.2(iii), it is enough to show that for a suitable finite extension k'' of k' in k_1 , $e_{k''/k'}(X)$ is semisimple. Let

$$0 = X_0 \subset X_1 \subset X_2 \subset \cdots \subset X_n = e_{k'/k}(X)$$

be a filtration of $e_{k'/k}(X)$ with simple successive quotients S_i . The extension of S_i by X_{i-1} defining X_i becomes split over k_1 . There is therefore a finite extension k'' of k' in k_1 over which these extensions split. The S_i remain simple because

$$\lg e_{k''/k}(X) = \lg e_{k'/k}(X).$$

It follows that $e_{k''/k'}(X)$ is semisimple.

For (ii), let X be semisimple with $\dim(X) < p$. It is enough to treat X simple. By 4.9, the center of $\text{End}(X)$ is separable over k ; hence, by 4.2(i), all finite scalar extensions occurring in the construction preserve semisimplicity, and so $e_{k_1/k}(X)$ is semisimple.

4.12. Proof of 0.1. Let T be a Tannakian category over k of characteristic p , and take for k_1 an algebraic closure $\bar{k}$ of k . Corollary 4.11 reduces 0.1 for T to 0.1 for $T_{\bar{k}}$, proved in 3.3.

6 Tensor products of exterior powers

The results that follow were suggested by the referee.

Theorem 0.7 makes it possible to generalize to objects of a Tannakian category Serre's theorem [5] on the semisimplicity of tensor products of exterior powers of representations.

Theorem 5.1. Let T be a Tannakian category over a field k of characteristic $p > 0$, let $(V_i)_{i \in I}$ be a finite family of semisimple objects of T , and let $(m_i)_{i \in I}$ be a family of integers ≥ 0 . If

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.1.1)$$

then the tensor product of the $\bigwedge^{m_i} V_i$ is semisimple.

Recall that the m -th exterior power of an object V of T is the image of the antisymmetrization

$$a : \bigotimes \text{olimits}^m V \longrightarrow \bigotimes \text{olimits}^m V.$$

For every fiber functor ω , one has

$$\omega(\bigwedge \text{olimits}^m V) = \bigwedge \text{olimits}^m \omega(V),$$

and the dual of $\bigwedge^m V$ is $\bigwedge^m (V^\vee)$.

We shall reduce the proof of 5.1 to the special case in which, for each i , $\dim(V_i) < p$. The arguments of 3.3, 4.11 and 4.12 then reduce further to supposing that k is algebraically closed, that T is the category of representations of an affine group scheme G over k , and that the family of representations (V_i) is faithful, i.e. that G is a subgroup scheme of $\prod \text{GL}(V_i)$, and hence of $\text{GL}(V)$, for $V := \bigoplus V_i$.

After a few preliminaries (5.3 to 5.10), we shall reduce in 5.11 to the case in which G is doubly saturated in $\text{GL}(V)$, and treat that case in 5.12.

5.2. Reduction to the case $\dim(V_i) < p$. We successively reduce to supposing that:

- (1) $m_i \leq \dim V_i$; otherwise $\bigwedge^{m_i} V_i = 0$.
- (2) $0 < m_i < \dim V_i$; otherwise $\bigwedge^{m_i} V_i$ has dimension one and the factor $\bigwedge^{m_i} V_i$ can be omitted. This changes neither hypothesis (5.1.1) nor the conclusion, since if L has dimension one, semisimplicity of X is equivalent to that of $X \otimes L$, the lattices of subobjects of X and $X \otimes L$ being isomorphic.
- (3) $m_i \leq (\dim V_i)/2$; otherwise replace $\bigwedge^{m_i} V_i$ by $\bigwedge^{\dim V_i - m_i} V_i^\vee$, which differs from it only by tensoring with the rank-one object $\bigwedge^{\dim V_i} V_i$.
- (4) $\sum m_i \geq 2$; otherwise the tensor product is 1 (if $\sum m_i = 0$) or is reduced to a V_i (if $\sum m_i = 1$).

If these conditions are satisfied, then $\dim V_i < p$. Indeed, if $|I| \geq 2$, (5.1.1) implies

$$\dim V_i - 1 = 1(\dim V_i - 1) \leq m_i(\dim V_i - m_i) < p - 1,$$

and if $|I| = 1$, then $m_i \geq 2$, $\dim V_i \geq 4$, and

$$\dim V_i \leq 2(\dim V_i - 2) \leq m_i(\dim V_i - m_i) < p.$$

5.3. Let V be a representation of $\mathrm{SL}(2)$. The action of the multiplicative group of diagonal matrices (a, a^{-1}) defines a grading

$$V = \bigoplus V^j$$

of V . The action of

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SL}(2)$$

exchanges V^j and V^{-j} . Hence $\dim V^j = \dim V^{-j}$. Let E be the action of the element

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

of the Lie algebra. This endomorphism of V has degree 2. If the weights of V are $< v$, i.e. if $V^j = 0$ for $|j| \geq v$, then $E^v = 0$.

If the weights of V are $< p$, one knows that the representation V is semisimple, a sum of Sym^i of the fundamental representation with $i < p$, and that the j -th iterate E^j of E induces an isomorphism

$$E^j : V^{-j} \xrightarrow{\sim} V^j. \quad (5.3.1)$$

Conversely, if V is a graded vector space, with $V^j = 0$ for $|j| \geq p$, and if E is an endomorphism of degree 2 of V satisfying (5.3.1), then there exists a unique action of $\mathrm{SL}(2)$ on V defining this grading and this endomorphism; one has $E^p = 0$, and the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

in $\mathrm{SL}(2)$ is given by the truncated exponential

$$\exp(tE) := \sum_{n < p} \frac{(tE)^n}{n!}. \quad (5.3.2)$$

5.4. Let E be a nilpotent endomorphism of a vector space V of dimension $\leq p$. Decomposing E into Jordan blocks, one checks that E comes from a representation of $\mathrm{SL}(2)$ as in 5.3, with weights $< p$. If E has only one Jordan block, the dominant weight is $\dim(V) - 1$.

Lemma 5.5. With the hypotheses and notation of 5.4, the weights of the representation $\bigwedge^m V$ of $\mathrm{SL}(2)$ are $\leq m(\dim V - m)$.

Lifting the representation V of $\mathrm{SL}(2)$ to characteristic 0, one reduces to checking in characteristic 0 that, for a representation W of $\mathrm{SL}(2)$ of dimension d , one has

$$\text{the weights of the representation } \rho_m \text{ of } \mathrm{SL}(2) \text{ on } \bigwedge^m W \text{ are } \leq m(d - m). \quad (5.5.1)$$

Since, in characteristic 0, every representation of $\mathrm{SL}(2)$ is a direct sum of Sym^j of the fundamental representation, (5.5.1) is equivalent to

$$(d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix})^{m(d-m)+1} = 0. \quad (5.5.2)$$

Let E be the nilpotent endomorphism of W giving the action of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$. If one identifies endomorphisms with actions of the one-dimensional Lie algebra, then

$$d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

is the natural extension E_m of E to $\bigwedge^m W$. Nilpotent endomorphisms with a single Jordan block are dense among nilpotent endomorphisms. Hence it is enough to check that

$$(E_m)^{m(d-m)+1} = 0$$

when E has a single Jordan block. In that case the weights of the representation W are

$$(d-1), (d-3), \dots, -(d-1),$$

each with multiplicity one. The largest weight of the representation $\bigwedge^m V$ is therefore

$$(d-1) + (d-3) + \dots + (d-(2m-1)) = md - m^2 = m(d-m),$$

and (5.5.2) follows.

5.6. Return to characteristic p . Let N be a nilpotent endomorphism of a vector space V of dimension $d \leq p$, and let N_m be its extension, in the sense of Lie algebras as in 5.5, to $\bigwedge^m V$. Since $d \leq p$, one has $N^p = 0$ and N defines an action

$$\rho(t) = \exp(tN)$$

of $\mathbb{G}_a$ on V . This action extends to an action of $\mathbb{G}_a$ on $\bigwedge^m V$.

Proposition 5.7. With the notation of 5.6, one has

$$(N_m)^{m(d-m)+1} = 0, \tag{5.7.1}$$

and, if $m(d-m) < p$, the action of $\mathbb{G}_a$ on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$.

Proof. As in 5.4, there exists an action ρ of $\mathrm{SL}(2)$ on V , with weights $< p$, such that

$$d\rho \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N.$$

It extends to an action ρ_m of $\mathrm{SL}(2)$ on $\bigwedge^m V$. By 5.5, $\bigwedge^m V$ has weights $\leq m(d-m)$. The vanishing (5.7.1) follows.

If $m(d-m) < p$, then by 5.4 the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}(2)$$

on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$.

5.8. Let (V_i) be a finite family of vector spaces of dimension $< p$, and let (m_i) be a family of integers ≥ 0 . For each i , let N_i be a nilpotent endomorphism of V_i . The truncated exponential $\exp(tN_i)$ is an action of $\mathbb{G}_a$ on V_i . These actions define an action of $\mathbb{G}_a$ on

$$\bigotimes_i \bigwedge^{m_i} V_i$$

and, by passage to the Lie algebra, an endomorphism N of this tensor product. It is deduced from the N_i , viewed as actions on the V_i of the one-dimensional Lie algebra.

Corollary 5.9. With the notation of 5.8, if

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.9.1)$$

then $N^p = 0$ and the action of $\mathbb{G}_a$ on $\bigotimes_i \bigwedge^{m_i} V_i$ is given by the truncated exponential $\exp(tN)$.

Proof. As in 5.4, choose an action ρ_i of $\mathrm{SL}(2)$ on V_i , with weights $< p$, such that

$$d\rho_i \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N_i.$$

By 5.5, the weights of $\mathrm{SL}(2)$ acting on $\bigotimes_i \bigwedge^{m_i} V_i$ are

$$\leq \sum_i m_i(\dim V_i - m_i) < p,$$

and one applies 5.4 as in 5.7. One could equally well apply 5.7 and 2.2.

5.10. The truncated exponential induces a bijection between nilpotent endomorphisms N with $N^p = 0$ and automorphisms g with $g^p = 1$, and the corresponding actions of $\mathbb{G}_a$ coincide:

$$\exp(N)^t = \exp(tN).$$

With the notation of 5.8, if the automorphism g_i of V_i satisfies $g_i^p = 1$, and if g is the automorphism of $\bigotimes_i \bigwedge^{m_i} V_i$ induced by the g_i , then condition (5.9.1) ensures that g^t is induced by the g_i^t .

Recall that, by the remarks following 5.1, 5.2 reduces the proof of 5.1 to the case in which k is algebraically closed and T is the category of representations of

$$G \subset \prod \mathrm{GL}(V_i) \subset \mathrm{GL}(V),$$

with $V := \bigoplus_i V_i$.

5.11. Reduction to the case in which G is doubly saturated in $\mathrm{GL}(V)$. Let G^* be the double saturation of G (2.1). As in 2.4, 5.9 and 5.10 ensure that, if (5.1.1) is satisfied, a subspace of $\bigotimes_i \bigwedge^{m_i} V_i$ stable under G is automatically stable under G^* . Semisimplicity as a representation of G^* therefore implies semisimplicity as a representation of G .

5.12. End of the proof of 5.1. As in 3.1, one reduces to supposing G reduced. One can then apply [5], p. 26, consequence of the “Main Theorem” on p. 20, to the representations V_i of $G(k)$. One could instead invoke [6].

5.13. In 5.1, we use the Schur functors

$$V_i \mapsto \bigwedge^{m_i} V_i,$$

associated with the representations of $\mathrm{GL}(\dim V_i)$ of dominant weight

$$(1, \dots, 1, 0, \dots, 0)$$

(m_i entries equal to 1). More generally one can consider Schur functors $S_{\lambda(i)}$, where $\lambda(i)$ is a dominant weight

$$(\lambda(i)_1, \dots, \lambda(i)_{\dim V_i})$$

of $\mathrm{GL}(\dim V_i)$. Regard $\lambda(i)$ as a partition. Let $\mu(i)$ be its dual and put

$$n(\lambda(i)) = \sum_j \mu(i)_j (\dim V_i - \mu(i)_j)$$

(cf. [6] 5.2). Condition (5.5.1) is to be replaced by

$$\sum_i n(\lambda(i)) < p. \quad (5.13.1)$$

That (5.13.1) and the semisimplicity of the V_i suffice to ensure the semisimplicity of

$$\bigotimes_i S_{\lambda(i)} V_i$$

can be deduced from 5.1 and from the fact that $S_{\lambda(i)}$ is a subquotient, and even, under the hypotheses made, a direct factor, of the functor

$$V \mapsto \bigotimes_j \bigwedge^{\lambda(i)_j} V$$

for V of dimension $\dim(V_i)$.

Acknowledgments. I thank J.-P. Serre, M. Raynaud, G. Prasad, and H. Esnault for their comments, which I hope have allowed me to improve the text, and the referee for a careful reading and for the suggestions that gave rise to Section 5.

References

- [1] P. Deligne, *Catégories tannakiennes*, in *The Grothendieck Festschrift*, vol. 2, pp. 111–194. Progress in Math. 87, Birkhäuser, 1990.
- [2] P. Deligne and J. S. Milne, *Tannakian categories*, in *Hodge Cycles, Motives and Shimura Varieties*, by P. Deligne, J. S. Milne, A. Ogus and K. Shih, pp. 101–228. Springer Lecture Notes in Mathematics 900.
- [3] N. Saavedra, *Catégories Tannakiennes*, Springer Lecture Notes in Mathematics 265.
- [4] J.-P. Serre, *Sur la semi-simplicité des produits tensoriels de représentations de groupes*, Invent. Math. 116 (1994), pp. 513–530.
- [5] J.-P. Serre, *Moursund Lectures 1998*, notes by W. E. Duckworth, available on arXiv: math/0305257.
- [6] J.-P. Serre, *Complète réductibilité*, Sémin. Bourbaki 2003–2004, exp. 932, in Astérisque 299 (Soc. Math. France, 2005).

SGA: Séminaire de Géométrie Algébrique du Bois-Marie.

SGA1: Séminaire 1960/61, directed by A. Grothendieck: *Revêtements étales et groupe fondamental*. Springer Lecture Notes in Mathematics 224.

SGA3: Séminaire 1962/64, directed by M. Demazure and A. Grothendieck: *Schémas en Groupes*. The exposé XVII of Michel Raynaud is in volume II, Springer Lecture Notes in Mathematics 152.

SGA4: Séminaire 1963/64, directed by M. Artin, A. Grothendieck and J. L. Verdier: *Théorie des topos et cohomologie étale des schémas*. The exposé XV of M. Artin is in volume III, Springer Lecture Notes in Mathematics 305.

Counting ℓ -adic Sheaves

Pierre Deligne
English working translation

To Gerard Laumon, on the occasion of his sixtieth birthday.

1 Introduction

1.1 1.1

Let X_0 be a smooth, projective, absolutely connected curve of genus g over a finite field $\mathbf{F}_q$ of characteristic p , and let X be obtained from it by extension of scalars to an algebraic closure $\mathbf{F}$ of $\mathbf{F}_q$:

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbf{F}) & \longrightarrow & \mathrm{Spec}(\mathbf{F}_q). \end{array} \quad (1.1.1)$$

The endomorphism of X_0 which is the identity on the underlying topological space and sends $f \mapsto f^q$ on the structure sheaf is an endomorphism of the $\mathbf{F}_q$ -scheme. We write Frob for the endomorphism of the $\mathbf{F}$ -scheme X obtained from it by extension of scalars. This is the Frobenius endomorphism of X . Fix a prime number $\ell \neq p$, an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$, and let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X . Inverse image by Frob induces a permutation of E . We denote it by V .

1.2 1.2

In the article [Dr], which remains as mysterious to me as it was thirty-one years ago, Drinfeld computes the number of fixed points of $V : E \rightarrow E$. A lisse $\overline{\mathbf{Q}}_\ell$ -sheaf L_0 of rank one on $\mathrm{Spec}(\mathbf{F}_q)$ is determined up to isomorphism by the unit $\lambda \in \overline{\mathbf{Q}}_\ell^*$ by which the geometric Frobenius $\mathrm{Fr} \in \mathrm{Gal}(\mathbf{F}/\mathbf{F}_q)$ acts on the fibre of L_0 at the geometric point $\mathrm{Spec}(\mathbf{F})$. The $\mathbf{F}_q$ -twist of F_0 on X_0 by L_0 is the tensor product with the inverse image of L_0 on X_0 . By abuse of language, we shall also say “ $\mathbf{F}_q$ -twist by λ .”

Drinfeld uses the fact that the isomorphism class of a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F on X is fixed by Frob^* if and only if F is the inverse image of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 , and that, if F is irreducible, F_0 is unique up to $\mathbf{F}_q$ -twist. The problem solved in [Dr] is thus to count lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 , taken up to $\mathbf{F}_q$ -twist, considering only those which are irreducible and remain so after inverse image on X .

In 1981 one almost had the correspondence between irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 and everywhere unramified cuspidal automorphic representations for $\mathrm{GL}(2, k(X_0))$. Thanks to this, Drinfeld reduced the problem to an application of the trace formula for $\mathrm{GL}(2)$. This reduction shows that the desired number is independent of ℓ . The formula obtained by Drinfeld has the following miraculous properties (A) and (B).

For $n \geq 1$, let N_n be the number of fixed points of the n -th iterate $V^n : E \rightarrow E$ of V . Computing N_n is the above problem with $X_0/\mathbf{F}_q$ replaced by $(X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n})/\mathbf{F}_{q^n}$.

(A) The function $n \mapsto N_n$ has the form

$$N_n = \sum a_i \beta_i^n, \quad (1.2.1)$$

for integers a_i and suitable Weil numbers β_i . The β_i are monomials in q and in the eigenvalues of the endomorphism Frob^* of $H^1(X, \overline{\mathbf{Q}}_\ell)$. Once the genus g is fixed, which monomials β_i occur, and with what multiplicities a_i , is independent of the curve under consideration.

The trace formula expresses N_n as the sum of several terms. Taken separately, these terms do not all have the form (1.2.1).

Let P be a compact Riemann surface of genus g . Consider complex local systems of vector spaces on P . Let M be the moduli space of those which are irreducible of rank 2. The space M and the set E are empty if $g = 0$ or 1 . Otherwise M is a connected complex symplectic variety. Its complex dimension is therefore an even integer $2N$.

(B) In (1.2.1), the dominant term is q^N : one of the β_i is q^N , its multiplicity a_i is 1, and the other β_i satisfy $|\beta_i| < q^N$.

1.3 1.3

Formula (1.2.1) is reminiscent of a Lefschetz trace formula in which, in good cases, the number of fixed points of the iterates T^n ($n \geq 1$) of an endomorphism T of a space S is

$$\text{Tr}(T^{*n}, H_{\mathbb{Q}}^*(S)) := \sum (-1)^i \text{Tr}(T^{*n}, H_{\mathbb{Q}}^i(S)). \quad (1.3.1)$$

The question mark is there to recall that one must consider a cohomology with support conditions. The support conditions to be imposed depend on the behaviour of T at infinity. For example, if for an exhaustion function f on S one has $f(T(x)) > f(x)$ (respectively $f(T(x)) < f(x)$) for $f(x)$ sufficiently large, the cohomology to consider is compactly supported cohomology (respectively ordinary cohomology).

Assume that $H_{\mathbb{Q}}^*$ is finite-dimensional so that (1.3.1) makes sense. The right hand side of (1.3.1) is then of the form (1.2.1): it is $\sum a(\beta)\beta^n$, where β ranges over the eigenvalues of T^* and the integer $a(\beta)$ is the alternating sum of the multiplicities of β as an eigenvalue of the endomorphisms T^* on the $H_{\mathbb{Q}}^i(S)$.

Unfortunately I have no idea how to regard the set E of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X as a “space” having a cohomology $H_{\mathbb{Q}}^*(E)$ for which one could hope that the number of fixed points of V^n is given by a formula of type (1.3.1).

1.4 1.4

Even though one does not know how to think geometrically of E , if $f \in E$ is the isomorphism class of F , one knows what the formal completion $\widehat{E}_f$ of E at f should be. A finite-dimensional local $\overline{\mathbf{Q}}_\ell$ -algebra Λ is augmented over $\overline{\mathbf{Q}}_\ell$. A deformation of F over $\text{Spec}(\Lambda)$ is a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F_Λ on X , endowed with a Λ -module structure and an isomorphism

$$F_\Lambda \otimes_\Lambda \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} F,$$

such that at one point x (and therefore at every point x of X), $(F_\Lambda)_x$ is a free Λ -module.

The functor on $\text{Spec}(\Lambda)$ of isomorphism classes of deformations of F over $\text{Spec}(\Lambda)$ is pro-representable. Let R_F be the complete local $\overline{\mathbf{Q}}_\ell$ -algebra with residue field $\overline{\mathbf{Q}}_\ell$ such that $\text{Spf}(R_F)$ pro-represents it. There are no obstructions to deformations, and R_F is therefore isomorphic to a formal power series algebra $\overline{\mathbf{Q}}_\ell[[t_1, \dots, t_m]]$.

Because F is irreducible, its only automorphisms are multiplications by $\lambda \in \overline{\mathbf{Q}}_\ell^*$. They extend to every deformation. If F and F' are two representatives of the isomorphism class f , two isomorphisms from F to F' therefore induce the same isomorphism between R_F and $R_{F'}$. Thus $\widehat{E}_f := \text{Spf}(R_F)$ depends, up to unique isomorphism, only on f .

The Zariski tangent space of $\widehat{E}_f$ at f is $H^1(X, \text{End}(F))$. The symmetric bilinear form $\text{Tr}(fg)$ on $\text{End}(F)$ is a self-duality. It induces on $H^1(X, \text{End}(F))$ a symplectic form with values

in $H^2(X, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell(-1)$. The same construction over $\mathbf{C}$ gives the symplectic structure of M . The formal completion $\widehat{E}_f$ is therefore smooth of even dimension, the same as that of M .

The inverse-image functor by $\text{Frob} : X \rightarrow X$ transforms deformations of F into deformations of $\text{Frob}^* F$. It therefore induces a morphism

$$\widehat{V}_f : \widehat{E}_f \longrightarrow \widehat{E}_{V(f)}.$$

The argument showing that, because Frob induces an equivalence of étale sites, V is bijective, also shows that the $\widehat{V}_f$ are isomorphisms. Another proof is as follows. Let ω be the symplectic form defined above on the tangent space at f . If $[t] \in H^1(X, \text{End}(F))$ is the class of the tangent vector t at F , the class of $d\widehat{V}_f(t)$ in $H^1(X, \text{End}(\text{Frob}^* F))$ is $\text{Frob}^*([t])$. For two tangent vectors t_1, t_2 , $\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2))$ is thus the inverse image, by Frob^* , of $\omega(t_1, t_2) \in \overline{\mathbf{Q}}_\ell(-1)$, identified with $H^2(X, \overline{\mathbf{Q}}_\ell)$. Since $\text{Frob} : X \rightarrow X$ has degree q , the endomorphism Frob^* of $H^2(X, \overline{\mathbf{Q}}_\ell)$ is multiplication by q , and

$$\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2)) = q \omega(t_1, t_2),$$

which is also written $(d\widehat{V}_f)^* \omega = q\omega$. The differential $d\widehat{V}_f$ is therefore an isomorphism.

The map $d\widehat{V}_f$ between Zariski tangent spaces is identified with

$$\text{Frob}^* : H^1(X, \text{End}(F)) \longrightarrow H^1(X, \text{End}(\text{Frob}^* F)).$$

If f is a fixed point of V , then F is the inverse image of F_0 on X_0 . One knows that F_0 may be chosen pure, and $\text{End}(F_0)$ is therefore pure of weight 0. The eigenvalues of $d\widehat{V}_f$ on the Zariski tangent space of $\widehat{E}_f$ at f are therefore Weil numbers of weight 1. In particular no eigenvalue of $d\widehat{V}_f$ at f is equal to 1. This is what makes it reasonable that a Lefschetz-type formula (1.3.1) might count the fixed points of V^n , all taken with multiplicity one.

1.5 1.5

If Y_0 is a scheme of finite type over $\mathbf{F}_q$, the Frobenius endomorphism Frob of $Y := Y_0 \otimes_{\mathbf{F}_q} \mathbf{F}$ induces a permutation Frob of $Y(\mathbf{F})$. The preceding discussion shows that the nature of (E, V) is different from that of (Y, Frob) :

- (i) E is “over $\overline{\mathbf{Q}}_\ell$,” hence an object of characteristic 0.
- (ii) The $\widehat{V}_f$ are isomorphisms. One would like to say that V has “degree one.”
- (iii) The fixed points of V are isolated because the eigenvalues of dV at a fixed point are Weil numbers of weight 1. The fixed points of Frob are isolated because $d\text{Frob} = 0$.
- (iv) If E has dimension $2N$, in the sense that the $\widehat{E}_f$ do, the order of magnitude of the number of fixed points of V^n is $(q^n)^N$, rather than $(q^n)^{2N}$.

1.6 1.6

The moduli space M is a complex algebraic variety. Contrary to what a hasty analogy might suggest, E is not naturally the space of $\overline{\mathbf{Q}}_\ell$ -points of an algebraic variety over $\overline{\mathbf{Q}}_\ell$. Let $x \in X(\mathbf{F})$. The set E is identified with the set of isomorphism classes of continuous irreducible representations V of $\pi_1(X, x)$, of dimension 2 over $\overline{\mathbf{Q}}_\ell$. As coordinates on E , one would like to take the $\text{Tr}(\gamma, V)$ for $\gamma \in \pi_1(X, x)$. In the complex case this construction gives M its “Betti” structure as a complex algebraic variety. Here, since $\pi_1(X, x)$ is a profinite group, every representation V admits an invariant lattice, and the $\text{Tr}(\gamma, V)$ take their values in the valuation ring $\mathbf{Z}_\ell$ of the valued field $\overline{\mathbf{Q}}_\ell$.

The point of view “ E as a space of representations of $\pi_1(X, x)$ ” also does not help one understand that the number of fixed points of V is independent of ℓ . Another mystery: why,

once the genus g is fixed, does the number of fixed points of V admit a uniform description in terms of $H^1(X, \overline{\mathbf{Q}}_\ell)$, of its symplectic structure with values in $\overline{\mathbf{Q}}_\ell(-1)$, and of its endomorphism Frob^* , whereas $\pi_1(X, x)$ is a quotient of the profinite completion of the topological π_1 of P , which quotient depends on X ?

1.7 1.7

Let E_r be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r on X . Thanks to Lafforgue [L], the number of fixed points of the bijection $\text{Frob}^* : E_r \rightarrow E_r$ has an automorphic interpretation. For $S_0 \subset X_0$ a finite set of closed points of X , and $S = S_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, one may also consider irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves on $X - S$, with prescribed ramification at each $s \in S$. In all cases that have been computed, the miracles (A) and (B) discovered by Drinfeld persist, mutatis mutandis. The aim of the talk is to give a panorama of what is known.

2 What is to be counted?

2.1 2.1

Fix an integer $r \geq 1$, and let $X_0, S_0/\mathbf{F}_q$ and $X, S/\mathbf{F}$ be as in 1.1 and 1.7. For $s \in S$, let $X_{(s)}$ be the henselization of X at s and $X_{(s)}^* := X_{(s)} - \{s\}$. Suppose given, for each $s \in S$, a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf of rank r , denoted $R(s)$, on $X_{(s)}^*$. Only its isomorphism class matters. The morphism $\text{Frob} : X \rightarrow X$ induces a morphism, again denoted Frob , from $X_{(s)}^*$ to $X_{(\text{Frob}(s))}^*$. We assume that, for every $s \in S$, $R(s)$ is isomorphic to the inverse image of $R(\text{Frob}(s))$ by Frob :

$$R(s) \simeq \text{Frob}^*(R(\text{Frob}(s))). \quad (2.1.1)$$

Let $R := (R(s))_{s \in S}$ be the family of the $R(s)$, and write $E(R)$ for the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves F of rank r on $X - S$ such that, for every $s \in S$, the inverse image $F|_{X_{(s)}^*}$ of F on $X_{(s)}^*$ is isomorphic to $R(s)$. Hypothesis (2.1.1) ensures that inverse image by Frob induces a permutation of $E(R)$. We denote this permutation by V .

Our choice of the letter V , the first letter of “Verschiebung,” is due to an analogy with the case of certain abelian varieties, explained in 2.2 below. In that analogy q is classically a prime number, rather than a power of a prime number.

2.2 2.2

For any scheme Y over $\mathbf{F}_q$, let Frob_Y be the endomorphism of the $\mathbf{F}_q$ -scheme Y which is the identity on the underlying topological space and which on the structure sheaf is given by $\text{Frob}_Y^*(f) = f^q$. This is the Frobenius endomorphism of Y .

Take for Y the abelian variety $\text{Pic}^0(X_0)$ over $\mathbf{F}_q$. We obtain

$$\text{Frob}_{\text{Pic}^0(X_0)} : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(X_0). \quad (2.2.1)$$

The Jacobian $\text{Pic}^0(X_0)$ represents the following functor on $\mathbf{F}_q$ -schemes: the sheaf (for the Zariski topology, or for the étale topology; it is the same) associated to the presheaf attaching to S the group $\text{Pic}^0(S \times_{\mathbf{F}_q} X_0)$ of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$, of degree zero on each fibre of the projection to S . If $f : Y_0 \rightarrow X_0$ is a morphism, inverse image of invertible sheaves by f defines a morphism $f^* : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(Y_0)$.

Take for f the endomorphism Frob_{X_0} of X_0 . The endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ deserves to be called V , for it satisfies

$$\text{Frob}_{X_0}^* \circ \text{Frob}_{\text{Pic}^0(X_0)} = \text{Frob}_{\text{Pic}^0(X_0)} \circ \text{Frob}_{X_0}^* = q. \quad (2.2.2)$$

Proof of (2.2.2). Interpret (2.2.2) as two equalities between endomorphisms of the functor represented by $\text{Pic}^0(X_0)$. For any contravariant functor G on the category of $\mathbf{F}_q$ -schemes, the Frobenius endomorphism Frob_G of G is defined by

$$\text{Frob}_G(S) : G(S) \rightarrow G(S), \quad \text{namely } G(\text{Frob}_S).$$

If G is represented by Y , Frob_G is induced by Frob_Y .

Take for G the functor $S \mapsto \text{Pic}(S \times_{\mathbf{F}_q} X_0)$ of isomorphism classes of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$. The endomorphism $\text{Frob}_G(S)$ of $\text{Pic}(S \times_{\mathbf{F}_q} X_0)$ is inverse image of isomorphism classes of invertible sheaves by $\text{Frob}_S \times \text{id}_{X_0} : S \times_{\mathbf{F}_q} X_0 \rightarrow S \times_{\mathbf{F}_q} X_0$. If one composes this inverse image with the inverse image by Frob_{X_0} , one obtains inverse image by $\text{Frob}_{S \times_{\mathbf{F}_q} X_0}$, which is nothing other than raising to the q -th tensor power. Restricting to invertible sheaves of degree zero on the fibres of the projection to S , and passing from the presheaf obtained to the associated sheaf, gives (2.2.2).

If now one extends scalars from $\mathbf{F}_q$ to $\mathbf{F}$, Frob_{X_0} becomes the endomorphism Frob of X , $\text{Pic}^0(X_0)$ becomes $\text{Pic}^0(X)$, and the endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ becomes the endomorphism of $\text{Pic}^0(X)$ “inverse image by Frob .”

2.3 2.3–2.6

The questions we ask are the following.

- (i) Compute, for $n \geq 1$, the number $N_n(R)$ of fixed points of the n -th iterate V^n of the permutation V of $E(R)$. Replacing (X_0, S_0) by $(X_0, S_0) \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$ replaces V by V^n .
- (ii) Determine whether the miracles 1.2 (A) and (B) persist.
- (iii) If so, why?

Let F be a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf on $X - S$. As in 1.2, one has:

Lemma 2.4. (i) For the isomorphism class of F to be fixed under Frob^* , it is necessary and sufficient that F be isomorphic to the inverse image on X of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 . Moreover, if F is irreducible,

- (ii) such an F_0 is unique up to $\mathbf{F}_q$ -twist, and
- (iii) the $\mathbf{F}_q$ -twists of F_0 by $\lambda \in \overline{\mathbf{Z}}_\ell^*$ are pairwise non-isomorphic.

According to 2.4, the number of fixed points of the permutation V of $E(R)$ is the number of classes, modulo $\mathbf{F}_q$ -twist, of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 of rank r on $X_0 - S_0$, such that

- (a) the inverse image F of F_0 on $X - S$ is irreducible;
- (b) for each $s \in S$, $F|_{X_{(s)}^*}$ is isomorphic to $R(s)$.

Let $D_0 = \sum n_i x_i$ be a divisor of degree one on $X_0 - S_0$. Regard F_0 as a Galois representation (see 2.7), let F_{x_i} be a Frobenius at x_i , and put

$$\langle D_0, F_0 \rangle := \prod \det(F_{x_i}, F_0)^{n_i}.$$

If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , then

$$\langle D_0, F'_0 \rangle = \lambda^r \langle D_0, F_0 \rangle.$$

By 2.4 (ii) and (iii), it is therefore equivalent either to count the F_0 satisfying (a) and (b) up to $\mathbf{F}_q$ -twist, or to count, up to isomorphism, only those which also satisfy $\langle D_0, F \rangle = 1$, and then divide by r . For an automorphic counterpart, see the remark following 2.6.

Put $K := k(X_0)$. For each closed point x of X_0 , let $k(x)$ be the residue field of x , $\deg(x) := [k(x) : \mathbf{F}_q]$ its degree, K_x the completion of K at x , and O_x the valuation ring of K_x . Let $\mathbf{A}$ be the adèle ring of K . The sum $v := \sum \deg(x)v_x : \mathbf{A}^* \rightarrow \mathbf{Z}$ is trivial on K^* . One has $\|a\| = q^{-v(a)}$.

By Lafforgue, F_0 corresponds to a cuspidal automorphic representation π of $\mathrm{GL}(r, \mathbf{A})$. If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , the corresponding representation π' is the $\mathbf{F}_q$ -twist of π by λ , defined as follows. As a representation, it is the tensor product of π by the one-dimensional representation $g \mapsto \lambda^{v \det g}$. As a space of functions on $\mathrm{GL}(r, \mathbf{A})$, it is the space of products of $f \in \pi$ with the character $g \mapsto \lambda^{v \det g}$.

Translated into automorphic language, problem 2.3 (i) is to count, up to $\mathbf{F}_q$ -twist, the cuspidal automorphic representations π of $\mathrm{GL}(r, \mathbf{A})$, unramified outside S_0 , and such that

- (a') for every $n \geq 1$, π remains cuspidal after base change from K to $K \otimes \mathbf{F}_{q^n}$. This condition expresses the irreducibility 2.4 (a) of F . It suffices to check it for n dividing r .
- (b') for each $s_0 \in S_0$ and $s \in S$ above s_0 , the local component π_{s_0} of π satisfies a condition dictated by $R(s)$ and by the local Langlands correspondence.

Lemma 2.6. If the cuspidal automorphic representation π satisfies (a'), then its $\mathbf{F}_q$ -twists are pairwise distinct. This is the translation of 2.4 (iii). If d is an idele satisfying $v(d) = 1$, it is therefore equivalent to count up to $\mathbf{F}_q$ -twist the π satisfying (a') and (b'), or to count only those whose central character ω_π satisfies $\omega_\pi(d) = 1$, and then divide by r .

In finite characteristic the notion of automorphic representation is purely algebraic. This makes it possible to take the automorphic representations π to have coefficients in $\overline{\mathbf{Q}}_\ell$, rather than in $\mathbf{C}$. If one does this, π_{s_0} is an irreducible admissible $\overline{\mathbf{Q}}_\ell$ -representation of $\mathrm{GL}(r, K_{s_0})$, and the local Langlands correspondence associates to it an isomorphism class of continuous F -semisimple $\overline{\mathbf{Q}}_\ell$ -representations of dimension r of the local Weil group $W(\overline{K}_{s_0}/K_{s_0})$. Condition (b') is that its restriction to the inertia subgroup be given by $R(s)$.

Almost all known results on question 2.3 (i) are obtained through the dictionary 2.5.

2.4 2.7–2.16

In 2.1 we used the language of $\overline{\mathbf{Q}}_\ell$ -sheaves. This saved us from having to choose base points. An equivalent language is that of Galois representations. Here is the translation. Let $\overline{k(X)}$ be an algebraic closure of the field of rational functions $k(X)$ of X , and let $k(X)^{\mathrm{nr}}$ be the largest subextension unramified outside S . The Galois group $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$ is the fundamental group of X at the geometric point $\bar{\eta} := \mathrm{Spec}(k(X))$ of X : the fibre functor at $\bar{\eta}$ is an equivalence from lisse $\overline{\mathbf{Q}}_\ell$ -sheaves (respectively irreducible ones of rank r) on $X - S$ to linear $\overline{\mathbf{Q}}_\ell$ -representations of $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$ (respectively irreducible representations of rank r).

The scheme $X_{(s)}^*$ is the spectrum of an extension $k(X_{(s)}^*)$ of $k(X)$. If one chooses an embedding of this extension into $\overline{k(X)}$, the corresponding inertia group $I_s := \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*))$ is likewise the fundamental group of $X_{(s)}^*$ at $\bar{\eta}$. It maps to $\pi_1(X, \bar{\eta})$:

$$\pi_1(X_{(s)}^*, \bar{\eta}) = I_s = \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*)) \rightarrow \mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X)) = \pi_1(X, \bar{\eta}). \quad (2.7.1)$$

The fibre at $\bar{\eta}$ of $R(s)$ is a representation ρ_s of I_s . The condition that $F|_{X_{(s)}^*}$ be isomorphic to $R(s)$ is equivalent to the restriction to I_s being isomorphic to ρ_s . Condition (2.1.1) is equivalent to saying that the ρ_s , for s over s_0 , come from a representation of a decomposition group $D_{s_0} \subset \mathrm{Gal}(k(X)/k(X_0))$ at s_0 . It implies that the representations ρ_s are quasi-unipotent.

I hope problem 2.3 (i) is more manageable when a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F such that $F|_{X_{(s)}^*} \simeq R(s)$ for every $s \in S$ is automatically irreducible. This is the case if the following condition holds. Consider an integer $0 < r' < r$ and, for each $s \in S$, a lisse sub- $\overline{\mathbf{Q}}_\ell$ -sheaf $R'(s)$ of rank r' of $R(s)$. Put $R''(s) := R(s)/R'(s)$. The condition is:

For all r' and all $R'(s)$, either there exists no lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F' of rank r' on $X - S$ with $F'|_{X_{(s)}^*} \simeq R'(s)$ for all s , or there exists no lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F'' of rank $r - r'$ on $X - S$ with $F''|_{X_{(s)}^*} \simeq R''(s)$ for all s . (2.8.1)

Condition (2.8.1) is satisfied if one of the $R(s)$ is irreducible. For other cases, see 2.10.4, 2.11, and 2.12.

For any field k of characteristic p or 0, containing all roots of unity of order prime to p , set

$$\widehat{\mathbf{Z}}_{p'}(1)(k) := \varprojlim \mu_n(k). \quad (2.9.1)$$

The projective limit is taken over the set of subgroups $\mu_n(k) \subset k^*$ (n prime to p), ordered by inclusion. The transition morphism from $\mu_m(k)$ to $\mu_n(k)$ is $x \mapsto x^{m/n}$. If no ambiguity is created, we omit “ p' ” and the mention of k .

For $k = \mathbf{F}$, and $\mathbf{F}_{q^k}$ the subfield of $\mathbf{F}$ with q^k elements, the subgroups $\mathbf{F}_{q^k}^*$ of $\mathbf{F}^*$ are cofinal among the $\mu_n(\mathbf{F})$. Since $(q^{kd} - 1)/(q^k - 1) = 1 + q^k + \dots + q^{k(d-1)}$, the transition morphism from $\mu_{q^{kd-1}} = \mathbf{F}_{q^{kd}}^*$ to $\mu_{q^{k-1}} = \mathbf{F}_{q^k}^*$ is the norm morphism, and

$$\widehat{\mathbf{Z}}(1) \xrightarrow{\sim} \varprojlim \mathbf{F}_{q^k}^* \quad (2.9.2)$$

(the projective limit being taken with respect to the norm morphisms between the $\mathbf{F}_{q^k}^*$). For k a product $\prod k_\alpha$ of finite extensions of $\mathbf{F}_q$ and $\iota : k \rightarrow k_\alpha \xrightarrow{\sim} \mathbf{F}_{q^k} \hookrightarrow \mathbf{F}$, write pr_ι for the composite

$$\text{pr}_\iota : \widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_{q^k}^* \xleftarrow{\sim} k_\alpha^* \hookrightarrow \bigoplus k_\alpha^* = k^*. \quad (2.9.3)$$

For $k = \mathbf{F}_{q^k}$ and ι the identical inclusion, $\text{Gal}(\mathbf{F}/\mathbf{F}_{q^k})$ acts on $\widehat{\mathbf{Z}}(1)$, and (2.9.3) induces an isomorphism of coinvariants with $\mathbf{F}_{q^k}^*$.

The local fundamental group $\pi_1(X_{(s)}^*, \bar{\eta}) = I_s$ is an extension of $\widehat{\mathbf{Z}}(1)$ by a pro- p -group P_s . The quotient $I_s/P_s \simeq \widehat{\mathbf{Z}}(1)$ is the tame fundamental group of $X_{(s)}^*$. The morphism $\text{Frob} : X_{(s)}^* \rightarrow X_{(\text{Frob}(s))}^*$ induces a morphism between tame fundamental groups. Via the identifications of these groups with $\widehat{\mathbf{Z}}(1)$, this morphism is multiplication by q .

Let $\mathbf{Z}_\ell(1)$ denote the quotient $\varprojlim \mu_{\ell^n}(\mathbf{F})$ of $\widehat{\mathbf{Z}}(1)$. Every quasi-unipotent $\overline{\mathbf{Q}}_\ell$ -representation ρ of $\widehat{\mathbf{Z}}(1)$ is isomorphic to a sum over finite-order characters $\chi : \widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$:

$$\rho \simeq \bigoplus_\chi \chi \otimes \nu_\rho(\chi), \quad (2.9.4)$$

where $\nu_\rho(\chi)$ is a unipotent representation of the quotient $\mathbf{Z}_\ell(1)$ of $\widehat{\mathbf{Z}}(1)$. Let $\lambda_\rho(\chi)$ be the following partition of the integer $\dim \nu_\rho(\chi)$: the decreasing sequence of sizes of the Jordan blocks of the image of a generator of $\mathbf{Z}_\ell(1)$ in the representation $\nu_\rho(\chi)$. The isomorphism class of ρ is determined by the function $\chi \mapsto \lambda_\rho(\chi)$.

Assume that the $R(s)$ are tame, i.e. that the corresponding representations ρ_s of the inertia groups I_s (2.7) factor through a representation of $\widehat{\mathbf{Z}}(1)$. We write $\chi \mapsto \lambda_s(\chi)$ for the function, constructed above, attached to this representation. With this notation, condition (2.1.1) is equivalent to

$$\lambda_{\text{Frob}(s)}(\chi) = \lambda_s(\chi^q). \quad (2.9.5)$$

If L is a $\overline{\mathbf{Q}}_\ell$ -sheaf of rank one on $X - S$, the restriction of L to $X_{(s)}^*$ defines a character l_s of the fundamental group I_s of $X_{(s)}^*$. Since $\ell \neq p$, the restriction of l_s to P_s is of finite order: there is a power P of p such that the l_s^P factor through characters, again denoted l_s^P , of $\widehat{\mathbf{Z}}(1) = I_s/P_s$. One has

$$\prod l_s^P = 1. \quad (2.10.1)$$

The analogue over $\mathbf{C}$ of (2.10.1) is that, for P as in 1.2 and S a finite subset of P , the sum over S of a small positive circle around each $s \in S$ is a boundary in $P - S$.

In rank r , (2.10.1) applied to $\bigwedge^r F$ provides a necessary condition for the existence of F on $X - S$ satisfying $F|_{X_{(s)}^*} \simeq R(s)$, and interesting cases in which the hypothesis of the irreducibility criterion (2.8.1) is satisfied. Let ρ_s be the representation of I_s corresponding to $R(s)$. Assume, as in 2.9, that the representations ρ_s are tame. Let $\mathcal{R}(s)$ be the multiset of characters of $\widehat{\mathbf{Z}}(1)$ forming the semisimplification of ρ_s . With the notation of 2.9, the multiplicity in $\mathcal{R}(s)$ of a character χ is $|\lambda_s(\chi)| = \dim \nu_{\rho_s}(\chi)$. Assume that

$$\prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (2.10.2)$$

By (2.10.1), this is a necessary condition for the existence of an F on $X - S$ such that $F|_{X_{(s)}^*} \simeq R_s$ for all $s \in S$. Consider the condition:

$$\text{For every } 0 < r' < r \text{ and every family } (\mathcal{R}'(s))_{s \in S} \text{ of sub-multisets with } r' \text{ elements} \\ \text{of the multisets } \mathcal{R}(s), \text{ one has } \prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}'(s)} \varepsilon \neq 1. \quad (2.10.3)$$

Lemma 2.11. Condition (2.10.3) implies (2.8.1), and hence the irreducibility of every lisse sheaf F of rank r on $X - S$ satisfying $F|_{X_{(s)}^*} \simeq R_s$ for all $s \in S$.

Remark 2.12. There exist $R(s)$ as above, giving rise to multisets $\mathcal{R}(s)$ satisfying (2.10.2), for which (2.10.3) is false but where condition (2.8.1) is nevertheless satisfied. Here is an example. Take $X = \mathbf{P}^1$, $r = 4$, $|S| = 3$, and semisimple $R(s)$, sums of multisets of characters $\mathcal{R}(s)$ of the following type

$$\{a_1, a_1, a_3, a_4\}, \quad \{b_1, b_1, b_3, b_4\}, \quad \{c_1, c_2, c_3, c_4\}.$$

Assume that the product of all these characters is 1 (2.10.2), and that $a_4 b_4 c_4 = 1$, so that (2.10.3) is false. If the a_i, b_i, c_i are “general,” condition (2.8.1) is satisfied, because there is no F of rank 3 with local monodromy given by

$$\{a_1, a_1, a_3\}, \quad \{b_1, b_1, b_3\}, \quad \{c_1, c_2, c_3\}.$$

The verification is by reduction to the complex case, by lifting (X, S) to characteristic 0. This reduction brings us to the following lemma.

Lemma 2.13. Let $u_1, u_3, v_1, v_3, w_1, w_2, w_3 \in \overline{\mathbf{Q}}_\ell^*$. Assume that $u_1 v_1 w_i \neq 1$ for $i = 1, 2, 3$. Under this hypothesis there do not exist elements U, V, W of $\text{GL}(3, \overline{\mathbf{Q}}_\ell)$, conjugate respectively to the diagonal matrices (u_1, u_1, u_3) , (v_1, v_1, v_3) , (w_1, w_2, w_3) , with $UVW = 1$.

Proof. Let L be the intersection of the kernels of $U - u_1$ and $V - v_1$. These kernels have dimension at least 2, so L is nonzero. On L , U is multiplication by u_1 , V is multiplication by v_1 , and, since $UVW = 1$, W is multiplication by $1/(u_1 v_1)$. Since $1/(u_1 v_1)$ is not an eigenvalue of W , this is impossible.

Let P be a compact Riemann surface of genus g . Suppose an injection of S into P has been chosen; we identify S with a finite set of points of P . If $P_{(s)}$ is a small disk around s , the fundamental group of $P_{(s)}^* := P_{(s)} - \{0\}$ is $\mathbf{Z}$, generated by one positive turn around s . More intrinsically, this local fundamental group is $\mathbf{Z}(1) := 2\pi i \mathbf{Z}$, with $2\pi i$ corresponding to the positive turn around s . For each $n \geq 1$, identify $\mathbf{Z}(1)/n$ with the group of n -th roots of unity in $\mathbf{C}$ by $z \mapsto \exp(z/n)$. Passing to the limit gives an isomorphism of $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$ with the projective limit of the $\mathbf{Z}(1)/n$ for n prime to p .

A complex representation ρ of $\mathbf{Z}(1)$ is isomorphic to a sum over characters $\chi : \mathbf{Z}(1) \rightarrow \mathbf{C}^*$:

$$\rho \simeq \bigoplus_{\chi} \chi \otimes \nu(\chi), \quad (2.14.1)$$

where $\nu(\chi)$ is a unipotent representation of $\mathbf{Z}(1)$. Define the partition $\lambda(\chi)$ of $\dim \nu(\chi)$ as in 2.9. The isomorphism class of ρ is determined by the function $\chi \mapsto \nu(\chi)$. We shall only have

to consider the case where the characters χ occurring in (2.14.1) are of finite order prime to p , and hence may be viewed as complex characters of $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$.

If T is a family $(T_s)_{s \in S}$ of isomorphism classes of complex representations of $\mathbf{Z}(1)$, we write $M_{\mathbf{C}}(T)$ for the moduli space of irreducible complex local systems of rank r on $P - S$, with local monodromy T_s at s . This moduli space has two natural structures of algebraic variety, under the names Betti and de Rham; only the underlying topological space will matter to us.

Choose

an isomorphism between the groups of roots of unity of order prime to p in $\overline{\mathbf{Q}}_{\ell}$ and in $\mathbf{C}$,

(2.14.2)

an isomorphism between the groups of roots of unity of order prime to p in $\mathbf{F}$ and in $\mathbf{C}$.

(2.14.3)

These data are exactly what is needed to attach to an isomorphism class of quasi-unipotent ℓ -adic representations of $\widehat{\mathbf{Z}}(1)$ an isomorphism class of complex representations of $\mathbf{Z}(1)$, as one checks by comparing 2.9 and 2.14.

If the $R(s)$ are tame, giving rise to representations ρ_s of $\widehat{\mathbf{Z}}(1)$, we write R^* for the corresponding family of complex representations $R^*(s)$ of $\mathbf{Z}(1)$, and we say that the moduli space $M_{\mathbf{C}}(R^*)$ corresponds to $E(R)$, or is the complex analogue of $E(R)$.

Conjecture 2.15. Under the hypotheses and with the notation of 2.14:

- (i) As a function of $n \geq 1$, the number of fixed points $N_n(R)$ of V^n acting on $E(R)$ is of the form

$$N_n(R) = \sum a_i \beta_i^n \quad (2.11.1)$$

for integers a_i and suitable numbers β_i .

- (ii) The sum of the a_i is equal to the Euler-Poincaré characteristic of the moduli space $M_{\mathbf{C}}(R^*)$ corresponding to $E(R)$.

- (iii) As in 1.2 (B), the dominant term in (2.15.1) is q to the power $\dim_{\mathbf{C}}(M_{\mathbf{C}}(R^*))/2$.

As will be explained in 6.6, if $g > 0$ then $\chi(M_{\mathbf{C}}(R^*)) = 0$. For a refinement of 2.15 (i) and (ii), see 6.3 and 6.7.

Using the analogy between the exponential and Artin-Schreier sheaves, it is possible to define complex analogues for certain wild representations ρ_s . Rather than moduli spaces of local systems on $P - S$, the complex analogues are moduli spaces of algebraic vector bundles with connection on $P - S$, with prescribed formal completions at each $s \in S$. I conjecture that 2.11 remains valid in this more general setting.

Variant 2.16. Let ρ and ρ' be two quasi-unipotent ℓ -adic representations $\rho, \rho' : \widehat{\mathbf{Z}}(1) \rightarrow \mathrm{GL}(r, \overline{\mathbf{Q}}_{\ell})$ (respectively two complex representations $\rho, \rho' : \mathbf{Z}(1) \rightarrow \mathrm{GL}(r, \mathbf{C})$). We say that the isomorphism class of ρ' is in the closure of the isomorphism class of ρ if ρ' is a limit of conjugates of ρ . With the notation of 2.9 (respectively 2.14), this means that, for each character χ , $\nu_{\rho}(\chi)$ and $\nu_{\rho'}(\chi)$ have the same dimension $r(\chi)$, and that the unipotent conjugacy class of $\mathrm{GL}(r(\chi))$ defined by the partition $\lambda_{\rho'}(\chi)$ of $r(\chi)$ is in the closure of that defined by $\lambda_{\rho}(\chi)$.

Transporting this definition to tame $\overline{\mathbf{Q}}_{\ell}$ -sheaves on $X_{(s)}^*$, one defines $\bar{E}(R)$ as the set of classes, up to semisimplification, of lisse $\overline{\mathbf{Q}}_{\ell}$ -sheaves F of rank r on $X - S$ such that for each $s \in S$, $F|_{X_{(s)}^*}$ lies in the closure of the isomorphism class of $R(s)$. One defines $\overline{M}_{\mathbf{C}}(R^*)$ as the moduli space of local systems of rank r on $P - S$, whose local monodromy at s lies in the closure of $R^*(s)$. These local systems are taken up to semisimplification.

One still has $V : \bar{E}(R) \rightarrow \bar{E}(R)$, and one can repeat Conjecture 2.15 for $\bar{E}(R)$ and for the corresponding $\overline{M}_{\mathbf{C}}(R^*)$. In some cases, the formulas giving the number of fixed points of V become simpler if $E(R)$ is replaced by $\bar{E}(R)$. See 4.5.

The precisions 2.17 to 2.22 that follow, added in July 2013, owe much to the talks I gave at IHES in March and April 2013.

2.5 2.17–2.22

Let us first place ourselves in the everywhere unramified case: S is empty, and E_r is the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r on X . Let $W(\mathbf{F})$ be the ring of Witt vectors of $\mathbf{F}$, let K be its fraction field, and let $\overline{K}$ be an algebraic closure of K . Let X_W be a lifting of X over W , and denote by X_K (respectively $X_{\overline{K}}$) the curve over K (respectively $\overline{K}$) obtained from it. Let M_K (respectively $M_{\overline{K}}$) be the scheme over K (respectively $\overline{K}$) which is the moduli space of geometrically irreducible vector bundles with connection of rank r on X_K (respectively $X_{\overline{K}}$). By extension of scalars from K to $\overline{K}$, M_K becomes $M_{\overline{K}}$.

Let ι be an embedding of $\overline{K}$ into $\mathbf{C}$, and in 2.14 take for P the Riemann surface deduced from $X_{\overline{K}}$ by extension of scalars from $\overline{K}$ to $\mathbf{C}$. For this choice of P , the corresponding space $M_{\mathbf{C}}$ defined in 2.14 is deduced from $M_{\overline{K}}$ by extension of scalars from K to $\mathbf{C}$, and, for every prime number ℓ' , the ℓ' -adic cohomology of $M_{\overline{K}}$ is therefore identified with the cohomology of $M_{\mathbf{C}}$ with coefficients in $\mathbf{Q}_{\ell'}$. The conjectures 2.15 (i) and (ii) (for S empty) are therefore consequences of the following.

Conjecture 2.18. The cohomology of $M_{\overline{K}}$ admits an endomorphism “which deserves to be called V^* ” such that, for each $n \geq 1$, the number N_n of fixed points of V^n acting on E_r is given by

$$N_n = \sum (-1)^i \text{Tr}(V^{*n}, H^i(M_{\overline{K}})). \quad (2.18.1)$$

In (2.18.1), H^i is ℓ' -adic cohomology for any prime number ℓ' . One should also be able to take de Rham cohomology of M_K/K . The clause in quotation marks has no precise sense, but suggests compatibilities. For instance, V^* should be an endomorphism of the cohomology algebra.

There will not generally exist an endomorphism $V : M_{\overline{K}} \rightarrow M_{\overline{K}}$ inducing V^* in cohomology, but I hope that in the analytic space $M_{\overline{K}}^{\text{an}}$ (in the sense of Berkovich) associated to $M_{\overline{K}}$ there is an open $M_{\overline{K}}^0$ such that

- (a) the restriction morphism $H^*(M_{\overline{K}}) = H^*(M_{\overline{K}}^{\text{an}}) \rightarrow H^*(M_{\overline{K}}^0)$ is an isomorphism; and
- (b) a crystalline interpretation defines $V = \text{Frob}^* : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$, inducing V^* in cohomology.

The conjectural morphism $V : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$ should send $M_{\overline{K}}^0$ into a proper open subset of $M_{\overline{K}}^0$, corresponding to vector bundles with connection for which parallel transport has a larger radius of convergence; this is what makes ordinary cohomology, rather than compactly supported cohomology, the natural choice in (2.18.1). A variant of Conjecture 2.18, for the completions 2.16 of E_r and $M_{\overline{K}}$, might be more natural.

Example 2.19. Suppose $r = 1$. In this case, $M_{\overline{K}}$ is the universal additive extension of the Jacobian $\text{Pic}^0(X_{\overline{K}})$, and

$$H^* \text{Pic}^0(X_{\overline{K}}) \xrightarrow{\sim} H^*(M_{\overline{K}}) \quad (2.19.1)$$

is an isomorphism. There is also a natural isomorphism

$$H^*(\text{Pic}^0(X)) \xrightarrow{\sim} H^*(\text{Pic}^0(X_{\overline{K}})). \quad (2.19.2)$$

Over $\mathbf{F}$, one has $V : \text{Pic}^0(X) \rightarrow \text{Pic}^0(X)$, deduced by functoriality from $\text{Frob} : X \rightarrow X$ (cf. 2.2). This endomorphism V of $\text{Pic}^0(X)$ gives

$$V^* : H^*(\text{Pic}^0(X)) \rightarrow H^*(\text{Pic}^0(X)). \quad (2.19.3)$$

Define the endomorphism

$$V^* : H^*(M_{\overline{K}}) \rightarrow H^*(M_{\overline{K}}) \quad (2.19.4)$$

to be transported from (2.19.3) by (2.19.2) and (2.19.1).

Proposition 2.20. If $r = 1$ and V^* is defined as above, then (2.18.1) is true.

Proof for $n = 1$. One reduces to this case by an extension of scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$. The endomorphisms Frob and V of $\text{Pic}^0(X)$ are transposes of one another by the autoduality of the Jacobian $\text{Pic}^0(X)$ of X . Hence

$$\text{Tr}(V^*, H^i(\text{Pic}^0(X))) = \text{Tr}(F^*, H^i(\text{Pic}^0(X)))$$

and

$$\sum (-1)^i \text{Tr}(V^*, H^i(\text{Pic}^0(X))) = |\text{Pic}^0(X_0)(\mathbf{F}_q)|. \quad (2.20.1)$$

Class field theory identifies the set of fixed points of V acting on E_1 with the Pontryagin dual of the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$, with values in the roots of unity of $\overline{\mathbf{Q}}_\ell$ (see 6.1). Thus

$$N_1 = |\text{Pic}^0(X_0)(\mathbf{F}_q)|, \quad (2.20.2)$$

and, by (2.20.1), this verifies (2.18.1). Notice that, again by the autoduality of the Jacobian, the subscheme $\text{Pic}^0(X)^V$ of $\text{Pic}^0(X)$ fixed by V is the Cartier dual of the finite abelian group $\text{Pic}^0(X)^{\text{Frob}=\text{Pic}^0(X_0)(\mathbf{F}_q)}$.

Now pass to the tamely ramified case. Choose over W a lifting (X_W, S_W) of X and of its etale divisor S . Let (X_K, S_K) and $(X_{\overline{K}}, S_{\overline{K}})$ be deduced from (X_W, S_W) by extension of scalars from W to K and to $\overline{K}$. To define in 2.14 a complex analogue $M_{\mathbf{C}}(R^*)$ of $E(R)$, we had to make the choices (2.14.2) and (2.14.3). To define a p -adic analogue we choose

$$\text{an isomorphism between the groups of roots of unity of order prime to } p \text{ in } \overline{\mathbf{Q}}_\ell \text{ and in } K. \quad (2.21.1)$$

There is no need to give an analogue of (2.14.3), because reduction modulo p supplies a canonical isomorphism between the groups of roots of unity of order prime to p in K or $\overline{K}$ and in $\mathbf{F}$.

Let $X_{K(s)}$ be the completion of X_K at (the lifting of) s , and $X_{K(s)}^* := X_{K(s)} - \{s\}$. The datum (2.21.1) gives a natural isomorphism between the finite-order characters $\widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$ and the prime-to- p part of $/\mathbf{Z}$. It is exactly what is needed to attach to $R(s)$ an isomorphism class $R(s)_K^*$ of vector bundles with connection on $X_{K(s)}^*$. Define $M_K(R^*)$ to be the scheme over K which is the moduli space of geometrically irreducible vector bundles with connection of rank r on $X_K - S_K$, with completion at $s \in S$ isomorphic to $R(s)_K$.

Let ι be an embedding of K into $\mathbf{C}$. If in 2.14 one takes P and $S \rightarrow P$ deduced from (X_K, S_K) by extension of scalars from K to $\mathbf{C}$, if (2.14.2) is deduced from (2.21.1) by ι , and if (2.14.3) is induced by ι , then $M_{\mathbf{C}}(R^*)$ defined in 2.14 is deduced from $M_K(R^*)$ by extension of scalars from K to $\mathbf{C}$.

Let $M_{\overline{K}}(R^*)$ be obtained from $M_K(R^*)$ by extension of scalars from K to $\overline{K}$. As in 2.18, I conjecture that the cohomology of $M_{\overline{K}}(R^*)$ admits an endomorphism “which deserves to be called V^* ” such that for $n \geq 1$, the number of fixed points of V acting on $E(R)$ is given by

$$N_n = \sum (-1)^i \text{Tr}(V^*, H^i(M_{\overline{K}}(R^*))), \quad (2.21.2)$$

and this conjecture implies 2.15 (i) and (ii).

Continue to assume that the $R(s)$ are tamely ramified, and assume condition (2.10.2). Write $\det$ for the functor “maximal exterior power” $\bigwedge^r$. It induces a map

$$\det : E(R) \rightarrow E(\det R). \quad (2.22.1)$$

Hypothesis (2.10.2) ensures that $\det R$ satisfies (2.10.1), and that $E(\det R)$ is nonempty, hence a torsor under the group E_1 of isomorphism classes of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank one on X . The action is tensor product.

In 2.16, for $r = 1$ and S empty, we defined a moduli space M_K . Here we denote it by M_{1K} . The functor $\bigwedge^r$ induces

$$\det : M_K(R^*) \rightarrow M_K(\det R^*), \quad (2.22.2)$$

and $M_K(\det R^*)$ is a principal homogeneous space under M_{1K} . Extend scalars from K to $\overline{K}$ and consider the Leray spectral sequence for the morphism $\det$:

$$E_2^{pq} = H^p(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \Rightarrow H^{p+q}(M_{\overline{K}}(R^*), \mathbf{Q}'_\ell).$$

I conjecture the existence of an endomorphism V^* of this spectral sequence, respecting its multiplicative structure. The E_2 -term is given by

$$H^p(M_{\overline{K}}(\det R^*)) \otimes H^0(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \xrightarrow{\sim} E_2^{pq},$$

and the cohomologies of $M_{\overline{K}}(\det R^*)$ and of $M_{1\overline{K}}$ are canonically isomorphic. The conjectured existence of V^* therefore implies 6.7.

3 Arinkin's Method

3.1 3.1–3.3

Let s_0 be a closed point of X_0 . One may identify the closed points s of X both with the $\mathbf{F}$ -points of the $\mathbf{F}$ -scheme X , and with the $\mathbf{F}$ -points of the $\mathbf{F}_q$ -scheme X_0 . In doing so, the s above s_0 are identified with the $\mathbf{F}$ -points of $\text{Spec}(k(s_0))$, i.e. with the embeddings of $k(s_0)$ into $\mathbf{F}$ extending the embedding of $\mathbf{F}_q$ into $\mathbf{F}$. If one chooses such an embedding s , a character $\varepsilon_0 : k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$ gives, by 2.9, a character $\varepsilon := \varepsilon_0 \text{pr}_s : \widehat{\mathbf{Z}}(1) \rightarrow k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$. If $s = \text{Frob}(s')$, the character ε' of $\widehat{\mathbf{Z}}(1)$ defined by ε_0 and s' is ε^q .

At each $s_0 \in S_0$, give a multiset $R(s_0)$ of r characters $k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$. By 3.1 one deduces, for every $s \in S$, a multiset of r characters $\mathcal{R}(s)$ of $\widehat{\mathbf{Z}}(1)$. Let ρ_s be the representation of $\widehat{\mathbf{Z}}(1)$ that is the sum of the one-dimensional representations defined by these characters. The corresponding $\overline{\mathbf{Q}}_\ell$ -sheaves $R(s)$ on the $X_{(s)}^*$ satisfy (2.1.1). Write R for the family $(R(s))_{s \in S}$. We assume that

$$\prod_s \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (3.2.1)$$

By (2.10.1), this is a necessary condition for the existence of a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F on $X - S$ with $F|_{X_{(s)}^*} \simeq R(s)$. If s_0 has degree d , if ε_0 is a character of $k(s_0)^*$, and if for every s above s_0 , $\varepsilon[s] := \varepsilon_0 \text{pr}_s$ is the corresponding character of $\widehat{\mathbf{Z}}(1)$, then the product of these $\varepsilon[s]$ is the character that 3.1 attaches to $\varepsilon_0^{(q^d-1)/(q-1)} = \varepsilon_0 \text{N}_{k(s_0)/\mathbf{F}_q}$ and to any s over s_0 . It is also the character that 3.1 attaches to $\varepsilon_0|_{\mathbf{F}_q^*}$ and to $\mathbf{F}_q \hookrightarrow \mathbf{F}$. Hypothesis (3.2.1) can therefore be rewritten as

$$\prod_{s_0 \in S_0} \prod_{\varepsilon_0 \in R(s_0)} \varepsilon_0|_{\mathbf{F}_q^*} = 1. \quad (3.2.2)$$

Assume further that the $\mathcal{R}(s)$ satisfy the general-position condition (2.10.3). I learned from Arinkin how, under this hypothesis, to construct an algebraic variety Z over $\mathbf{F}_q$ such that, for every $n \geq 1$, the number $N_n(R)$ of fixed points of the permutation V^n of $E(R)$ is equal to the number of $\mathbf{F}_{q^n}$ -points of Z :

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.3.1)$$

Under the hypotheses of 3.3, Conjecture 2.15 (i) is therefore true. Beware that Z will only be defined after cutting into locally closed pieces.

3.2 3.4–3.6

For each $s_0 \in S_0$, let $\varepsilon[s_0](i)$ ($0 \leq i \leq a(s_0)$) be the characters of $k(s_0)^*$ occurring in $R(s_0)$, arranged in any order. Let $n[s_0](i)$ be the multiplicity of $\varepsilon[s_0](i)$ in the multiset $R(s_0)$, and

let $n[s_0]$ be the family of integers $(n[s_0](i))_{0 \leq i \leq a(s_0)}$. For $s \in S$ above s_0 , write $\varepsilon[s](i)$ for the character $\varepsilon[s_0](i) \circ \text{pr}_s$ of $\widehat{\mathbf{Z}}(1)$ (3.1), and set $n[s] := n[s_0]$. The multiset $\mathcal{R}(s)$ is

$$\mathcal{R}(s) := \{\text{the } \varepsilon[s](i), \text{ with multiplicities } n[s](i)\}. \quad (3.4.1)$$

Let E_0 be a vector bundle of rank r on X_0 . A parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$ on E_0 is the datum, for each $s_0 \in S_0$, of a finite filtration $F(s_0)$ of the $k(s_0)$ -vector space $(E_0)_{s_0}$, such that $\text{Gr}_{F(s_0)}^i((E_0)_{s_0})$ has dimension $n[s_0](i)$. Parabolic structures for E on X are defined similarly. If E_0 on X_0 is endowed with a parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$, its inverse image E on X inherits a parabolic structure α of type $(n(s))_{s \in S}$. We say that (E_0, α_0) is geometrically indecomposable if its inverse image E on X admits no nontrivial decomposition $E = E' \oplus E''$ compatible with the parabolic structure α , i.e. inducing decompositions of the $F^i E_s$. Let $\text{End}(E_0, \alpha_0)$ (respectively $\text{End}(E, \alpha)$) be the algebra of endomorphisms of E_0 (respectively E) respecting the filtrations $F(s_0)$ of the $(E_0)_{s_0}$ (respectively $F(s)$ of the E_s). Each of the following conditions is equivalent to geometric indecomposability:

$$\text{End}(E, \alpha) = \text{End}(E_0, \alpha_0) \otimes_{\mathbf{F}_q} \mathbf{F} \text{ has no idempotent other than 0 and 1.} \quad (3.4.2)$$

$$\text{End}(E_0, \alpha_0) / \text{rad } \text{End}(E_0, \alpha_0) = \mathbf{F}_q. \quad (3.4.3)$$

Fix an integer d . Consider vector bundles E_0 of rank r and degree d on X_0 , endowed with a parabolic structure α of type $(n[s_0])_{s_0 \in S_0}$. Let $Z(\mathbf{F}_q)$ be the set of isomorphism classes of such (E_0, α) that are geometrically indecomposable. Define $Z(\mathbf{F}_{q^n})$ similarly, replacing $X_0/\mathbf{F}_q$ by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}/\mathbf{F}_{q^n}$. I learned from Arinkin the following theorem.

Theorem 3.5. Under the hypotheses of 3.2 and 3.3 and with the notation of 3.4, the number $N_n(R)$ of fixed points of V^n (2.1) is

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.5.1)$$

Replacing X_0 by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$, one reduces to the case $n = 1$. Beware that the proof does not suggest a natural bijection between $E(R)^V$ and $Z(\mathbf{F}_q)$. It gives a family of one-dimensional vector spaces indexed by $E(R)^V$, with direct sum V , another family, with direct sum V' , indexed by $Z(\mathbf{F}_q)$, and an isomorphism of vector spaces $V \simeq V'$.

Sketch of proof of 3.5 $\Rightarrow$ (3.3.1). For every $\mathbf{F}_q$ -scheme T , let $X_T := X_0 \times_{\mathbf{F}_q} T$ and $S_T := S_0 \times_{\mathbf{F}_q} T$. Define $Z(T)$ as the set of isomorphism classes of vector bundles of rank r on X_T , endowed with a parabolic structure along S_T , and such that the conditions of 3.2 and 3.3 are satisfied on each geometric fibre of $X_T \rightarrow T$: degree d , type of the parabolic structure, indecomposability.

Naively one would like to take for the scheme Z of (3.3.1) a scheme representing the functor Z . For several reasons this functor is not representable:

- (a) The objects considered have nontrivial automorphisms, for instance homotheties; thus this functor is not a sheaf, even for the Zariski topology. For the functor Pic , the same problem is bypassed by passing to the associated sheaf. Here the situation is worse, because, for (E_T, α_T) a vector bundle with parabolic structure on X_T , and for (E_t, α_t) the induced bundle on the fibre X_t of $X_T \rightarrow T$ at $t \in T$,
- (b) the dimension of the automorphism group of (E_t, α_t) (equal to $\dim \text{End}(E_t, \alpha_t)$) is not always locally constant in t .
- (c) The set of t for which (E_t, α_t) is geometrically indecomposable is not always locally closed.

Because of (c), one cannot even hope that Z comes from an algebraic stack C (i.e. that $Z(T)$ be the set of isomorphism classes in $C(T)$). Nevertheless:

- (d) The automorphism groups are connected algebraic groups. If $k' \subset k''$ are finite extensions of $\mathbf{F}_q$, this ensures that

$$Z(k') \xrightarrow{\sim} Z(k'')^{\text{Gal}(k''/k')}.$$

- (e) In (b) above, the function $t \mapsto \dim \text{End}(E_t, \alpha_t)$ is constructible; in (c), the set of t such that (E_t, α_t) is geometrically indecomposable is constructible.

Properties (d) and (e) make it possible, by “cutting Z into pieces” and taking the associated (representable) sheaves, to obtain Z as promised by (3.3.1).

3.3 3.7–3.16

Sketch of proof of 3.5 (for $n = 1$). One applies the strategy 2.5. We must show that $|Z(\mathbf{F}_q)|$ is the number of cuspidal automorphic representations π of $\text{GL}(r, \mathbf{A})$, counted up to $\mathbf{F}_q$ -twist, that are unramified outside S_0 and whose local component π_{s_0} at $s_0 \in S_0$ is of the following type. By the local Langlands correspondence, the representation π_{s_0} of $\text{GL}(r, K_{s_0})$ corresponds to an F -semisimple $\overline{\mathbf{Q}}_\ell$ -representation ρ of the Weil group $W(\overline{K}_{s_0}/K_{s_0})$. We first require ρ to be semisimple (in the language of the Weil–Deligne group: $N = 0$; in Arthur’s language: trivial action of the local $\text{SL}(2)$ or $\text{SU}(2)$). The abelianization $W(\overline{K}_{s_0}/K_{s_0})^{\text{ab}}$ is $K_{s_0}^*$. Its inertia group $O_{s_0}^*$ has quotient $k(s_0)^*$. We require the restriction of ρ to inertia to factor through the representation of $k(s_0)^*$ which is the sum of the ε_0 in $R(s_0)$.

Let $P[s_0] \subset \text{GL}(r, O_{s_0})$ be the parahoric subgroup inverse image of the standard parabolic $\overline{P}[s_0]$ of $\text{GL}(r, k(s_0))$ of type $n(s_0)$. The reductive quotient $L[s_0]$ of $\overline{P}[s_0]$ is $\prod \text{GL}(n[s_0](i), k(s_0))$. Let $U[s_0]$ be the kernel of $P[s_0] \rightarrow L[s_0]$, and let $\varepsilon[s_0]$ be the character

$$\varepsilon[s_0] = \prod \varepsilon[s_0](i)(\det g_i) : P[s_0] \rightarrow \overline{P}[s_0] \rightarrow L[s_0] \rightarrow \overline{\mathbf{Q}}_\ell^*$$

of the parahoric $P[s_0]$. By construction, the $\varepsilon[s_0](i)$ are pairwise distinct. We shall admit the following assertion.

Assertion 3.8. The irreducible admissible representations of $\text{GL}(r, K_{s_0})$ of the type defined in 3.7 are those which admit a line stable under P on which P acts through the character $\varepsilon[s_0]$. This line is unique.

Let A' be the set of cuspidal automorphic representations π of $\text{GL}(r, \mathbf{A})$ admitting a line $L(\pi)$ fixed by the $\text{GL}(r, O_x)$ for $x \notin S_0$, and stable by $P[s_0]$ for $s_0 \in S$, with the action of $P[s_0]$ given by $\varepsilon[s_0]$. The line $L(\pi)$ is unique if it exists. By Lafforgue [L] and 3.8, $N_1(R)$ is the number of classes modulo $\mathbf{F}_q$ -twist in A' . Fix an idele d of valuation 1, and let A be the set of π in A' whose central character ω_π is equal to 1 at d . As observed after 2.3.1, one has

$$N_1(R) = |A' \bmod \mathbf{F}_q\text{-twist}| = |A \bmod \mathbf{F}_q\text{-twist by } \zeta \in \mu_r(\overline{\mathbf{Q}}_\ell)| = \frac{1}{r} |A|. \quad (3.9.1)$$

Let $\mathcal{A}$ be the vector space of functions on $\text{GL}(r, \mathbf{A})$ which are

- (a) invariant on the left by $\text{GL}(r, K) \cdot d^{\mathbf{Z}}$;
- (b) invariant on the right under $\text{GL}(r, O_x)$, for $x \notin S_0$;
- (c) transformed by $\varepsilon[s_0]$ for the right action of $P[s_0]$, for $s_0 \in S_0$.

A key point of the proof of 3.5 is the following lemma.

Lemma 3.11. The vector space $\mathcal{A}$ is the direct sum of the lines $L(\pi)$, for $\pi \in A$. It is therefore of dimension $rN_1(R)$.

This lemma is the automorphic counterpart, and a consequence, of the fact that irreducibility of the $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 under consideration follows from the local hypotheses made. One must check that the functions in $\mathcal{A}$ are automatically cuspidal. This done, if one writes the space

of cuspidal functions on $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$ as a direct sum of irreducible representations π of $\mathrm{GL}(r, \mathbf{A})$, the π in A contribute $L(\pi)$ to $\mathcal{A}$, and the others contribute nothing. That the functions in $\mathcal{A}$ are cuspidal follows from the fact that, in the spectral decomposition of functions on $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$, the induced representations giving rise to Eisenstein series admit no nonzero vector satisfying 3.10 (b) and (c).

The space $\mathcal{A}$, the sum of the $L(\pi)$ for $\pi \in A$, is identified with the space of functions f on

$$\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A}) / \prod_{x \notin S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.1)$$

such that, for every $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$. Let v be the map from (3.12.1) to $\mathbf{Z}/r$ induced by $v \det : \mathrm{GL}(r, \mathbf{A}) \rightarrow \mathbf{Z}$. Let $\bar{A}$ be the quotient of A by the $\mathbf{F}_q$ -twist action of $\mu_r(\mathbf{Q}_\ell)$, and for each orbit B , let V_B be the subspace of $\mathcal{A}$ which is the sum of the $L(\pi)$ for $\pi \in B$. By (2.3.1), $\dim(V_B) = r$. For $m \in \mathbf{Z}/r$, write $(3.12.1)_m$ for the part of (3.12.1) where $v = m$, let φ_m be its characteristic function, and put $L(\pi, m) = \{\varphi_m f \mid f \in L(\pi)\}$. The dimension of $L(\pi, m)$ is at most one. For each character η of $\mathbf{Z}/r$, if π' is the $\mathbf{F}_q$ -twist of π by $\eta(1)$, multiplication by η^v transforms $L(\pi)$ into $L(\pi')$. For $\pi \in B$, $L(\pi, m)$ is therefore independent of π ; write $L(B, m) := L(\pi, m)$. The space V_B is stable under the multiplications by the η^v , hence under multiplication by the φ_m : for $\pi \in B$,

$$V_B = \bigoplus_{m \in \mathbf{Z}/r} L(B, m).$$

Each $L(B, m)$ is therefore one-dimensional. The sum $\mathcal{A}_m$ of the $L(B, m)$ is of dimension $|\bar{A}| = N_1(R)$. It is identified with the space of functions f on $(3.12.1)_m$ such that, for each $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$.

Let $\mathrm{GL}(r, \mathbf{A})^{(d)}$ be the set of $g \in \mathrm{GL}(r, \mathbf{A})$ such that $v \det(g) = d$. If $\bar{d}$ is the class of d modulo r , the set

$$\mathrm{GL}(r, K) \backslash \mathrm{GL}(r, \mathbf{A})^{(d)} / \prod_{x \in S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.2)$$

maps bijectively onto $(3.12.1)_{\bar{d}}$. Hence:

Lemma 3.13. $N_1(R)$ is the dimension of the space, again denoted $\mathcal{A}_d$, of functions f on (3.12.2) which, for the right action of the $P[s_0]$ ($s_0 \in S_0$), satisfy $f(gp) = \varepsilon[s_0](p)f(g)$.

The set (3.12.2) is interpreted as the set of isomorphism classes of triples (E_0, α, β) , where E_0 is a vector bundle of rank r and degree d on X_0 , and where α and β are structures of the following types on E_0 :

- (a) α is a parabolic structure of type $(n[s_0])_{s_0 \in S_0}$;
- (b) β is the datum of bases of the vector spaces $\mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0})$.

We identify a basis as in (b) with an isomorphism

$$\beta_{s_0, i} : k(s_0)^{n[s_0](i)} \xrightarrow{\sim} \mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0}).$$

With this interpretation, the space $\mathcal{A}_d$ of 3.13 becomes the space of functions $f(E_0, \alpha, \beta)$, for E_0, α, β as above, such that

- (i) $f(E_0, \alpha, \beta)$ depends only on the isomorphism class of (E_0, α, β) ;
- (ii) for $s_0 \in S_0$ and $g \in L[s_0] = \prod \mathrm{GL}(n_i, k(s_0))$, one has

$$f(E_0, \alpha, \beta g) = \varepsilon[s_0](g)f(E_0, \alpha, \beta).$$

For an automorphism T of (E_0, α) , let $\text{Gr}_F^i(s_0)(T_{s_0})$ be the automorphism of $\text{Gr}_F^i(s_0)((E_0)_{s_0})$ induced by T , and put

$$\chi(T) := \prod_{s_0} \prod_i \varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})). \quad (3.14.1)$$

For any β , hypothesis (i) gives $f(E_0, \alpha, T(\beta)) = f(E_0, \alpha, \beta)$. Hypothesis (ii) gives $f(E_0, \alpha, T(\beta)) = \chi(T)f(E_0, \alpha, \beta)$. If there exists an automorphism T with $\chi(T) \neq 1$, then $f(E_0, \alpha, \beta) = 0$. Conversely, if $\chi(T) = 1$ for every automorphism T of (E_0, α) , then f may be prescribed freely at one (E_0, α, β) , and its value there determines its value at all (E_0, α, β') . Thus the dimension of $\mathcal{A}_d$ is the number of isomorphism classes of (E_0, α) as above such that for every automorphism T , one has $\chi(T) = 1$. Theorem 3.5 follows from the characterization (3.4.3) of geometric indecomposability and the following proposition.

Proposition 3.15. Let E_0 be a vector bundle of rank r on X_0 , endowed with a parabolic structure of type $(n(s_0))_{s_0 \in S_0}$. The following conditions are equivalent:

- (i) The quotient of the $\mathbf{F}_q$ -algebra $\text{End}(E_0, \alpha)$ by its radical is reduced to $\mathbf{F}_q$.
- (ii) For every automorphism T of (E_0, α) , $\chi(T)$, defined by (3.14.1), is equal to 1.

We shall deduce that (i) implies (ii) from (3.2.2), and that the negation of (i) implies the negation of (ii) from the general-position hypothesis (2.10.3).

Proof of (i) $\Rightarrow$ (ii). If λ is the image in $\mathbf{F}_q^*$ of the automorphism $T \in \text{End}(E_0, \alpha)^*$, then

$$\varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})) = \varepsilon(s_0)(i)^{n[s_0](i)}(\lambda),$$

so that (3.2.2) implies $\chi(T) = 1$.

Proof of (ii) $\Rightarrow$ (i). The quotient of the $\mathbf{F}_q$ -algebra $\text{End}(E_0, \alpha)$ by its radical is a product of matrix algebras over finite extensions of $\mathbf{F}_q$. If it is not reduced to $\mathbf{F}_q$, it contains a commutative separable $\mathbf{F}_q$ -algebra of dimension > 1 . This algebra admits a lifting into $\text{End}(E_0, \alpha)$. Suppose then that $\text{End}(E_0, \alpha)$ contains a commutative separable subalgebra k of dimension > 1 . We must show that the character (3.14.1) of $\text{End}(E_0, \alpha)^*$ is nontrivial. We shall prove that its restriction χ to k^* is nontrivial.

Split case. Suppose first that k is a product of copies of $\mathbf{F}_q$, say $k \simeq \mathbf{F}_q^I$, and that the closed points $s_0 \in S_0$ have degree 1, so that $\mathbf{F}_q \xrightarrow{\sim} k(s_0)$. The $\mathbf{F}_q^I$ -module structure of E_0 gives a decomposition

$$E_0 = \bigoplus_{\iota \in I} E_0^\iota \quad (3.15.1)$$

such that $(\lambda_\iota)_{\iota \in I} \in \mathbf{F}_q^I$ acts on E_0^ι by multiplication by λ_ι . The decomposition (3.15.1) is compatible with the parabolic structure. For $\iota \in I$, put

$$n^\iota[s_0](i) = \dim \text{Gr}_{F(s_0)}^i((E_0^\iota)_{s_0}).$$

The sum over i of the $n^\iota[s_0](i)$ is the rank of the bundle E_0^ι . The restriction of χ to the factor with index ι of $(\mathbf{F}_q^I)^* = \bigoplus_{\iota \in I} \mathbf{F}_q^*$ is

$$\lambda \longmapsto \prod_{s_0 \in S_0} \prod_i \varepsilon[s_0](i)^{n^\iota[s_0](i)}(\lambda). \quad (3.15.2)$$

Its composite with the morphism $\widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_q^*$ of 3.1 is the character

$$\prod_{s \in S} \prod_i \varepsilon[s](i)^{n^\iota[s_0](i)}$$

of $\widehat{\mathbf{Z}}(1)$. Hypothesis (2.10.3) ensures that this character is nontrivial.

Reduction of the general case to the split case. It suffices to show that nontriviality of χ is invariant under extension of scalars from $\mathbf{F}_q$ to a finite extension $\mathbf{F}_{q^n}$. Under such an extension of scalars, X_0 becomes a curve X' over $\mathbf{F}_{q^n}$, S_0 becomes a divisor S' of X' , and k becomes $k' := k \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$. For $s' \in S'$ above $s_0 \in S_0$, put $n[s'] := n[s_0]$, and let $\varepsilon[s'](i)$ be the composite of $\varepsilon[s_0](i)$ with the norm $N_{k(s')/k(s_0)}$. The bundle E' on X' inverse image of E_0 inherits a parabolic structure α' of type $(n(s'))_{s' \in S'}$. Repeating for E' on $X'/\mathbf{F}_{q^n}$ the construction (3.14.1), one obtains a character χ' of k'^* . The curve $X/\mathbf{F}$, the integers $n[s](i)$, and the characters $\varepsilon[s](i)$ of $\hat{\mathbf{Z}}(1)$, come equally from $X_0/\mathbf{F}_q$ and from $X'/\mathbf{F}_{q^n}$.

To prove that triviality of χ is equivalent to triviality of χ' , it suffices to prove that

$$\chi' = \chi \circ N_{k'/k}. \quad (3.15.3)$$

Indeed, the norm $N_{k'/k}$ is surjective. Identity (3.15.3) follows from identity (3.16.1) below, applied to each $\mathrm{Gr}_F^i(s_0)(E_{s_0})$, endowed with its $(k, k(s_0))$ -bimodule structure.

Let k_0 be a finite extension of $\mathbf{F}_q$, N_0 a finite-dimensional k_0 -vector space, k a product of finite extensions of $\mathbf{F}_q$, and $\rho : k \rightarrow \mathrm{End}_{k_0}(N_0)$. The (k, k_0) -bimodule N defines a homomorphism

$$[N] : k^* \rightarrow k_0^*, \quad \lambda \mapsto \det(\rho(\lambda)).$$

Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$. One obtains k'_0 , k' , and a (k', k'_0) -bimodule N' . This bimodule defines

$$[N'] : k'^* \rightarrow k'_0^*, \quad \lambda \mapsto \det_{k'_0}(\rho(\lambda)).$$

Lemma 3.16. The diagram

$$\begin{array}{ccc} k'^* & \xrightarrow{[N']} & k'_0^* \\ N_{k'/k} \downarrow & & \downarrow N_{k'_0/k_0} \\ k^* & \xrightarrow{[N]} & k_0^* \end{array} \quad (3.16.1)$$

is commutative.

Proof. Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}$. Diagram (3.16.1) embeds in an analogous diagram (3.16.2), where k and k_0 are replaced by $k_F := k \otimes_{\mathbf{F}_q} \mathbf{F}$, $k_{0F} := k_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, where N_0 is replaced by a (k_F, k_{0F}) -bimodule N , and where the extension $\mathbf{F}_{q^n}$ of $\mathbf{F}_q$ is replaced by a product $\mathbf{F}^J$ of n copies of $\mathbf{F}$:

$$\begin{array}{ccc} (k_F \otimes \mathbf{F}^J)^* & \xrightarrow{[N \otimes \mathbf{F}^J]} & (k_{0F} \otimes \mathbf{F}^J)^* \\ \downarrow & & \downarrow \\ k_F^* & \xrightarrow{[N]} & k_{0F}^*. \end{array} \quad (3.16.2)$$

One has $(k_F \otimes \mathbf{F}^J)^* = (k_F^*)^J$, $(k_{0F} \otimes \mathbf{F}^J)^* \simeq (k_{0F}^*)^J$, and the vertical “norm” maps are the product maps. The first horizontal line is the product of copies, indexed by J , of the second horizontal line. The commutativity of (3.16.2) follows.

4 Trace Formula

4.1 4.1–4.5

In [DF], the counting problem 2.3 (i) is solved when $|S_0| \geq 2$ and the imposed local monodromies are unipotent, with a single Jordan block. In automorphic language this means asking the local components π_{s_0} ($s_0 \in S_0$) to be of the form

$$\text{special representation} \otimes \chi \det g, \quad (4.1.1)$$

for χ an unramified character of $K_{s_0}^*$.

The hypothesis $|S_0| \geq 2$ allows one to pass from automorphic representations for $\mathrm{GL}(r, K)$ to automorphic representations for the multiplicative group of a division algebra D of dimension r^2 over K . It is reassuring to use the trace formula only in a case where the quotient is compact (modulo the centre), but it should be possible instead to use the trace formula for $\mathrm{GL}(n)$, for a suitable test function, with the simplification brought by the fact that one wants to detect automorphic representations which at two places are in the discrete series. After all, this is how one relates $\mathrm{GL}(r, K)$ and D^* . The explanations that follow will be simplified by staying with $\mathrm{GL}(r, K)$, but admitting that the only nonzero terms in the required trace formula are those associated with conjugacy classes of elliptic elements of finite order in $\mathrm{GL}(r, K)$.

Choose an embedding $\mathbf{F}_{q^r} \hookrightarrow M_r(\mathbf{F}_q)$. It defines a morphism $\mathbf{F}_{q^r}^* \hookrightarrow \mathrm{GL}(r, \mathbf{F}_q) \hookrightarrow \mathrm{GL}(r, K)$, through which the set of orbits of $\mathrm{Gal}(\mathbf{F}_{q^r}/\mathbf{F}_q)$ in $\mathbf{F}_{q^r}^*$ maps bijectively to the set of conjugacy classes under consideration.

Let $T(\gamma)$ be the term of the trace formula (giving the number of fixed points of V) thus associated to $\gamma \in \mathbf{F}_{q^r}^*$. It depends only on the subextension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ generated by γ . Put $T(m) := T(\gamma)$. The number c_m of elements of $\mathbf{F}_{q^m}^*$ which generate the extension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ is

$$c_m = \sum_{a|m} \mu(a)(q^{m/a} - 1).$$

The number of fixed points of V is

$$N_1 = \sum_{m|r} \frac{c(m)}{m} T(m) \tag{4.1.2}$$

$$= \sum_{m|r} \sum_{ab=m} \frac{1}{m} \mu(a)(q^b - 1) T(ab) \tag{1}$$

$$= \sum_b \sum_{ab|r} \frac{1}{m} \mu(a)(q^b - 1) T(ab). \tag{2}$$

The number of fixed points of V^n ($n \geq 1$) is obtained similarly, after extending scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$.

Surprise 4.2. When n varies, each $T(m)$ is generally not, as a function of n , of the form (1.2.1). On the other hand, in (4.1.2), each term of the sum over b is of this form.

I have no explanation, neither geometric nor automorphic, of 4.2, only a proof. I likewise have no explanation for the following.

Surprise 4.3. Under the hypotheses of 4.1, the number N_1 of fixed points of V is divisible by q .

If $r = 2$, the trace formula is a little less frightening. Drinfeld [Dr] used it for the case where S_0 is empty. Flicker (unpublished) treated the case where $|S_0| = 1$ and where the local monodromy at each $s \in S$ is unipotent with a single Jordan block, i.e. since $r = 2$, nontrivial. Gathering the information thus obtained, one gets:

Surprise 4.5. Let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy (both nontrivial and trivial). The number of fixed points of $V := \mathrm{Frob}^*$ on E depends only on $X_0/\mathbf{F}_q$ and on $|S|$.

This is the surprise to which 2.16 alludes. If, as in [DF], one requires the local monodromy at each $s \in S$ to be unipotent with a single Jordan block, the number of fixed points depends on X_0 , on $|S|$, and also on the action of Frobenius on S (viewed as a conjugacy class in the symmetric group $S_{|S|}$).

5 Examples

5.1 5.1–5.9

Take $X_0 = \mathbf{P}^1$, $|S| = 4$, $r = 2$, and let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with local monodromy at each $s \in S$ nontrivial unipotent. In this case, the number of fixed points of the iterates of the permutation $V = \text{Frob}^*$ of E is given by

$$N_n = q^n. \quad (5.1.1)$$

If $|S_0| \geq 2$, (5.1.1) is an application of the main result of [DF]. The following curious result ([DF] §7) allows one to deduce the case where S_0 is reduced to one closed point of degree 4.

Proposition 5.2. Each permutation σ of type $(2, 2)$ of S extends to a projectivity of $\mathbf{P}^1$. The action of this projectivity on E is trivial.

The proof is by reduction to an analogous theorem over $\mathbf{C}$.

It is natural to complete the set E of 5.1 to $\bar{E}$, the set of isomorphism classes, up to semisimplification, of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy at each $s \in S$. Since $X_0 = \mathbf{P}^1$ and $|S| = 4$, this completion $\bar{E}$ differs from E only by the adjunction of one point corresponding to the constant $\overline{\mathbf{Q}}_\ell$ -sheaf $\overline{\mathbf{Q}}_\ell^2$. We again write V for the permutation of $\bar{E}$ induced by Frob^* . The number $\bar{N}_n$ of fixed points of V^n acting on $\bar{E}$ is

$$\bar{N}_n = q^n + 1 \quad (\text{fixed points on } \bar{E}). \quad (5.3.1)$$

The complex analogue of $\bar{E}$ is the moduli space $\bar{M}$ of complex local systems of rank 2 on $\mathbf{P}^1 - S$, for $|S| = 4$, with unipotent local monodromy. These local systems are taken up to semisimplification. If one chooses a base point $o \in \mathbf{P}^1 - S$, the functor $F \mapsto F_o$ is an equivalence from local systems to representations of $\pi_1(\mathbf{P}^1 - S, o)$, and one can take as coordinates on $\bar{M}$ the traces of elements of π_1 . The space $\bar{M}$ is affine, purely of dimension 2, with a unique singular point corresponding to the constant local system $\mathbf{C}^2$. The singularity is of type D_4 . The nonzero Betti numbers of $\bar{M}$ are

$$b_0 = 1, \quad b_2 = 1. \quad (5.4.1)$$

I have not made a complete verification, but the situation seems exactly the same for $X_0 = \mathbf{P}^1$, $|S| = 3$, $r = 3$, with unipotent local monodromy. Again, if one of the local monodromies is not a single Jordan block, the $\overline{\mathbf{Q}}_\ell$ -sheaf is an iterated extension of constant local systems $\overline{\mathbf{Q}}_\ell$. The analogue of 5.3 holds for σ a cyclic permutation of the three points of S . The number of fixed points of V^n is again $q^n + 1$, with the “1” given by reducible local systems with unipotent local monodromy, all having a constant local system as semisimplification. The unique singular point of the complex analogue is of type E_6 .

Take $X_0 = \mathbf{P}^1$, $|S| = 4$, $r = 2$, and assume that, as in 3.4, the imposed local monodromy is, at each $s_0 \in S_0$, given by two distinct characters $\alpha'[s_0]$, $\alpha''[s_0]$ of $k(s_0)^*$. Assume (3.2.2) and the general-position hypothesis (2.10.3). Using (3.5.1), one obtains:

Proposition 5.7. Under the hypotheses of 5.6, one has

$$N_1 = q + 1 + |S_0(\mathbf{F}_q)|. \quad (5.7.1)$$

Arinkin has checked that for $X_0 = \mathbf{P}^1$ and $r = 2$, if a vector bundle E_0 of rank 2 on X_0 with parabolic structure α_0 at S_0 is indecomposable, then $\text{End}(E_0, \alpha)$ is reduced to the scalars. If one defines Z as in 3.6, and passes to the associated sheaf for the Zariski topology, one obtains a functor representable by an algebraic space. In the case $|S| = 4$, if one takes the degree $d = 1$, it is represented by the sum of two copies of X_0 , glued along $X_0 - S_0$. The number of points of this scheme over $\mathbf{F}_q$ is given by (5.7.1).

Pass to the complex analogue. On $\mathbf{P}^1 - S$, with $|S| = 4$, consider the moduli space of complex local systems of rank 2 with prescribed local monodromy at each $s \in S$. At $s \in S$, require the local monodromy to be conjugate to $A_s = \begin{pmatrix} a_s & 0 \\ 0 & b_s \end{pmatrix}$. Assume $\prod_s a_s b_s = 1$, and that if, for every s , c_s is one of a_s or b_s , one never has $\prod c_s = 1$. Put $A := (A_s)_{s \in S}$, and write $M(A)$ for the moduli space, 2.10. It is smooth and purely of dimension 2. One may view $M(A)$, for A_s close to 1, as a deformation of $\bar{M}$ considered in 5.4. In this deformation, the space of vanishing cycles has dimension 4, since the singularity of $\bar{M}$ is of type D_4 . At infinity the geometry does not change, and the nonzero Betti numbers are therefore

$$b_0 = 1, \quad b_2 = 5. \quad (5.8.1)$$

Take $X_0 = \mathbf{P}^1$, $|S| = 3$, $r = 3$, and assume that, as in 3.4, the imposed local monodromy is at each $s_0 \in S_0$ given by three distinct characters of $k(s_0)^*$. Assume (3.2.2) and the general-position hypothesis 2.10.3. Using (3.5.1), one obtains here

$$N_1 = q + 1 + 2|S_0(\mathbf{F}_q)|. \quad (5.9.1)$$

The sheaf associated to Z as in 3.6 is indeed, for $d = 1$, represented by the sum of three copies of X_0 , glued along $X_0 - S_0$. If S_0 consists of three rational points over $\mathbf{F}_q$, (5.9.1) says that $N_1 = (q + 1) + 6$, as suggested by the E_6 singularity of the complex analogue for unipotent local monodromy (cf. 5.5).

6 Rank 1

6.1 6.1–6.7

Let $E(1, \emptyset)$ denote the set $E(R)$ of 2.1 for $r = 1$ and $S_0 = \emptyset$. Tensor product gives it the structure of an abelian group. Class field theory identifies $E(1, \emptyset)^V$ with the group of $\bar{\mathbf{Q}}_\ell^*$ -valued characters of the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$. Indeed, this theory identifies the abelianized fundamental group of X_0 with the profinite completion of $\text{Pic}(X_0)$. The group $\text{Pic}(X_0)$ maps to $\mathbf{Z}$ by the degree map, with kernel the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$ of $\mathbf{F}_q$ -points of the Jacobian $\text{Pic}^0(X_0)$. The isomorphism classes of lisse $\bar{\mathbf{Q}}_\ell^*$ -sheaves F_0 of rank one on X_0 are thus identified with characters χ of $\text{Pic}(X_0)$ with values in $\bar{\mathbf{Z}}_\ell^*$; $\mathbf{F}_q$ -twist by λ corresponds to multiplication by the character λ^n of $\mathbf{Z}$; and, by attaching to F_0 the restriction of χ to $\text{Pic}^0(X_0)(\mathbf{F}_q)$, one obtains a bijection between $\mathbf{F}_q$ -twist classes of lisse F_0 of rank 1 on X_0 and characters of $\text{Pic}^0(X_0)(\mathbf{F}_q)$.

For arbitrary $r \geq 1$ and R as in 2.1, tensor product induces an action of the abelian group $E(1, \emptyset)$ on $E(R)$, and of $E(1, \emptyset)^V$ on $E(R)^V$. If $r = 1$, and R satisfies the compatibility (2.10.1), class field theory shows that $E(R)^V$ is a principal homogeneous space under $E(1, \emptyset)^V$. Thus

$$|E(R)^V| = |E(1, \emptyset)^V| = |\text{Pic}^0(X_0)(\mathbf{F}_q)| \quad (r = 1 \text{ and } (2.10.1)). \quad (6.2.1)$$

If $r > 1$, taking the exterior power $\wedge^r$ gives a map

$$\det : E(R) \rightarrow E(\bigwedge^r R)$$

such that, for the action of $E(1, \emptyset)$, denoted $\otimes$, one has

$$\det(l \otimes F) = l^r \otimes \det(F).$$

Similarly, after passage to the fixed points of $V = \text{Frob}^*$, for $\det : E(R)^V \rightarrow E(\bigwedge^r R)^V$ and the action of $E(1, \emptyset)^V$.

The action of $E(1, \emptyset)^V$ on $E(R)^V$ is not necessarily free, and the fibres of

$$\det : E(R)^V \rightarrow E(\bigwedge^r R)^V$$

do not necessarily all have the same number of elements. Nevertheless, in all cases that have been computed, $|E(1, \emptyset)^V|$ divides $|E(R)^V|$. As a function of n , $|\mathrm{Pic}^0(X_0)(\mathbf{F}_{q^n})|$ has the form (2.15.1):

$$|\mathrm{Pic}^0(X_0)(\mathbf{F}_{q^n})| = \sum (-1)^i \mathrm{Tr}(\mathrm{Frob}^*, H^i(\mathrm{Pic}^0(X), \mathbf{Q}_\ell)) = \sum (-1)^i \mathrm{Tr}(\mathrm{Frob}^*, \bigwedge^i H^1(X, \mathbf{Q}_\ell)).$$

I conjecture the following precise form of Conjecture 2.15 (i).

Conjecture 6.3. As a function of n , the number of fixed points $N_n(R)$ of V^n is of the form

$$N_n(R) = |\mathrm{Pic}^0(X_0)(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n \quad (6.3.1)$$

for integers c_k and suitable numbers γ_k .

Here is an analogue of 6.3 in algebraic geometry. Let A_0 be an abelian variety over $\mathbf{F}_q$, and let $f_0 : X_0 \rightarrow A_0$ be a proper smooth morphism. Although the fibres of $f : X_0(\mathbf{F}_{q^n}) \rightarrow A_0(\mathbf{F}_{q^n})$ need not have the same number of elements (example: $X_0 = A_0$ and f_0 multiplication by an integer prime to p), $|A_0(\mathbf{F}_{q^n})|$ divides $|X_0(\mathbf{F}_{q^n})|$. Indeed, the $R^i f_! \mathbf{Q}_\ell$ are lisse $\mathbf{Q}_\ell$ -sheaves on A , and the (abelian) fundamental group of A acts on these sheaves. One knows that this action is semisimple. Let $(R^i f_! \mathbf{Q}_\ell)_0$ be the subsheaf of invariants. It comes from a subsheaf $(R^i f_{0!} \mathbf{Q}_\ell)_0$ of $R^i f_{0!} \mathbf{Q}_\ell$ on A_0 , and this last sheaf is the inverse image on A_0 of a sheaf on $\mathrm{Spec}(\mathbf{F}_q)$, denoted $H^i(X_0/A_0)$. One knows that

$$H^i(A, (R^j f_! \mathbf{Q}_\ell)_0) \xrightarrow{\sim} H^i(A, R^j f_! \mathbf{Q}_\ell). \quad (6.4.1)$$

Grothendieck's trace formula for the number of $\mathbf{F}_q$ -points of X_0 can therefore be rewritten as

$$|X_0(\mathbf{F}_{q^n})| = \mathrm{Tr} \left(\mathrm{Frob}^*, \left(\sum (-1)^i H^i(A, \mathbf{Q}_\ell) \right) \otimes \left(\sum (-1)^j H^j(X/A) \right) \right). \quad (6.4.2)$$

If the γ_k are the eigenvalues of Frobenius on the $H^j(X/A)$, and the c_k are the alternating sums of their multiplicities, one deduces from (6.4.2) that

$$|X_0(\mathbf{F}_{q^n})| = |A_0(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n, \quad (6.4.3)$$

a formula analogous to (6.3.1). The promised divisibility of $|X_0(\mathbf{F}_{q^n})|$ by $|A_0(\mathbf{F}_{q^n})|$ follows from the fact that the Frobenius eigenvalues on $H^j(X/A)$, contained in the cohomology of a fibre of $X_0 \rightarrow A_0$, are algebraic integers.

In 6.4 the crucial point is not that f_0 is proper and smooth, but that the $R^i f_! \mathbf{Q}_\ell$ are lisse. One can avoid having to assume that the $R^i f_! \mathbf{Q}_\ell$ are semisimple by considering, rather than the subsheaf of invariants for the action of $\pi_1(A)$, the largest subsheaf on which the action is unipotent. Assume that there is an action of A_0 on X_0 , and an integer r , such that f_0 is equivariant for the action of A_0 on A_0 by $x : y \mapsto rx + y$. Under this hypothesis, the inverse images of the $R^i f_! \mathbf{Q}_\ell$ on A by multiplication by $r : A \rightarrow A$ are constant sheaves. This implies both their lissity and their semisimplicity.

If $g \geq 1$, Conjecture 6.3 implies that in the conjectural expression (2.11.1) for $N_n(R)$, one has $\sum a_i = 0$. Pass to the complex analogues as in 2.14. Let $M(1, \emptyset)$ be the group of isomorphism classes of complex local systems of rank 1 on P . It is isomorphic to $(\mathbf{C}^*)^{2g}$, and, if the compatibility $\prod \det R^*(s) = 1$ is satisfied, $M(\det R^*)$ is a principal homogeneous space under $M(1, \emptyset)$. The group $M(1, \emptyset)$ acts on $M(R^*)$ and on $M(\det R^*)$. Denote the action by $\otimes$. The map induced by $\bigwedge^r$,

$$\det : M(R^*) \rightarrow M(\det R^*),$$

satisfies $\det(l \otimes F) = l^r \otimes \det(F)$. This implies that $\det$ is a fibration, that the $R^i \det_! \mathbf{Z}$ are locally constant, and that the action of $\pi_1 M(\det R^*)$ on the $R^i \det_! \mathbf{Z}$ factors through its quotient $\pi_1 / r\pi_1 \simeq (\mathbf{Z}/r\mathbf{Z})^{2g}$. If $g \geq 1$, one deduces by arguments analogous to those of 6.4 that $\chi(M(R^*)) = 0$, in agreement with 2.15 (ii). Let $(R^i \det_! \mathbf{Q})_0$ be the local subsystem of $R^i \det_! \mathbf{Q}$ of invariants under the action of π_1 , and let $\chi(M(R^*)/M(\det R^*))$ be the alternating sum of the ranks of the $(R^i \det_! \mathbf{Q})_0$.

Conjecture 6.7. With the notation of 6.3 and 6.6, one has

$$\sum c_k = \chi(M(R^*)/M(\det R^*)).$$

Bibliography

- [DF] P. Deligne and Y. Z. Flicker. Counting local systems with principal unipotent local monodromy, *Annals of Math.* (to appear).
- [Dr] V. Drinfeld. The number of two dimensional irreducible representations of the fundamental group of a curve over a finite field, *Funk. anal. i Prilozhen.* **15** 4 (1981), 75–76.
- [L] L. Lafforgue. Chtoucas de Drinfeld et correspondance de Langlands, *Inv. math.* **147** 1 (2002), 1–241.